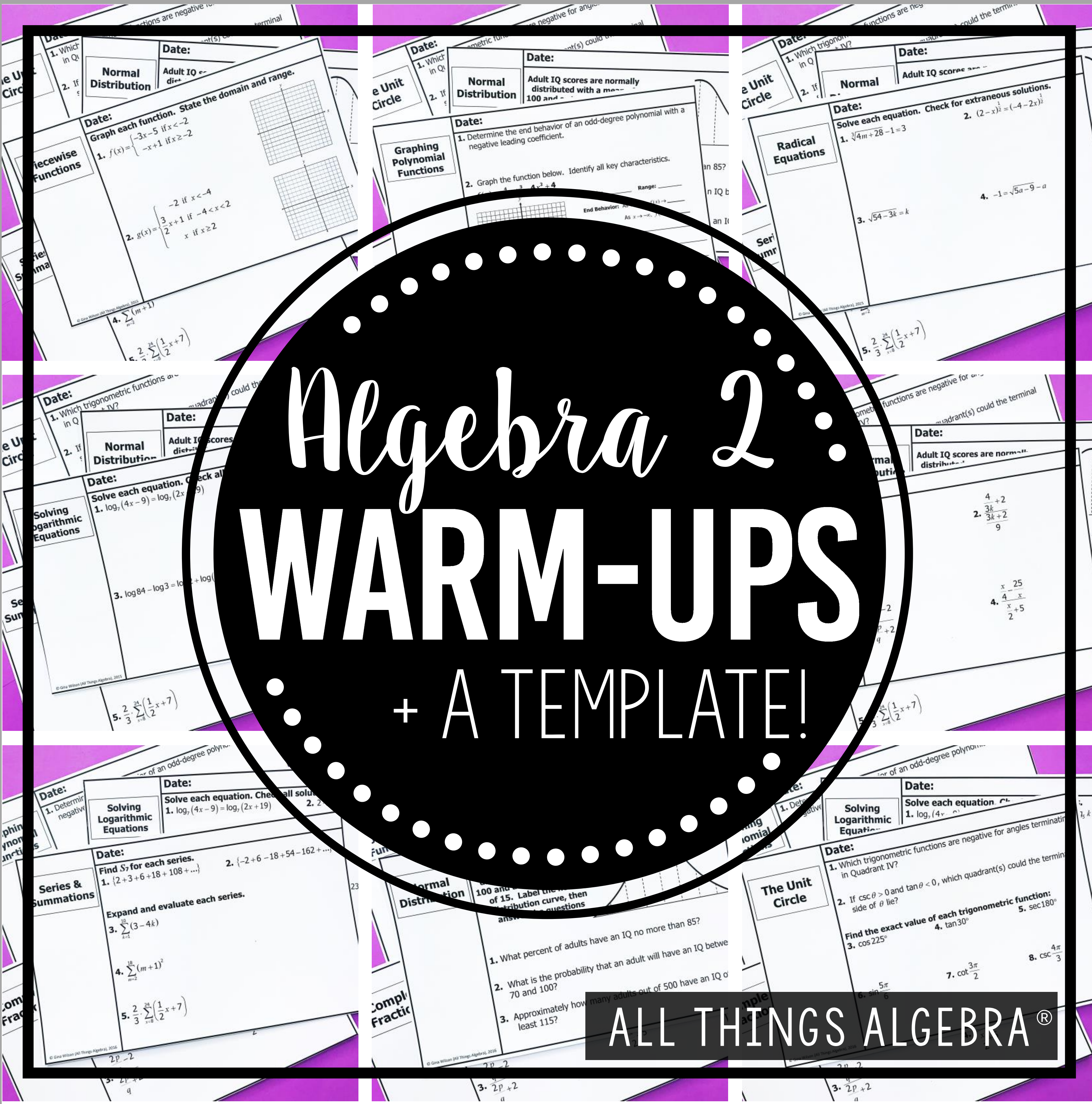
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Gina

Simplifying Radicals & Classifying Numbers	**Date:**
	1. Simplify each expression. **a)** $\sqrt{208}$ **b)** $-\sqrt{578}$ **c)** $\sqrt{153}$
	2. Name all sets to which each value belongs. **a)** $\dfrac{-\sqrt{400}}{\sqrt{16}}$ **b)** $1.58\overline{3}$ **c)** $\sqrt{95}$

Order of Ops & Evaluating Expressions	**Date:**
	Simplify each expression. **1.** $6^3 \div \left\{ (12+5^2) - (\lvert -7 \rvert - 15)^2 \right\}$ **2.** $\dfrac{3^3 - 6 + \sqrt{-40+11^2}}{18 - 6^2 \cdot 2}$
	Evaluate the expressions below if a = 8, b = -2 and c = -9. **3.** $\lvert -a^2 - 2bc \rvert$ **4.** $-\dfrac{7}{6}c + \dfrac{3}{4}ab$

Multi-Step Equations

Date:

Solve each equation.

1. $14a-(2a+9)=\frac{2}{3}(12a-18)$

2. $-3(10-x)=11x-2(4x+15)$

3. $\frac{3}{8}=\frac{6w-7}{2w+14}$

4. Solve $F=\frac{9}{5}C+32$ for C

Word Problems

Date:

1. The leg of an isosceles triangle is two less than three times the length of its base. If the perimeter of the triangle is 45 meters, find the length of the leg.

2. Find three consecutive numbers such that five times the largest number is 17 less than three times the sum of the smallest and median number.

Absolute Value Equations

Date:

Solve each equation. ***Be sure to check for extraneous solutions. ***

1. $|4x+6|=26$

2. $\dfrac{|-8-5r|}{6}=2$

3. $8-|7y-9|=3$

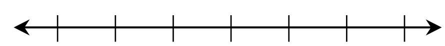
4. $|8k-5|=4k-23$

Multi-Step Inequalities

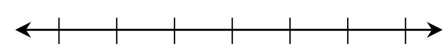
Date:

Solve, graph, and write the solution in interval notation.

1. $\dfrac{3(2x+16)}{-8} \geq x-1$



2. $-\dfrac{5}{4}(24-6m) > 14-\dfrac{1}{2}(16-7m)$



Compound Inequalities

Date:

Solve, graph, and write the solution in interval notation.

1. $2(2-3c) \le -2$ or $4c+5 < -3$

2. $4-7a \le 67$ and $\dfrac{5a+2}{-9} > 2$

3. $-50 \le 9n-2 \le 25$

More Compound Inequalities

Date:

Solve, graph, and write the solution in interval notation.

1. $4-9x < -68$ and $-6x-5 \le -11$

2. $10n-9 < 51$ or $7-n \ge 8$

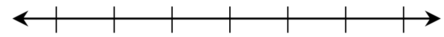
3. $-2v-2 > -8$ and $4v-7 > 9$

Absolute Value Inequalities

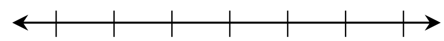
Date:

Solve, graph, and write the solution in interval notation.

1. $|6x - 10| \leq 34$



2. $-6|2v - 6| + 5 < -79$



Relations & Functions

Date:

For each relation below: a) State the domain and range, and **b)** Determine whether it is a function.

1. {(-1, 4), (0, 5), (1, 4), (2, 7)}

a) **b)**

2. {(-5, -2), (2, 1), (0, 3), (-5, -4)}

a) **b)**

3. $y = 2x - 7$

a) **b)**

4. $y = -x^2 + 3$

a) **b)**

Evaluating Functions

Date:

Given $f(x) = 8x - 9$, $g(x) = -x^2 + 7x$ and $h(x) = |2 - 4x|$, find each value.

1. $g(8)$

2. $h(13) - f(-1)$

3. $f(3y - 1)$

4. $g(a + 2)$

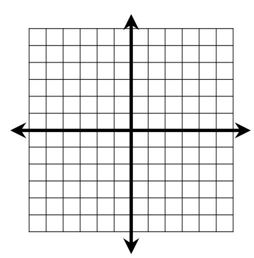
5. If $f(x) = -23$, find x.

Graphing Linear Functions

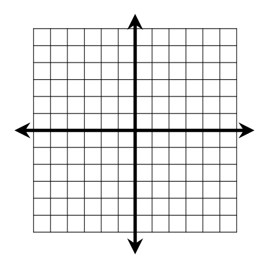
Date:

Write the equation in slope-intercept form, if necessary. Then graph.

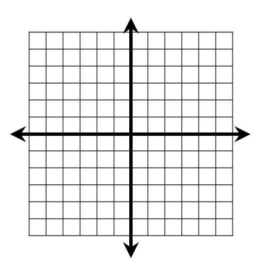
1. $12y = 9x - 12$



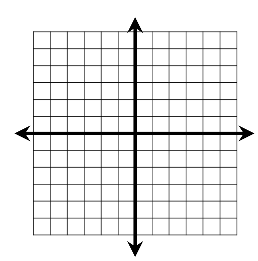
2. $x - 5y = 20$



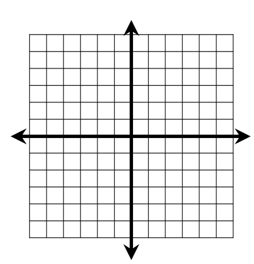
3. $15x = -10y$



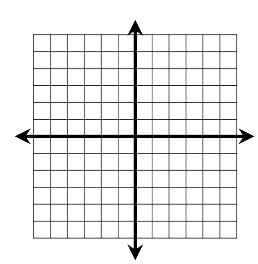
4. $-2x - y = 5$



5. $x = -4$



6. $y = 2$

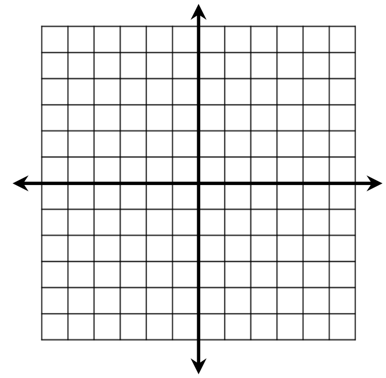


x- and y-Intercepts

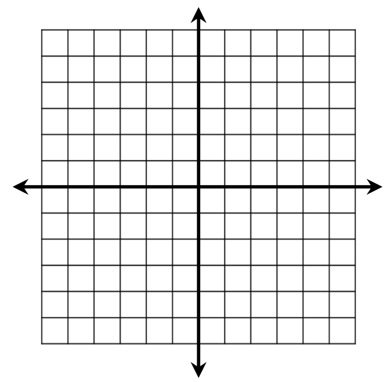
Date:

Find the x- and y-intercepts of each equation, then graph.

1. $y = -6x + 2$



2. $5x = 8y + 20$



Writing Linear Functions

Date:

Write a linear equation in slope-intercept form with the given information.

1. Passes through (-2, 11) with a slope of -4

2. Passes through (-6, 4) with a slope of $\frac{1}{2}$

3. Passes through the points (-4, 8) and (8, -1)

Parallel vs. Perpendicular Lines

Date:

Determine whether the functions are parallel, perpendicular, or neither.

1. $x - y = 5$ and $y = -x + 9$

2. $4x - 3y = 9$ and $8y = 6x - 8$

3. $y = 2$ and $y = -2$

Write a linear equation in slope-intercept form with the given information.

4. Passes through (2, -6) and parallel to the line $2x - 3y = 15$.

5. Passes through (8, -5) and perpendicular to the line passing through the points (-3, 9) and (4, -5).

Linear Equations Applications

Date:

Define variables, set up an equation, then solve.

1. Caitlyn is going away to college and will need to rent a truck to help move. The cost of the truck is \$35 plus \$0.79 per mile. If her college is 85 miles away and she budgeted \$100 for the rental, will she have enough money?

2. Max works two jobs, one at the zoo and the other at the library. He drives 24 miles round-trip to the zoo and 9 miles round-trip to the library. He never works two jobs in the same day. His car has a 15-gallon gas tank and gets 20 miles per gallon. If his tank is now empty and he has worked 12 days at the library since last filling up, how many days did he work at the zoo?

Linear Regression

Date:

On October 3rd, Holly invested \$2500 in some stocks. The table below shows the value of her investment on certain days since investing.

day	Oct. 3	Oct. 6	Oct. 11	Oct. 16	Oct. 20
value	\$2500	\$2509.12	\$2525.74	\$2539.02	\$2548.55

1. Use linear regression to find an equation of best fit.

2. Find the approximate value of her investment on Oct. 31st.

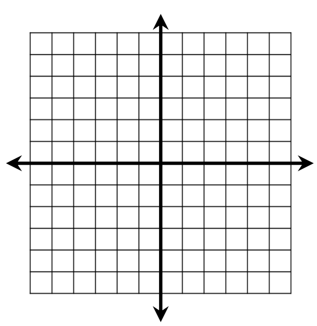
3. How many days will it take her investment to reach a value of \$3,000?

Solving Systems:
by Graphing & Substitution

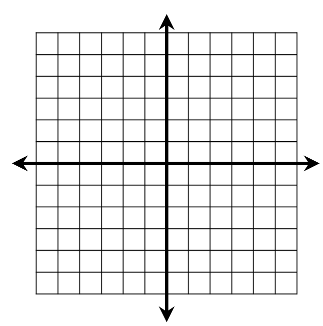
Date:

Solve questions 1 and 2 by graphing.

1. $\begin{cases} 3x + 4y = -16 \\ 2x = y - 7 \end{cases}$



2. $\begin{cases} 2x - 3y = 15 \\ -6y = 6 - 4x \end{cases}$



3. Solve by substitution.

$\begin{cases} 2x + 7y = -23 \\ 5x - y = -39 \end{cases}$

Solving Systems:
by Elimination

Date:

Solve each system by elimination.

1. $\begin{cases} x - 4y = -14 \\ 6x + 8y = -12 \end{cases}$

2. $\begin{cases} 6y = 2x + 4 \\ 7x + 14 = 21y \end{cases}$

Systems Applications

Date:

Kate bought 5 pounds of hamburger and 2 pounds of hot dogs and paid $28.50. Eric bought half the amount of hamburger and a fourth of the amount of hot dogs that Kate did and paid $12.50. Find the cost per pound of hamburger and hot dogs.

Systems with Three Variables

Date:

Solve the following system:

$$\begin{cases} x + 2y - 4z = 7 \\ 3x - y + 6z = -7 \\ 2x + 3y - z = 10 \end{cases}$$

Three Variable Application

Date:

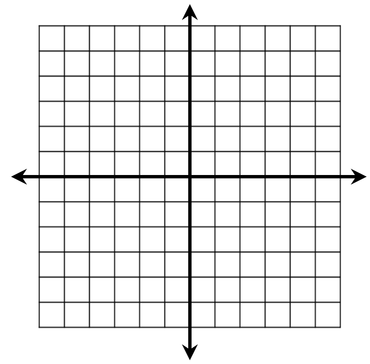
Jack has 63 pennies, dimes, and quarters worth $6.30. If the number of dimes is three less than the number of quarters, how many of each coin does he have?

Linear Inequalities & Systems

Date:

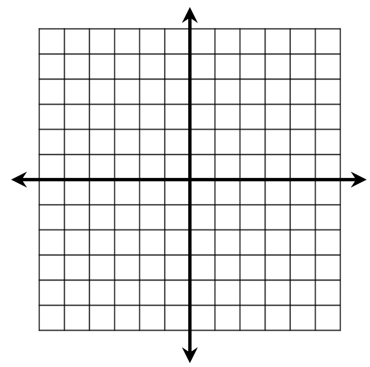
1. Graph the linear inequality.

$4x - 5y \geq 10$



2. Graphing the system of linear inequalities.

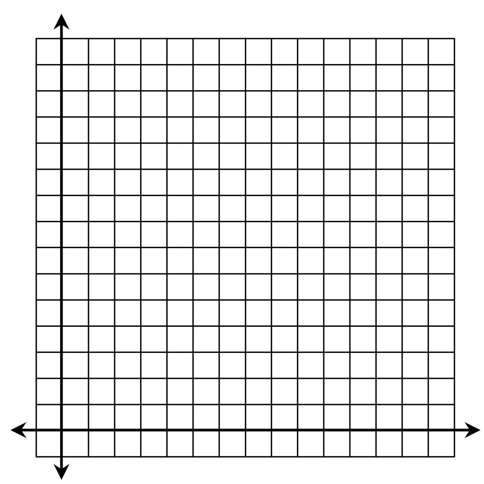
$$\begin{cases} 6x + 3y > 15 \\ y > -2 \end{cases}$$



Linear Inequalities Application

Date:

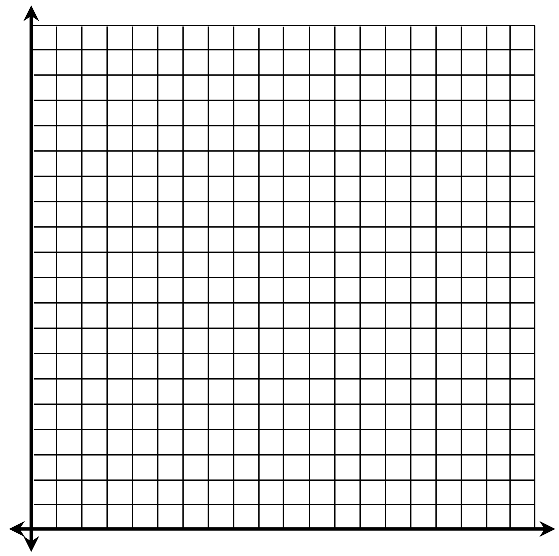
It costs $20 per day to board a dog and $16 per day to board a cat. The kennel needs to make at least $800 per day to cover expenses but can hold no more than 80 total animals. Write a system of inequalities, graph, and give two possible solutions to the number of dogs and cats they can board each day.



Linear Programming

Date:

Erica is making necklaces and bracelets to sell at the craft show. It takes her 3 hours to make a necklace and 2 hours to make a bracelet. She sells the necklaces for \$18 and the bracelets for \$15. She has 60 hours available this coming week to make the jewelry and would like to make at least 25 total pieces of jewelry that includes a minimum of 2 necklaces. How many of each should she make to maximize her profit?

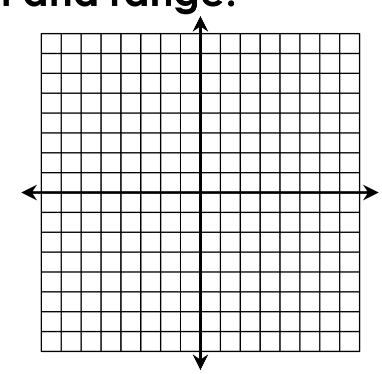


Piecewise Functions

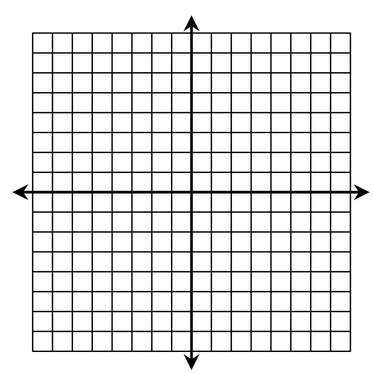
Date:

Graph each function. State the domain and range.

1. $f(x) = \begin{cases} -3x - 5 & \text{if } x < -2 \\ -x + 1 & \text{if } x \geq -2 \end{cases}$



2. $g(x) = \begin{cases} -2 & \text{if } x < -4 \\ \dfrac{3}{2}x + 1 & \text{if } -4 < x < 2 \\ x & \text{if } x \geq 2 \end{cases}$

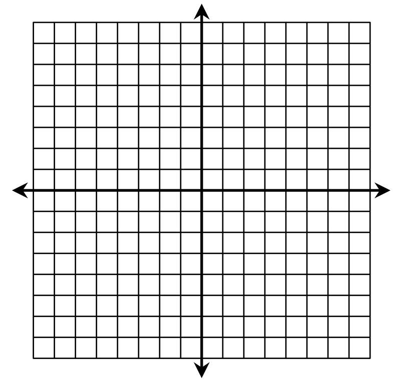


Absolute Value Functions

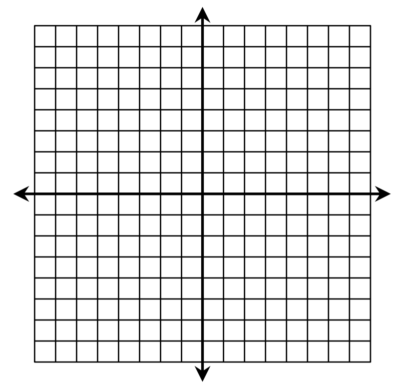
Date:

Graph function using a table of values.

1. $f(x) = -2|x+4|$



2. $f(x) = \left|\dfrac{4}{3}x - 1\right|$

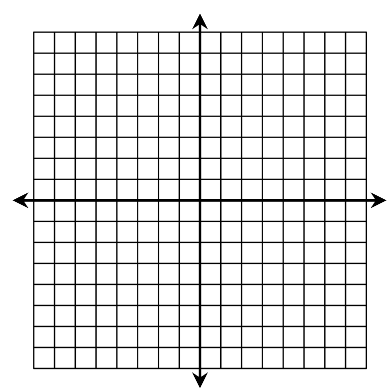


Absolute Value Inequalities

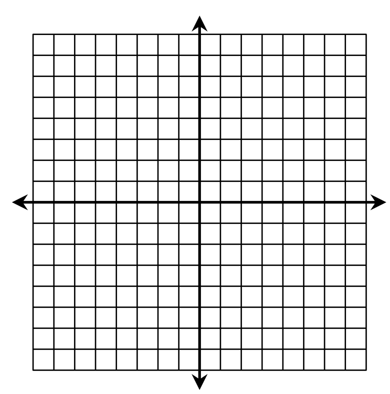
Date:

Graph inequality using a table of values.

1. $y < |3x-5|$



2. $y \geq -|2x+3| + 9$



Parent Functions and Transformations

Date:

Identify the parent function for each function family, then sketch the graph.

1. Linear

2. Absolute Value

Describe the transformations in each function.

3. $f(x) = |x+8|$

4. $f(x) = -|x| - 3$

5. $f(x) = 4|x-1|$

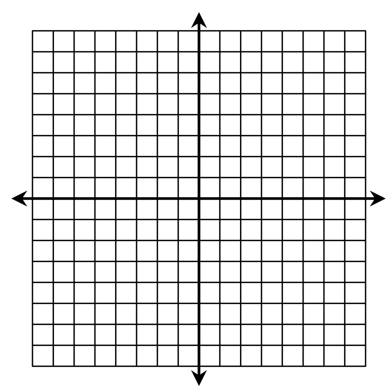
6. $f(x) = \frac{4}{5}|x+5| + 2$

Absolute Value Functions:
Vertex Form

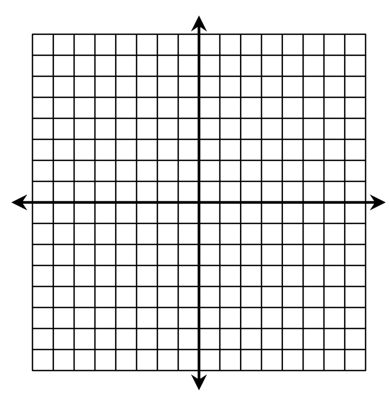
Date:

Identify the vertex, then graph.

1. $f(x) = |x+3| - 6$



2. $f(x) = -\frac{2}{3}|x| + 5$

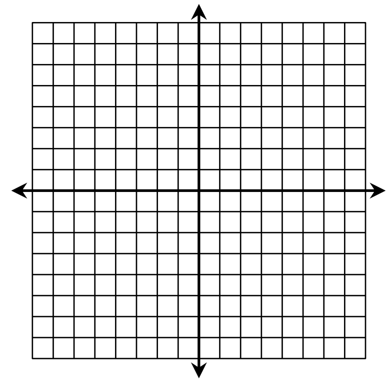


Quadratic Functions: Standard Form

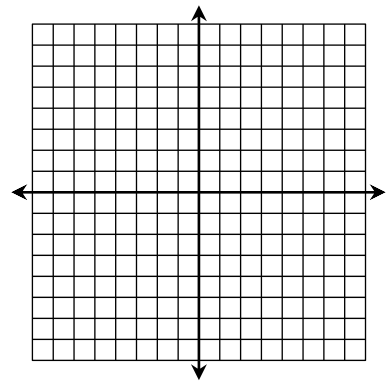
Date:

Give the axis of symmetry and vertex of each function, then graph using a table of values.

1. $f(x) = x^2 + 6x + 8$



2. $f(x) = -\frac{1}{2}x^2 + 4x - 1$

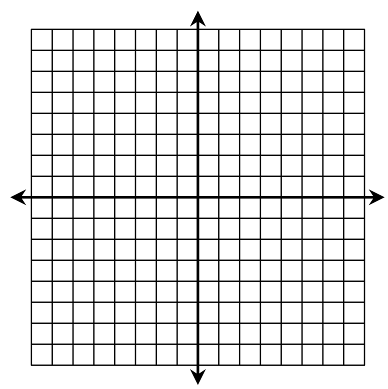


Quadratic Inequalities

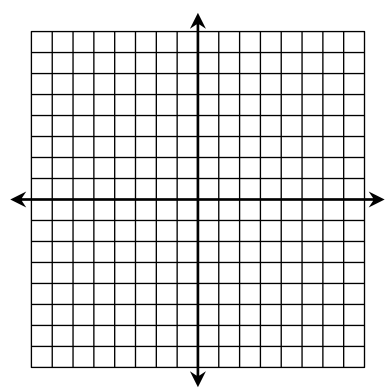
Date:

Give the axis of symmetry and vertex of each function, then graph using a table of values.

1. $y \le -2x^2 - 4x + 5$



2. $y > x^2 + 8x + 18$

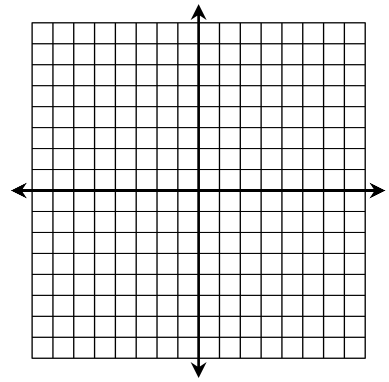


Quadratic Functions:
Vertex Form

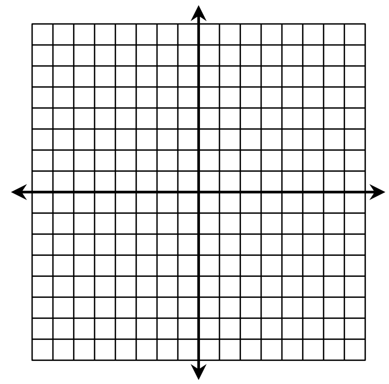
Date:

Give the axis of symmetry and vertex of each function. Graph using a table of values.

1. $f(x) = (x-2)^2 - 7$



2. $f(x) = -3(x+1)^2 + 4$

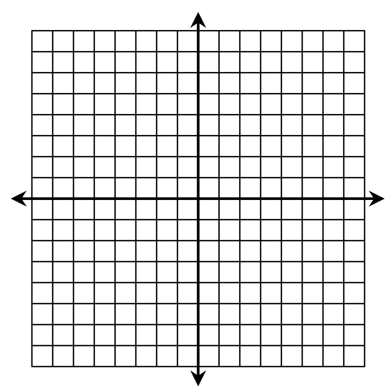


More Piecewise Functions Practice

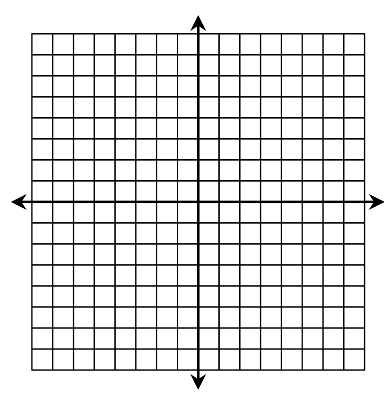
Date:

Graph each function. State the domain and range.

1. $f(x) = \begin{cases} |x-1| - 5 & \text{if } x < 3 \\ -2x + 7 & \text{if } x \geq 3 \end{cases}$



2. $f(x) = \begin{cases} x^2 - 1 & \text{if } x < -1 \\ -x + 4 & \text{if } x > 1 \end{cases}$



Date:

Converting Standard Form to Vertex Form

Write each function in vertex form. Identify the vertex.

1. $f(x) = x^2 - 8x + 19$

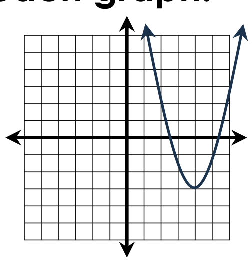
2. $f(x) = -2x^2 - 4x + 14$

Date:

End Behavior & Increasing Decreasing Intervals

Identify the increasing/decreasing intervals and end behavior of each graph.

1.

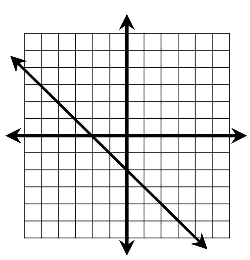


Inc. Intervals: ________ **Dec. Intervals:** ________

End Behavior: As $x \to \infty$, $f(x) \to$ ______

As $x \to -\infty$, $f(x) \to$ ______

2.

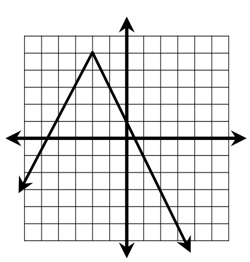


Inc. Intervals: ________ **Dec. Intervals:** ________

End Behavior: As $x \to \infty$, $f(x) \to$ ______

As $x \to -\infty$, $f(x) \to$ ______

3.



Inc. Intervals: ________ **Dec. Intervals:** ________

End Behavior: As $x \to \infty$, $f(x) \to$ ______

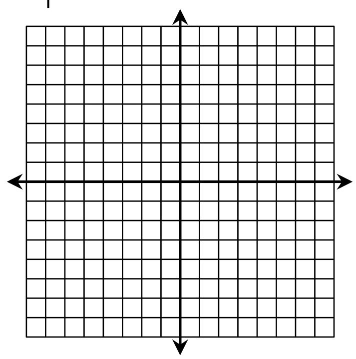
As $x \to -\infty$, $f(x) \to$ ______

Function Families Review

Date:

Graph each function and give its characteristics.

1. $f(x) = -|x+4| - 3$



Function Family: ______________________

D: ______________; **R:** ______________

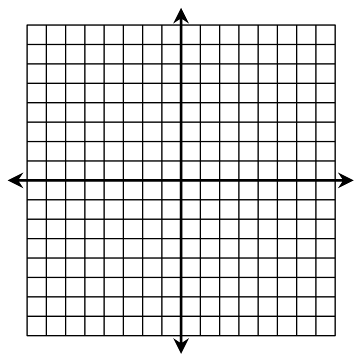
End Behavior: As $x \to \infty$, $f(x) \to$ ______

As $x \to -\infty$, $f(x) \to$ ______

Increasing Interval(s): ______________

Decreasing Interval(s): ______________

2. $f(x) = \dfrac{6}{5}x - 1$



Function Family: ______________________

D: ______________; **R:** ______________

End Behavior: As $x \to \infty$, $f(x) \to$ ______

As $x \to -\infty$, $f(x) \to$ ______

Increasing Interval(s): ______________

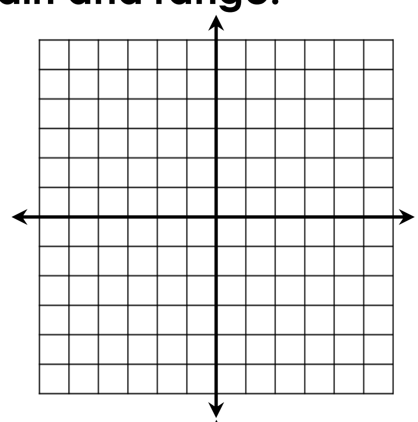
Decreasing Interval(s): ______________

Greatest Integer Functions

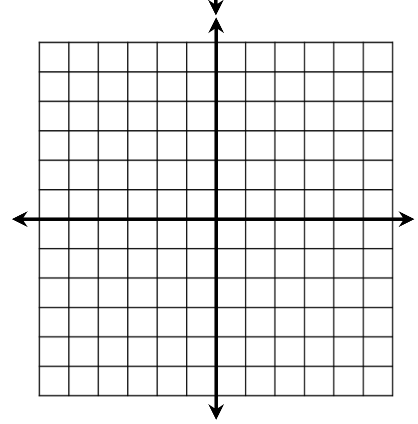
Date:

Graph each function. State the domain and range.

1. $f(x) = [\![x+5]\!]$



2. $f(x) = 2[\![x-3]\!] + 1$

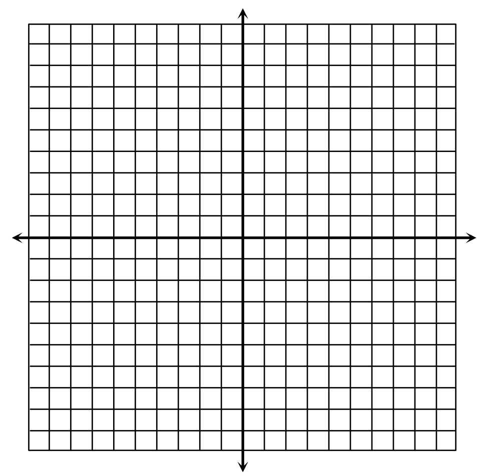


Quadratic Roots	**Date:**

1. What are quadratic roots?

2. What other names are quadratic roots called?

3. Find the roots by graphing: $f(x) = -x^2 + 2x + 8$



Factoring Review	**Date:**

Factor each polynomial completely:

1. $x^2 - 14x - 95$

2. $5x^2 - 40x + 80$

3. $4x^2 - x - 14$

4. $16x^2 - 49$

5. $5 - 20x^2$

6. $24x^2 - 10x$

Solving Quadratics by Factoring	**Date:**

Solve each quadratic equation by factoring:

1. $x^2 + 11x + 30 = 0$

2. $2x^2 + x = 2x + 6$

3. $4x^2 - 74 = x^2 + 1$

4. $8x^2 + 43x = 7x$

Factored Form vs. Vertex Form	**Date:**

1. What are the roots of the quadratic equation below?

$$f(x) = (3x + 2)(x - 4)$$

2. Write the equation below in factored form and vertex form. Give the axis of symmetry, vertex, and roots.

$$f(x) = -x^2 - 12x - 27$$

Solving Quadratics by Square Roots

Date:

Solve each quadratic equation by square roots:

1. $x^2 - 10 = 159$

2. $36x^2 - 1 = 0$

3. $2x^2 + 7 = 41$

4. $-\frac{2}{3}x^2 - 19 = -35$

Imaginary Numbers

Date:

Simplify each expression:

1. $\sqrt{-16}$

2. $\sqrt{-5} \cdot \sqrt{-12} \cdot \sqrt{-3}$

3. $14i \cdot 3i$

4. i^{67}

Solve the equations below:

5. $x^2 + 100 = 0$

6. $7 - 3x^2 = 211$

Complex Numbers

Date:

Simplify each expression:

1. $(-9-i)-(4-3i)$

2. $(2+5i)(-4-3i)$

3. $\dfrac{14}{2i}$

4. $\dfrac{4-5i}{2+4i}$

Simplify, then name all sets to which the value belongs:

5. $\sqrt{-6}\cdot\sqrt{-3}$

6. i^{22}

Completing the Square

Date:

Solve by completing the square:

1. $x^2+12x+47=0$

2. $-4x^2+408=20-8x$

The Quadratic Formula

Date:

Solve by the quadratic formula:

1. $10x^2 - 9x = 2x + 6$

2. $-x^2 = 8x + 26$

The Discriminant

Date:

Find the discriminant, then determine the number and type of roots:

1. $-2x^2 + 14x - 23 = 0$

2. $16x^2 - 8x + 1 = 0$

3. $x^2 - 10x + 31 = 0$

4. $3x^2 - 12 = 0$

Choosing the Best Method

Date:

Solve one equation by factoring and the other by completing the square. Explain your reasoning.

1. $x^2 - 14x + 37 = 0$

2. $-x^2 - 9x + 136 = 0$

Quadratic Applications

Date:

1. The dimensions of a square are altered so that 10 inches is added to one side while 1 inch is subtracted from the other. The area of the resulting rectangle is 80 in^2. How much larger is the area of the rectangle compared to the square?

2. Find three consecutive positive even integers such that the product of the median and largest integer is 6 less than 21 times the smallest integer.

Projectile Motion

Date:

The football coach threw a football from a platform to his quarterback below. The height of the football, h, at time t seconds is modeled by the equation $h(t) = -16t^2 + 28t + 15$.

1. What is the maximum height of the ball?

2. If the quarterback caught the ball at a height of 6 feet, how many seconds was the ball in the air?

3. Give the domain and range of the function.

Quadratic Regression

Date:

The table below shows the population in a town from 2018 to 2024, in thousands of people.

year	2018	2019	2020	2021	2022	2023	2024
population	24.8	26.2	27.3	27.8	28.1	27.5	27.2

1. Use quadratic regression to write an equation for the population, y, in years since 2018, x.

2. Find the approximate population of the town in 2027.

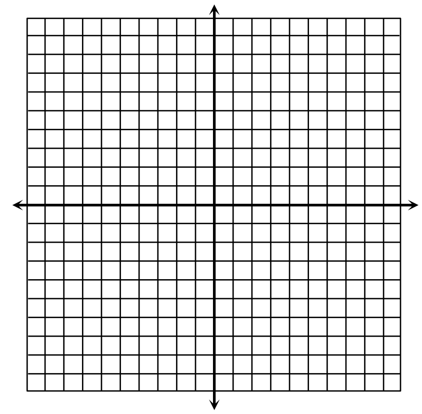
3. If the population continues the same trend, in what year will the population reach 15,000 people?

Nonlinear Systems
(Solve Graphically)

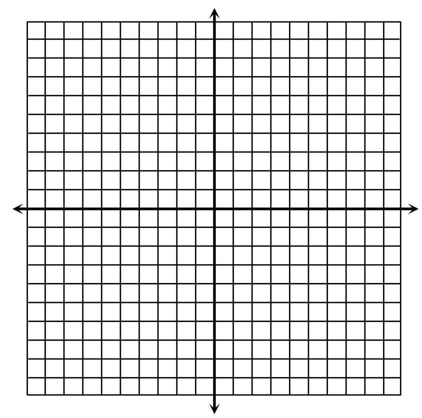
Date:

Solve each system by graphing.

1. $\begin{cases} y = -x^2 + 3x + 1 \\ x - y = 7 \end{cases}$



2. $\begin{cases} y = 2x^2 + 4x - 3 \\ y = -x^2 + 10x - 17 \end{cases}$



Nonlinear Systems
(Solve Algebraically)

Date:

Solve each system algebraically.

1. $\begin{cases} y = 2x^2 - 12x + 11 \\ 4x + y = 3 \end{cases}$

2. $\begin{cases} y = x^2 + 10x + 18 \\ y = -2x^2 - 8x + 3 \end{cases}$

Simplifying Monomials

Date:

Simplify the expressions below.

1. $-3xy^3 \cdot (2x^2y^4)^3 + 11x^7y^{15}$

2. $(-9a^3b^5)^2 \cdot \left(\frac{1}{3}a^2b\right)^3$

3. $\frac{5m^3n^7 \cdot 2m^2n}{12m^3n^9}$

4. $\left(\frac{60c^4d^2}{5cd^3}\right)^2 \cdot \frac{2}{3}c^{-6}d^{-1}$

Polynomials

Date:

Simplify the expressions below. Classify your answer.

1. $(-5x - 2x^3 - 4x^2) + (x^3 - 7x + 4x^2)$

2. $(a^4 + 10a^2 - 9a) - (2a^2 - 3a^4 + a - 8a^3)$

3. $m^2(m-1)^3$

4. $\frac{-36k^4 + 8k^3 - 4k^2}{4k^2}$

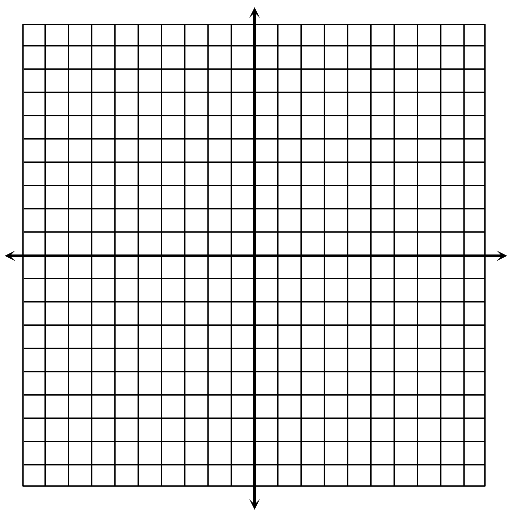
Graphing Polynomial Functions

Date:

1. Determine the end behavior of an odd-degree polynomial with a negative leading coefficient.

2. Graph the function below. Identify all key characteristics.

$$f(x) = x^4 - x^3 - 4x^2 + 4$$



Domain: __________ **Range:** _________

End Behavior: As $x \to \infty$, $f(x) \to$ ______

As $x \to -\infty$, $f(x) \to$ ______

Rel. Minimum(s): ____________________________

Rel. Maximum(s): ____________________________

Inc. Intervals: ______________________________

Dec. Intervals: ______________________________

Zeros of Polynomial Functions

Date:

Give the zeros, their multiplicity, and the effect on the graph for each function below.

1. $f(x) = x(x+5)^4(2x+7)^5$

Zero	Multiplicity	Effect

2. $f(x) = x^5 - 2x^4 + x^3$

Zero	Multiplicity	Effect

3. The graph of a polynomial function has zeros at 4 (multiplicity 2) and 9 (multiplicity 1). Write an equation in standard form to represent this function.

Factoring Polynomials
(2 Terms)

Date:

Factor each expression. Be sure to check for a GCF.

1. $49a^6 - 9b^4$

2. $r^3 + 64s^3$

3. $125w^3 - 27$

4. $8n^5 - 32n$

5. $16x^4 - 128x$

6. $7m^5n^2 + 7m^2n^2$

Factoring Polynomials
(3-4 Terms)

Date:

Factor each expression. Be sure to check for a GCF.

1. $3x^3 - 12x^2 - 36x$

2. $m^4 + 14m^2 + 45$

3. $y^5 - 20y^3 + 64y$

4. $4a^4 + 3a^2 - 1$

5. $k^3 - 2k^2 - 7k + 14$

6. $2x^5 + x^4 - 18x - 9$

Solving Polynomial Equations

Date:

Solve each polynomial equation below by factoring.

1. $15x^3 + 6x^2 = 0$

2. $20x^3 - 45x = 0$

3. $x^3 + 125 = 0$

Solving Polynomial Equations

Date:

Solve each polynomial equation below by factoring.

1. $x^4 + x^2 - 72 = 0$

2. $6x^3 - 13x^2 - 15x = 0$

3. $3x^3 + x^2 - 54x - 18 = 0$

Long Division

Date:

Find each quotient using long division.

1. $\dfrac{x^2 - 3x - 20}{x + 7}$

2. $\dfrac{w^4 - 8w^3 - 2w + 20}{w - 8}$

Synthetic Division

Date:

Find each quotient using synthetic division.

1. $\dfrac{8n^2 - 10n + 7}{n - 1}$

2. $\dfrac{x^4 - 4x^2 - 15x - 9}{x + 3}$

Remainder Theorem

Date:

Evaluate each function using the Remainder Theorem.

1. $f(x) = 3x^3 + 7x^2 + 6x + 13;\ f(-2)$

2. $f(x) = x^4 - 13x^2 + 10x - 1;\ f(3)$

Function Operations

Date:

Given $f(x) = x^2 - x$, $g(x) = x - 4$, and $h(x) = x^2 + 6x - 16$, find each function. Indicate any restrictions in the domain.

1. $(f+g)(x)$

2. $(g \cdot h)(x)$

3. $\left(\dfrac{g}{f}\right)(x)$

4. $(h \circ g)(x)$

Using the same functions, find each function value.

5. $(h - f)(2)$

6. $(g \circ f)(-7)$

Regression

Date:

The amount of money in Carmen's savings account on the last day of certain years is given below:

Year	2003	2007	2009	2012	2015
Balance	\$4,002	\$3,209	\$2,975	\$3,371	\$4,129

1. Write a **quartic function** to represent the balance of Carmen's savings account.

2. Predict the balance of Carmen's account at the end of 2020.

Simplifying Radicals

Date:

1. $3\sqrt{507}$

2. $-2\sqrt[4]{176}$

3. $\sqrt[3]{\frac{54}{128}}$

4. $-3\sqrt{126a^9b^3}$

5. $\sqrt[3]{-448x^{17}y^4}$

6. $\sqrt[4]{625m^{21}n^5}$

Rational Exponents

Date:

Write in radical form. Simplify if possible.

1. $(9x)^{\frac{1}{2}}$ **2.** $y^{\frac{7}{4}}$ **3.** $m^{\frac{10}{3}}n^{\frac{4}{3}}$

Write in exponential form:

4. $\sqrt[4]{3k}$ **5.** $\sqrt[3]{p^{11}}$ **6.** $\sqrt[4]{a^5b^8}$

Simplify:

7. $y^{\frac{7}{8}} \cdot y^{\frac{3}{8}}$ **8.** $\dfrac{(2m)^{\frac{11}{6}}}{(2m)^{\frac{1}{2}}}$ **9.** $\left(4^{-\frac{3}{4}}\right)^2$

Add, Subtract, & Multiply Radicals

Date:

1. $2\sqrt[4]{768} - 3\sqrt{20} + 7\sqrt[4]{3}$ **2.** $2b\sqrt{147a^5} - 3a^2\sqrt{12ab^2}$

3. $3\sqrt[3]{4p^5} \cdot -2\sqrt[3]{14p^6}$ **4.** $\sqrt{8}\left(\sqrt{2} - 5\sqrt{3}\right)$

5. $(6 - 4\sqrt{5})(1 + \sqrt{5})$ **6.** $\left(2\sqrt{3} - 8\right)^2$

Dividing Radicals

Date:

1. $\dfrac{24\sqrt[4]{240}}{3\sqrt[4]{5}}$

2. $\dfrac{\sqrt[3]{486m^{12}}}{\sqrt[3]{3m}}$

3. $\sqrt{\dfrac{28x}{3}}$

4. $\dfrac{3-\sqrt{3}}{\sqrt{12}}$

5. $\dfrac{4}{2-2\sqrt{2}}$

6. $\dfrac{2-3\sqrt{5}}{5-4\sqrt{5}}$

Solving Radical Equations

Date:

Solve each equation. Check for extraneous solutions.

1. $\sqrt[3]{4m+28}-1=3$

2. $(2-x)^{\frac{1}{2}}=(-4-2x)^{\frac{1}{2}}$

3. $\sqrt{54-3k}=k$

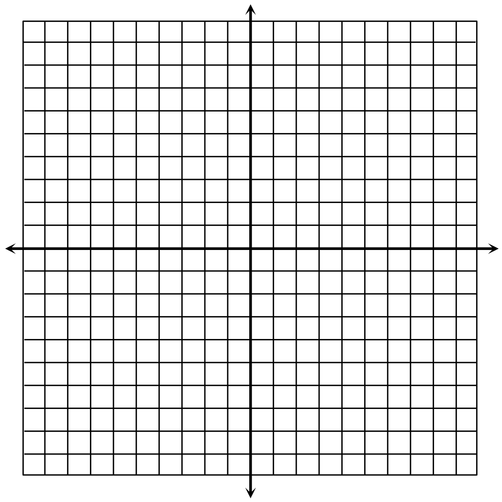
4. $-1=\sqrt{5a-9}-a$

Graphing Square Root Functions

Date:

1. The square root parent function is vertically compressed by a factor of 1/3, reflected in the x-axis, then translated 4 units up. Write an equation to represent this function. Give the coordinates of the endpoint.

2. Graph $f(x) = 2\sqrt{x+7} - 1$ and identify all key characteristics.



Domain: ____________________

Range: ______________________

End Behavior:
As $x \to \infty,\ f(x) \to$ ________

As $x \to -\infty,\ f(x) \to$ ________

Endpoint: ______________

Inc. Interval: __________________

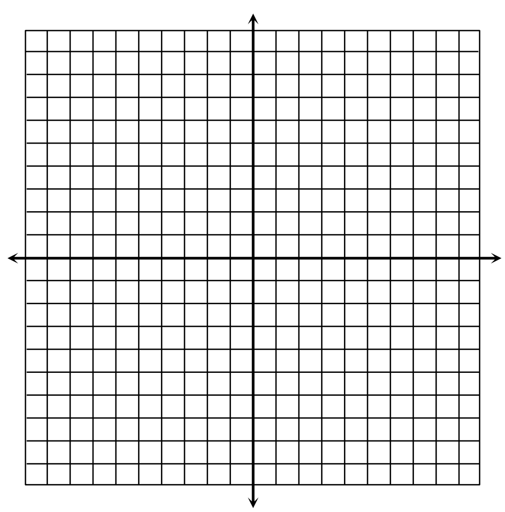
Dec. Interval: __________________

Graphing Cube Root Functions

Date:

1. The cube root parent function is reflected across the x-axis, then translated 9 units right and 2 units down. Write an equation to represent this function. Give the coordinates of the point of inflection.

2. Graph $f(x) = -\sqrt[3]{x-2} + 3$ and identify all key characteristics.



Domain: ____________________

Range: ______________________

End Behavior:
As $x \to \infty,\ f(x) \to$ ________

As $x \to -\infty,\ f(x) \to$ ________

Point of Inflection: ______________

Inc. Interval: __________________

Dec. Interval: __________________

Inverse Relations & Functions

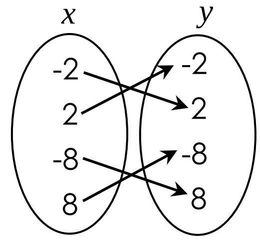
Date:

Which relations represent one-to-one functions?

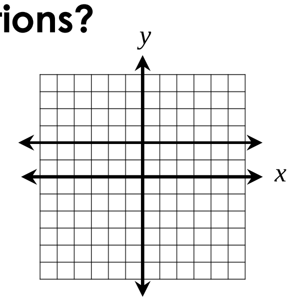
1.

x	y
-9	4
-5	-7
-1	1
0	4

2.

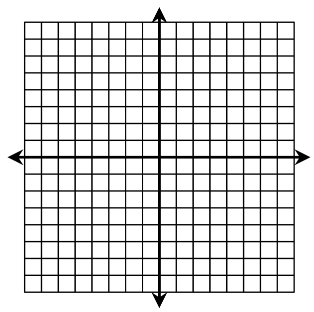


3.

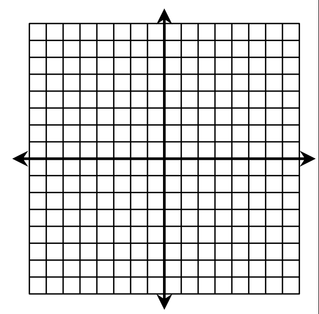


Write the inverse of each function. Graph to verify.

4. $f(x) = -\frac{4}{3}x$



5. $f(x) = 3x + 6$



Verifying Inverses

Date:

Determine whether the pair of equations are inverse functions.

1. $f(x) = \frac{-2x-8}{3}$ and $g(x) = -4 - \frac{3}{2}x$

2. $f(x) = -x - 1$ and $g(x) = 5x + 5$

3. $f(x) = \sqrt{x} + 3$ and $g(x) = x^2 - 6x + 9$ (if $x \geq 3$)

Graphing Exponential Functions

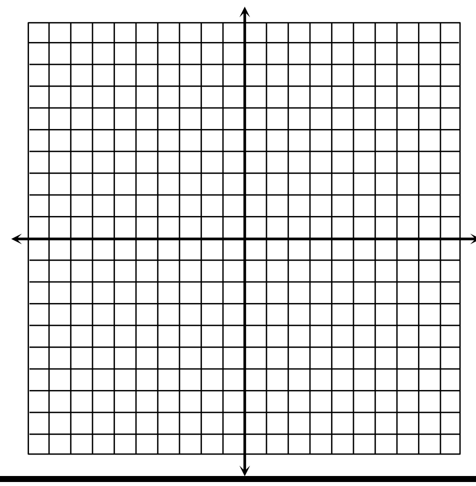
Date:

1. Determine whether the function below represents an exponential growth or decay:

a. $f(x) = 2 \cdot \left(\frac{5}{6}\right)^x$

b. $f(x) = 0.8 \cdot \left(\frac{4}{3}\right)^x$

2. Graph $\boldsymbol{f(x) = \frac{2}{3} \cdot 3^{x+1} - 9}$ and identify all key characteristics.



Domain: ____________

Range: ______________

End Behavior:

As $x \to \infty$, $f(x) \to$ _______

As $x \to -\infty$, $f(x) \to$ _______

***y*-intercept:** ____________

Asymptote: _____________

Solving Exponential Functions
(with a Common Base)

Date:

Solve each equation.

1. $6^{7+3x} = 6^{x-11}$

2. $2 \cdot 2^{a+7} = 2^{5a-4}$

3. $\left(\frac{1}{9}\right) = 3^{5n-12}$

4. $4^{3p} \cdot \frac{1}{64} = 16^{3p-9}$

Logarithms

Date:

Write in exponential form:

1. $\log_3 729 = 6$ **2.** $\log_2 x = 7$ **3.** $\log 65 = x + 2$

Write in logarithmic form:

4. $2^9 = 512$ **5.** $8^{x+3} = 36$ **6.** $\sqrt{49} = 7$

Evaluate. Round to the nearest ten-thousands if necessary.

7. $\log_6 216$ **8.** $\log_5 1$ **9.** $\log_{16} 64$

10. $\log 100$ **11.** $\log_9 5$ **12.** $\log_4 97$

Properties of Logarithms

Date:

Condense each expression into a single logarithm:

1. $\frac{1}{2} \cdot \log_3 16 + \log_3 5$ **2.** $7 \cdot \log_5 a - 2 \cdot \log_5 b^4$

3. $2 \cdot \log 3 + \log(x - 8)$ **4.** $\frac{1}{2}(\log_8 48 - \log_4 3) + 3 \cdot \log_8 3$

Expand:

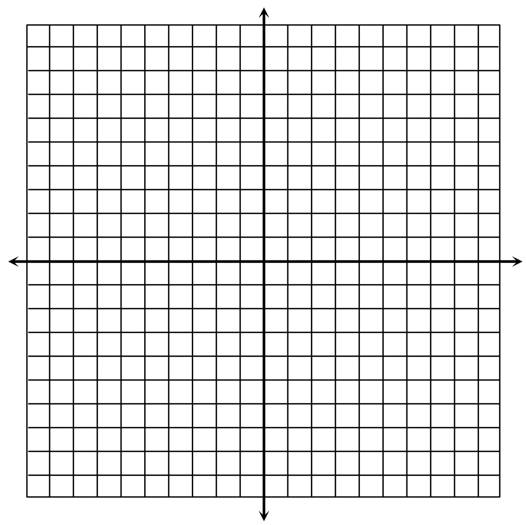
5. $\log_4 \left(\frac{p^5}{q^2}\right)^3$ **6.** $\log \sqrt[4]{m^2 n^5}$

Graphing Logarithmic Functions

Date:

1. What is the relationship between exponential and logarithmic functions?

2. Graph $f(x) = \log_2(x+7) - 3$ and identify all key characteristics.



Domain: ________________

Range: ________________

End Behavior:

As $x \to$ ______, $f(x) \to$ _____

As $x \to$ ______, $f(x) \to$ _____

***x*-intercept:** ________________

Asymptote: ________________

Solving Logarithmic Equations

Date:

Solve each equation. Check all solutions.

1. $\log_7(4x-9) = \log_7(2x+19)$

2. $2 \cdot \log_5 k = \log_5(k+42)$

3. $\log 84 - \log 3 = \log 2 + \log(m-5)$

4. $\log_3(8y-23) = 4$

Solving Exponential Functions
(with Logarithms)

Date:

Solve each equation.

1. $3^{x-2} = 40$

2. $-4 \cdot 5^{2x-1} = -128$

3. $\frac{1}{2} \cdot 9^{w} - 3 = 14$

4. $2 \cdot 3^{p+5} + 27 = 163$

Exp/Logs Equations Review

Date:

Solve each equation.

1. $2 \cdot \log_4(m+6) = \log_4 1$

2. $4 = \log_2(3y - 26)$

3. $8^{2w-1} = 32^{2w+5}$

4. $4 \cdot 5^{p-2} - 10 = 82$

Base e & Natural Logs

Date:

Write in logarithmic form:

1. $e^7 = x$

2. $e^{2x+1} = 9$

Write in exponential form:

3. $\ln 10 = 2$

4. $\ln 71 = x$

Solve.

5. $3 \cdot \ln 2 + \ln(m-1) = 5$

6. $2 \cdot e^{m+7} - 1 = 103$

Exponential Growth & Decay

Date:

1. The current population of a town is 8,200. If the population increases by 15% each year, find the population of the town in 12 years.

2. Emmanuel bought a new boat in 2011 for $36,000. Each year, it depreciates at a rate of 6.5%. Find the value of the boat in 2020.

Compound Interest

Date:

1. Mr. Jameson invested $3,200 in a savings account that earns 4% interest compounded semiannually. Find the total amount of money he will have in 5 years.

2. Kate moved $20,000 from her savings to a new account that earns 7.5% interest compounded monthly. How much will the account earn in interest after 10 years?

Exponential & Logarithmic Regression

Date:

1. The table below shows the revenue of a company in thousands of dollars since 2005. Use an **exponential function** to predict the revenue of the company in 2025.

Year	Revenue
2005	261
2006	318
2007	404
2008	492
2009	575

2. The table below shows the average freshman class GPA of a university during certain years. Use a **logarithmic function** to predict the year the GPA will reach 4.0.

Year	Revenue
1996	3.12
2000	3.49
2004	3.68
2008	3.71
2012	3.82

Choosing the Best Model

Date:

1. The table below shows the number of trees (in thousands) planted at a tree farm. Which function is the better model: quadratic, cubic, or exponential? Write an equation and approximate the number of trees planted in 2015.

Year	Trees
1992	0.8
1994	10.2
1996	45.1
1998	115.2
2000	241.9

2. The table below shows the number of days it takes to build a house based on the number of workers. Which function is the better model: linear, quadratic, or logarithmic? Write an equation and approximate the number of days it will take to build a house if there are 40 workers.

Workers	Days
6	178
10	121
12	115
18	100
24	86

Simplifying Rational Expressions

Date:

Simplify.

1. $\dfrac{16mn^3}{2m \cdot 5n^7}$

2. $\dfrac{18x^4 + 27x^3}{24x + 36}$

3. $\dfrac{35k - 14}{4 - 25k^2}$

4. $\dfrac{p^2 - p - 72}{3p^2 - 28p + 9}$

Mult/Div Rational Expressions

Date:

Find each product or quotient.

1. $\dfrac{4ab}{3a^2b} \cdot \dfrac{9b^3}{10a}$

2. $\dfrac{9-3w}{4w} \cdot \dfrac{20w^2}{w^2-11w+24}$

3. $\dfrac{15pq}{2q^2} \div (3p^5)$

4. $\dfrac{2n^2-11n+14}{6n^2-24} \div \dfrac{3n-9}{2n^2-2n-12}$

Add/Sub Rational Expressions

Date:

Find each sum or difference.

1. $\dfrac{13x+1}{2x^3-6x^2} - \dfrac{x+37}{2x^3-6x^2}$

2. $\dfrac{3}{2a^2} + \dfrac{5}{3a}$

3. $\dfrac{7}{y-5} + \dfrac{4}{y-2}$

4. $\dfrac{r^2-6r+19}{2r^2-15r+7} - \dfrac{2}{r-7}$

Complex Fractions

Date:

Simplify.

1. $$\dfrac{\dfrac{18w}{4w^3}}{\dfrac{15w^5}{28w^2}}$$

2. $$\dfrac{\dfrac{4}{3k}+2}{\dfrac{3k+2}{9}}$$

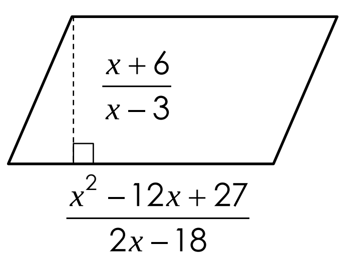
3. $$\dfrac{\dfrac{2p}{q}-2}{\dfrac{2p}{q}+2}$$

4. $$\dfrac{\dfrac{x}{4}-\dfrac{25}{x}}{\dfrac{x}{2}+5}$$

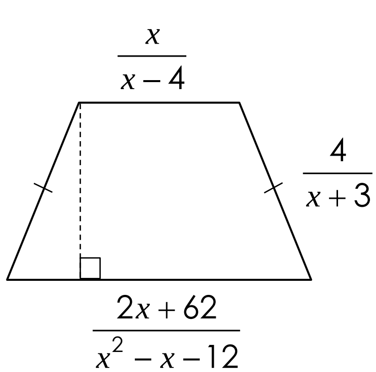
Applications

Date:

1. Find the **area** of the parallelogram below.



2. Find the **perimeter** of the trapezoid below.



Reciprocal Functions

Date:

1. The reciprocal parent function is reflected across the x-axis, then translated four units left. Write an equation that could represent this function, then identify the vertical and horizontal asymptotes.

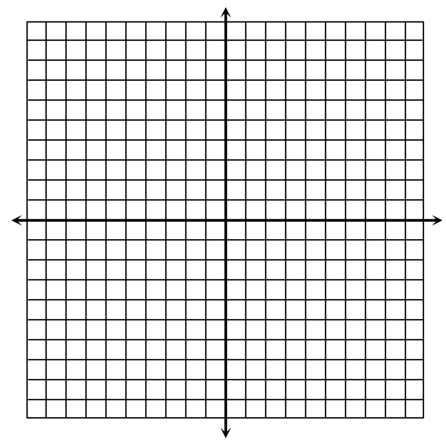
2. Graph $f(x) = \dfrac{2}{x-1} - 4$ and identify all key characteristics.

Domain: ____________________

Range: ____________________

VA: ____________________

HA: ____________________



Rational Functions

Date:

1. Identify the key characteristics of the function below.

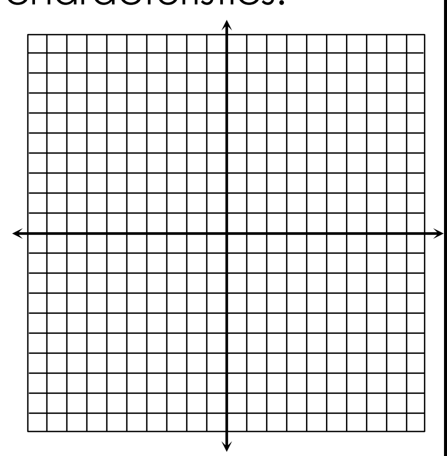
$$f(x) = \frac{x^3}{x^2 + 2x}$$

x-int: ____________________ **Holes:** ____________________

VA: ____________________ **HA:** ____________________

D: ____________________ **R:** ____________________

2. Graph $f(x) = \dfrac{x^2 - 36}{x^2 + 7x + 6}$ and all key characteristics.



x-int: ____________________ **Holes:** ____________________

VA: ____________________ **HA:** ____________________

D: ____________________ **R:** ____________________

Solving Rational Functions

Date:

Solve. Be sure to check all solutions.

1. $\frac{x-2}{x+2}=\frac{6}{x+9}$

2. $\frac{a+1}{2a}-\frac{a-1}{4a}=\frac{1}{3}$

2. $\frac{5}{w+2}+\frac{3}{w-2}=\frac{12}{w^2-4}$

4. $\frac{3}{r+1}=\frac{2r-1}{r^2-5r+6}+\frac{3}{r-2}$

Variation

Date:

1. Determine whether the equations represent a direct, joint, or inverse variation. Identify the constant.

a) $\frac{6y}{z}=10x$ **b)** $\frac{8y}{x}=4$ **c)** $\frac{12}{x}=\frac{y}{6}$

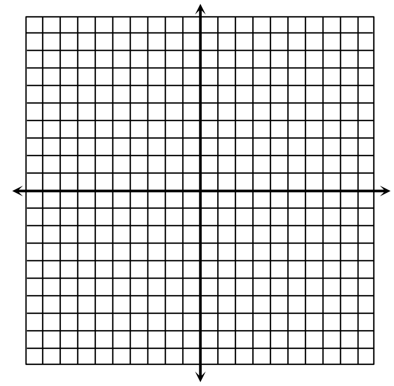
2. The time, t, it takes a child to go down a slide varies directly with the length, l, of the slide and inversely with the square root of the height, h, of the slide. If it takes 12 seconds to go down a slide 15 feet long and 9 feet high, how long will it take a child to go down a slide 25 feet long and 16 feet high?

Circles

Date:

Write in standard form, then graph the circle:

1. $x^2 + y^2 + 4x - 12y = -24$



Center: ______________ **Radius:**_______________

Write the equation of the following circles:

2. center: (-9, 0);
point on circle: (-5, 3)

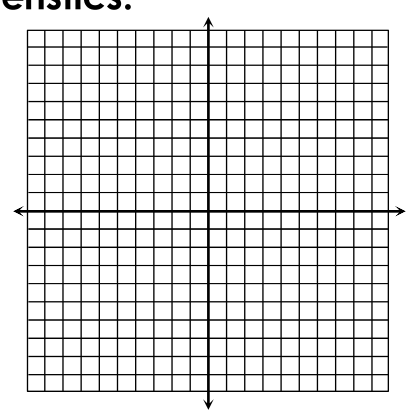
3. Endpoints of diameter:
(4, -7) and (-2, -1)

Ellipses (Graphing)

Date:

Graph each ellipse. Identify key characteristics.

1. $\dfrac{(x-3)^2}{49} + \dfrac{(y-2)^2}{9} = 1$



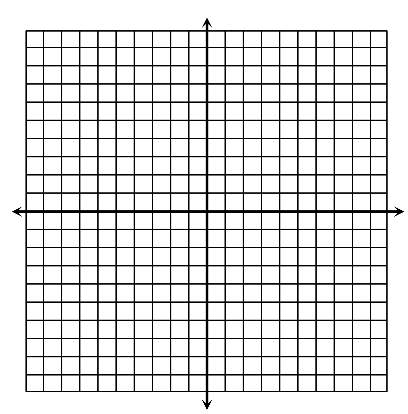
Center: _______________

Vertices: ______________________________

Co-Vertices: __________________________

Foci: __________________________________

2. $16x^2 + y^2 + 10y = -9$



Center: _______________

Vertices: ______________________________

Co-Vertices: __________________________

Foci: __________________________________

Ellipses
(Writing Equations)

Date:

Write the equation of the ellipse in standard form with the given characteristics.

1. Vertices: (-8, 9), (-8, -5)
Co-Vertices: (-5, 2), (-11, 2)

2. Vertices: (-3, 12), (-3, -6)
Co-Vertices: (2, 3), (-8, 3)

3. Vertices: (18, -7), (-8, -7)
Foci: (10, -7), (0, -7)

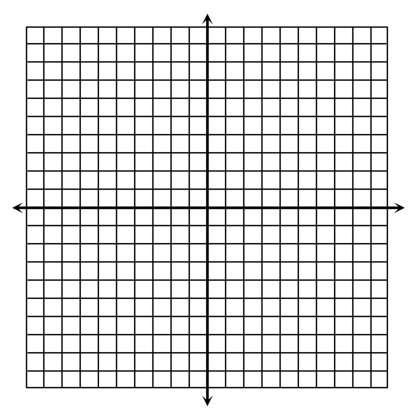
4. Co-Vertices: (3, 4), (-1, 4)
Foci: $(1, 4 \pm 2\sqrt{3})$

Hyperbolas
(Graphing)

Date:

Graph each hyperbola. Identify key characteristics.

1. $\dfrac{(x+2)^2}{16} - \dfrac{(y+4)^2}{36} = 1$



Center: ______________

Vertices: ____________________________

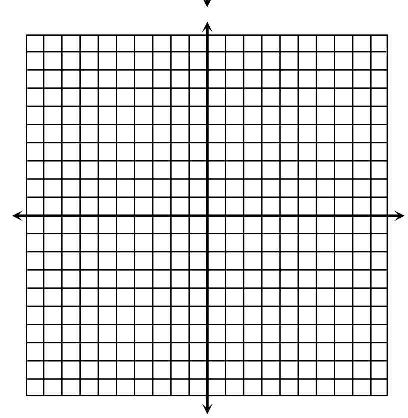
Co-Vertices: ____________________________

Foci: ____________________________

Asymptotes: ________________________

____

2. $-9x^2 + y^2 - 2y - 80 = 0$



Center: ______________

Vertices: ____________________________

Co-Vertices: ____________________________

Foci: ____________________________

Asymptotes: ________________________

____

Hyperbolas
(Writing Equations)

Date:

Write the equation of the hyperbola in standard form with the given characteristics.

1. Vertices: (10, 5), (-4, 5)
Co-Vertices: (3, 15), (3, -5)

2. Vertices: (8, 4), (8, -12)
Co-Vertices: (13, -4), (3, -4)

3. Vertices: (3, 10), (3, 2)
Foci: (3, 11), (3, 1)

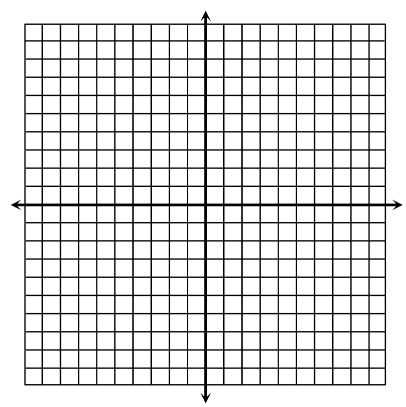
4. Co-Vertices: (-7, 10), (-7, -8)
Foci: $(-7 \pm \sqrt{130}, 1)$

Parabolas
(Graphing)

Date:

Graph each parabola. Identify key characteristics.

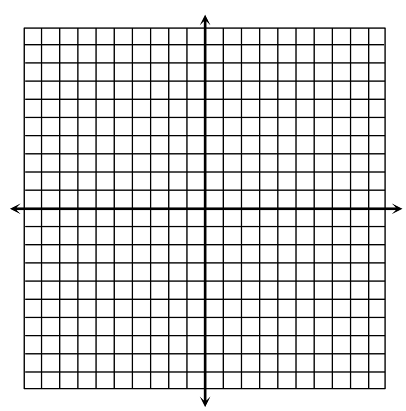
1. $-\frac{1}{12}(y+5)^2 = x$



Vertex: ______________ **Axis of Sym.:** ______________

Focus: ______________ **Directrix:** ______________

2. $x^2 - 4x - 6y = 32$



Vertex: ______________ **Axis of Sym.:** ______________

Focus: ______________ **Directrix:** ______________

Parabolas
(Writing Equations)

Date:

Write the equation of the parabola in standard form with the given characteristics.

1. Vertex: (1, -4),
Focus: (1, -1)

2. Vertex: (-3, -8)
Directrix: $x = 2$

3. Vertex: (0, 6)
Directrix: $y = 6.5$

4. Focus: (5, 2)
Directrix: $x = -9$

Identifying Conics

Date:

Classify the conic, then write the equation in standard form.

1. $-16x^2 + y^2 + 14y + 33 = 0$

2. $y^2 + 8x + 6y + 49 = 0$

3. $x^2 + y^2 - 8x - 18y + 72 = 0$

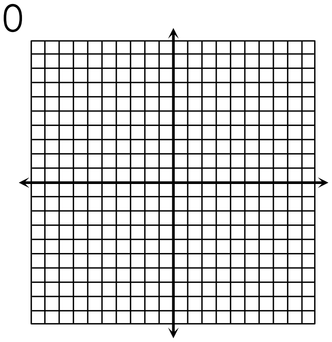
4. $9x^2 + 4y^2 - 18x + 16y - 11 = 0$

Non-Linear Systems
(Solve by Graphing)

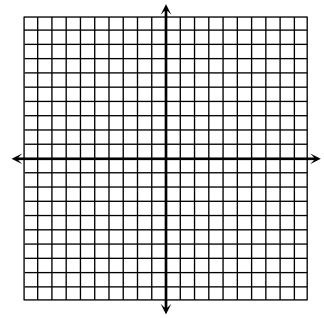
Date:

Solve each system by graphing.

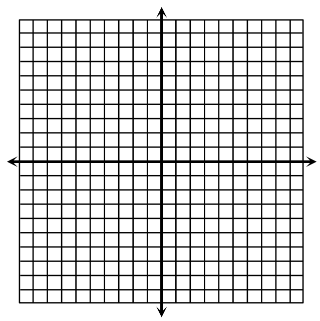
1. $\begin{cases} y = -x^2 - 8x - 13 \\ x - 2y = -10 \end{cases}$



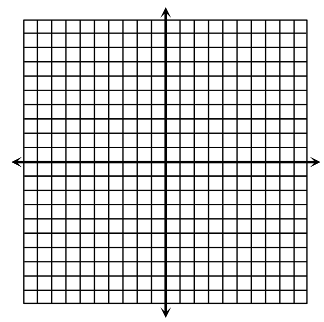
2. $\begin{cases} y = 2x^2 - 5 \\ x + y = -8 \end{cases}$



3. $\begin{cases} y = x^2 - 6x + 10 \\ y = -x^2 + 2x + 4 \end{cases}$



4. $\begin{cases} y^2 = -8(x - 1) \\ (x + 1)^2 + y^2 = 16 \end{cases}$



Non-Linear Systems
(Solve Algebraically)

Date:

Solve each system algebraically.

1. $\begin{cases} y = x^2 - 9x + 15 \\ 3x + y = 7 \end{cases}$

2. $\begin{cases} y = x^2 + 3x - 18 \\ y = x^2 - 2x + 2 \end{cases}$

3. $\begin{cases} y = 2x^2 + 10x + 7 \\ y = -x^2 - 5x - 5 \end{cases}$

4. $\begin{cases} 5x^2 + 5y^2 = 85 \\ x - y = 3 \end{cases}$

Sequences

Date:

Write the first five terms of each sequence.

1. $a_1 = -2; a_n = 5a_{n-1} + 3$ (for $n \geq 2$)

2. $a_1 = 3, a_2 = 7; a_n = a_{n-1} + 2a_{n-2}$ (for $n \geq 3$)

3. $a_n = n^3 - 4n^2$

4. $a_n = \frac{1}{n} + n$

5. Write a recursive formula: $\left\{1, 2, \frac{1}{2}, 4, \frac{1}{8}, 32, \ldots\right\}$

6. Write an explicit formula: $\{4, 10, 28, 82, 244, \ldots\}$

Series & Summations

Date:

Find S_7 for each series.

1. $\{2+3+6+18+108+\ldots\}$

2. $\{-2+6-18+54-162+\ldots\}$

Expand and evaluate each series.

3. $\sum_{k=1}^{10}(3-4k)$

4. $\sum_{m=2}^{18}(m+1)^2$

5. $\frac{2}{3} \cdot \sum_{x=8}^{24}\left(\frac{1}{2}x+7\right)$

Arithmetic Sequences

Date:

Write a rule for each sequence, then find the indicated term.

1. $\{19, 12, 5, -2, -9, \ldots\}; a_{32}$

2. $\left\{1, \frac{8}{3}, \frac{13}{3}, 6, \frac{23}{3}, \ldots\right\}; a_{25}$

If the terms belong to an arithmetic sequence, write the rule and find the indicated value.

3. $a_1 = -37$ and $a_{15} = 19$; Find a_7

Arithmetic Series

Date:

Find S_9 for each series.

1. $\{26 + 34 + 42 + 50 + \ldots\}$

2. $\{(-5) + (-8) + (-11) + (-14) + \ldots\}$

Evaluate each series.

3. $\sum_{n=1}^{19} (5n - 17)$

4. $\sum_{y=1}^{20} \left(8 - \frac{3}{2}y\right)$

5. $\sum_{p=2}^{37} (9 - n)$

6. $\sum_{i=5}^{24} \left(\frac{1}{4}i + 3\right)$

Geometric Sequences

Date:

Write a rule for each sequence, then find the indicated term.

1. $\{7, 21, 63, 189, \ldots\}; a_{10}$

2. $\{5120, -1280, -320, 80, \ldots\}; a_8$

If the terms belong to a geometric sequence, write the rule and find the indicated value.

3. $a_3 = -35.2$ and $a_7 = -9011.2$; Find a_1

Geometric Series

Date:

Find S_7 for each series.

1. $\{480 - 240 + 120 - 60 + \ldots\}$

2. $\left\{\frac{1}{2} + \frac{3}{2} + \frac{9}{2} + \frac{27}{2} + \ldots\right\}$

Evaluate each series.

3. $\sum_{k=1}^{9} 4^{k-1}$

4. $\sum_{m=1}^{7} -2187 \cdot \left(\frac{1}{3}\right)^{m-1}$

5. $\sum_{i=3}^{10} -\frac{1}{8} \cdot (-2)^{i-1}$

6. $\sum_{k=2}^{8} 2.56 \cdot \left(\frac{5}{2}\right)^{k-1}$

Infinite Geometric Series

Date:

Determine if the series is convergent or divergent. Find the sum when possible.

1. $\{1-2+4-8+\ldots\}$

2. $\{80+60+45+33.75+\ldots\}$

3. $\sum_{n=1}^{\infty} 30\cdot\left(\frac{3}{5}\right)^{k-1}$

4. $\sum_{y=1}^{\infty} \frac{2}{5}\cdot(-4)^{y-1}$

5. $\sum_{x=3}^{\infty} -6\cdot\left(\frac{4}{3}\right)^{x-1}$

6. $\sum_{m=1}^{\infty} -\frac{27}{8}\cdot\left(-\frac{2}{3}\right)^{m-1}$

Arithmetic vs. Geometric

Date:

Determine whether the sequence is arithmetic, geometric, or neither. Identify the common difference or ratio.

1. $\left\{\frac{5}{32}, \frac{5}{8}, \frac{5}{2}, 10, \ldots\right\}$

2. $\{-14, -15, -16, -17, \ldots\}$

3. $\left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \ldots\right\}$

4. $\left\{600, -180, 54, -\frac{81}{5}, \ldots\right\}$

Determine arithmetic or geometric, then find the sum.

5. $\{(-9)+(-1)+7+15+\ldots\}; S_{29}$

6. $\left\{\frac{1}{2}-\frac{1}{4}+\frac{1}{8}-\frac{1}{16}+\ldots\right\}; S_8$

Sequences Applications

Date:

1. Dan invested $2500 in stock that is estimated to increase in value by 6% each year. Determine whether this represents an arithmetic or geometric situation, then write a formula and find the value of the stock in 30 years.

2. Angela is at the library signing and giving away 400 books to promote a book she recently published. After the first hour, she had 362 books left. After the second hour, she had 324 books left. Determine whether this represents an arithmetic or geometric situation, then write a formula to find the number of books left after 8 hours.

Series Applications

Date:

1. The table below shows the number of miles that Sophie ran for the first three days of her new exercise routine. Determine whether this represents an arithmetic or geometric situation, then find the total miles ran after 16 days.

Day	Miles
1	1.8
2	2.2
3	2.6

2. During a flu outbreak, the hospital recorded 12 cases the first week, 30 cases the second week, and 75 cases the third week. Determine whether this represents an arithmetic or geometric sequence, then find the total number of cases reported after 6 weeks.

Counting Principle, Permutations & Combinations

Date:

1. Initials are created by the first letter of the first name, middle name, and last name. How many 3-letter initials are possible?

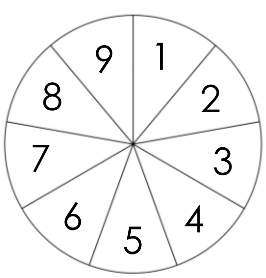
Determine if the situation represents a permutation or combination, then solve.

2. How many ways are there to select five employees from 24 to send to a conference?

3. There are 14 violinists in the orchestra. How many ways can the conductor choose first, second, and third chair?

4. How many different ways can you arrange the letters in the word FIREFIGHTER?

Theoretical Probability



Date:

1. The spinner to the left is spun then a card is randomly selected from a standard deck. Find the probability of spinning a number less than 5, then getting a jack or a diamond.

2. A bag of Skittles has 5 red, 11 orange, 7 yellow, 8 green, 14 blue, and 9 purple skittles. A skittle is randomly selected, eaten, then another is selected. Find each probability.

a) P(yellow then blue) **b)** P(purple then red)

c) P(two green) **d)** P(two orange)

Conditional Probability

Date:

1. A number 1-25 is randomly chosen. What is the probability that it is a multiple of 3, given that it is at most 16?

2. Out of 32 people surveyed, 15 own a dog and 8 own a cat. Four of those who own a dog also own a cat. If a person is chosen at random, find the probability that they own a dog, if it is known that they do not own a cat.

3. A taste test was conducted at the mall to see whether people prefer Coke or Pepsi. The results are shown in the table below. Find each probability.

	Males	Females
Coke	108	92
Pepsi	76	124

a) P(male **|** prefers Pepsi)

b) P(prefers Coke **|** female)

Frequencies

Date:

Four math classes took the same unit test. The joint relative frequencies are shown in the table to the left. Give the marginal joint frequencies. Then find each probability.

	Pass	Fail	Total
Class A	0.192	0.064	
Class B	0.256	0.024	
Class C	0.184	0.04	
Class D	0.208	0.032	
Total			

1. P(failed | Class B)

2. P(Class D | passed)

The Binomial Theorem

Date:

Expand each binomial:

1. $(x - 4)^5$

2. $(3k - 1)^6$

3. Find the third term of $(2m + n)^8$.

Binomial Probability

Date:

1. Type O blood is the most frequently occurring blood type found in 43% of the population. If 12 people are randomly selected, find the probability that exactly 7 people have this blood type.

2. A coin is tossed four times. What is the probability of getting tails no more than one time?

3. A die is rolled eight times. What is the probability that an odd number shows up at least six times?

Measures of Variation	**Date:**

Find the mean absolute deviation, standard deviation, and variance for each data set.

1. {16, 4, 18, 7, 9, 23, 5, 19, 11, 8}

MAD = ____________

Variance: σ^2 = ____________

Standard Deviation: σ = ____________

2. {146, 150, 144, 152, 147, 143}

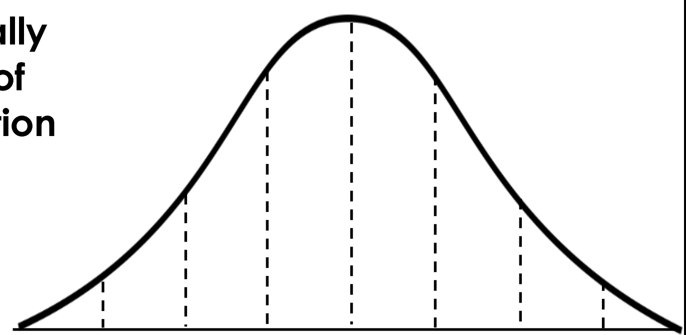
MAD = ____________

Variance: σ^2 = ____________

Standard Deviation: σ = ____________

Normal Distribution	**Date:**

Adult IQ scores are normally distributed with a mean of 100 and a standard deviation of 15. Label the normal distribution curve, then answer the questions



1. What percent of adults have an IQ no more than 85?

2. What is the probability that an adult will have an IQ between 70 and 100?

3. Approximately how many adults out of 500 have an IQ of at least 115?

z-Scores

Date:

1. A group of test scores are normally distributed with a mean of 85 and a standard deviation of 4. Find each z-score.

a) 82 **b)** 96 **c)** 73

2. Human body temperature is normally distributed with a mean of 98.2° and a standard deviation of 0.5°. If Nate's z-score is 2.2, find his temperature.

3. The number of pages in a book is normally distributed with a standard deviation of 25. If a particular book has 175 pages with a z-score of -0.4, find the mean number of pages.

Standard Normal Distribution

Date:

Credit scores for a group of 60 people are normally distributed with a mean of 680 and a standard deviation of 42.

1. If X represents the credit score, find each probability:

a. $P(X < 600)$ **b.** $P(X > 750)$ **c.** $P(700 < X < 825)$

2. Approximately how many people have a credit score of at least 720?

3. Approximately how many people have a credit score between 625 and 790?

Trigonometric Functions

Date:

1. If $\tan\theta = \dfrac{12}{35}$, find the remaining function values.

$\sin\theta =$ ________ $\csc\theta =$ ________

$\cos\theta =$ ________ $\sec\theta =$ ________

$\tan\theta =$ ________ $\cot\theta =$ ________

2. If $\sec\theta = \dfrac{\sqrt{6}}{2}$, find the remaining function values.

$\sin\theta =$ ________ $\csc\theta =$ ________

$\cos\theta =$ ________ $\sec\theta =$ ________

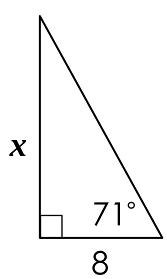
$\tan\theta =$ ________ $\cot\theta =$ ________

Finding Missing Sides & Angles

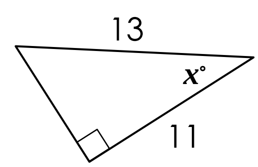
Date:

Find each missing side or angle:

1.



2.



3. A ladder placed against a building reaches a window that is 28 feet above the ground. If the angle of elevation from the bottom of the ladder to the window is 44°, find the length of the ladder.

4. A helicopter flying at an altitude of 5,600 feet spots a landing pad below. Find the angle of depression from the helicopter to the pad if the horizontal distance between them is 8,000 feet.

Degrees & Radians

Date:

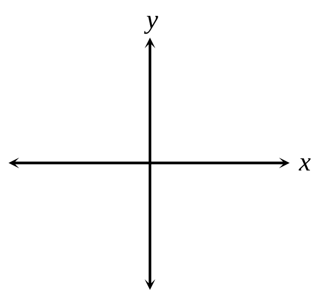
Convert each degree measure to radians and radian measure to degrees:

1. $63°$ **2.** $-320°$ **3.** $\frac{13\pi}{10}$ **4.** $-\frac{7\pi}{15}$

Give two coterminal angles (one positive and one negative) and a reference angle for each angle:

5. $155°$ **6.** $-\frac{2\pi}{9}$

7. Sketch angle θ in standard form with a terminal side that passes through the point (5, -4). Find the exact value for each function.



$\sin\theta =$ ________ $\csc\theta =$ ________

$\cos\theta =$ ________ $\sec\theta =$ ________

$\tan\theta =$ ________ $\cot\theta =$ ________

The Unit Circle

Date:

1. Which trigonometric functions are negative for angles terminating in Quadrant IV?

2. If $\csc\theta > 0$ and $\tan\theta < 0$, which quadrant(s) could the terminal side of θ lie?

Find the exact value of each trigonometric function:

3. $\cos 225°$ **4.** $\tan 30°$ **5.** $\sec 180°$

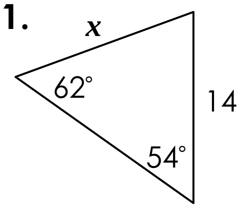
6. $\sin\frac{5\pi}{6}$ **7.** $\cot\frac{3\pi}{2}$ **8.** $\csc\frac{4\pi}{3}$

Law of Sines

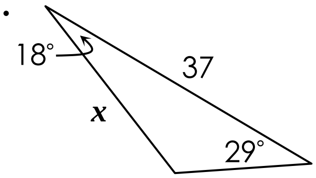
Date:

Find each missing measure to the nearest tenth:

1.



2.



Information about ΔABC is given below. Give all possibilities.

3. $m\angle A = 58°$, $AB = 35$, $BC = 31$, find $m\angle C$.

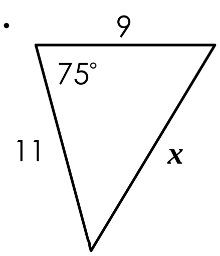
4. $m\angle B = 22°$, $BC = 31$, $AC = 9$, find $m\angle A$.

Law of Cosines

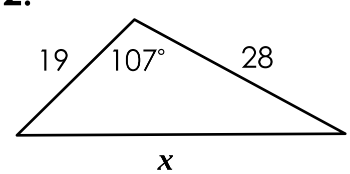
Date:

Find each missing measure to the nearest tenth:

1.



2.



3. In ΔQRS, $QR = 23$, $RS = 26$, and $QS = 21$. Find $m\angle R$.

Solving Triangles

Date:

Use the information given to solve each triangle:

1. In ΔJKL, $JK = 12$, $KL = 14$, and $JL = 7$.

$m\angle J =$ ________

$m\angle K =$ ________

$m\angle L =$ ________

2. In ΔABC, $m\angle C = 103°$, $m\angle A = 64°$, and $AB = 26$.

$m\angle B =$ ________

$AC =$ ________

$BC =$ ________

3. In ΔXYZ, $m\angle Y = 118°$, $YZ = 9$ and $XY = 15$.

$m\angle X =$ ________

$m\angle Z =$ ________

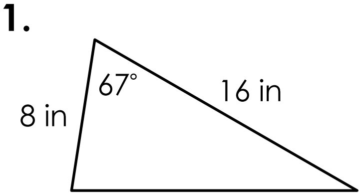
$XZ =$ ________

Area of Triangles

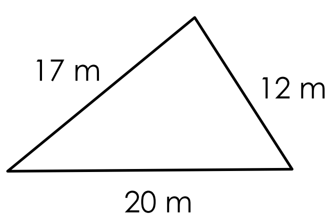
Date:

Find the area of each triangle to the nearest tenth:

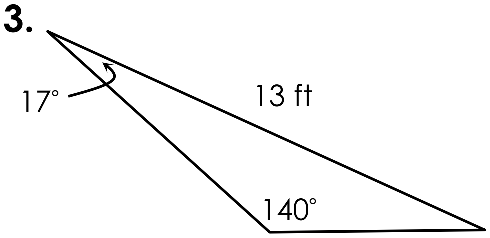
1.



2.



3.



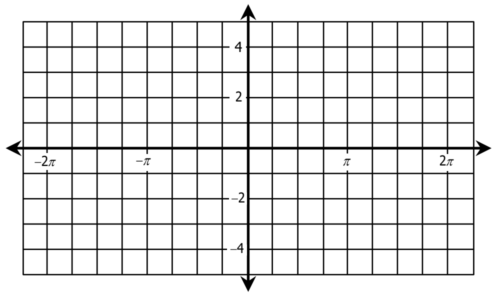
Graphing sin/cos/tan Functions

Date:

Identify the amplitude and period for each function, then graph.

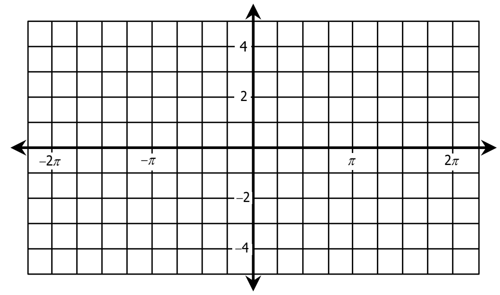
1. $y = 3 \cdot \sin\frac{4}{3}\theta$

Amp: _________ **Period:** __________



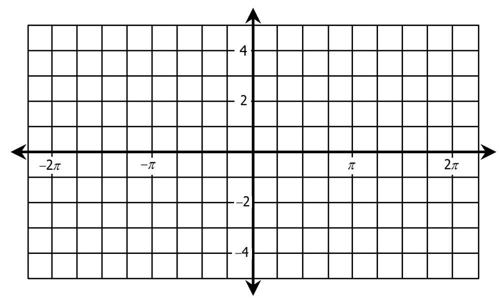
2. $y = \frac{9}{2} \cdot \cos\theta$

Amp: _________ **Period:** __________



3. $y = 2 \cdot \tan 2\theta$

Amp: _________ **Period:** __________

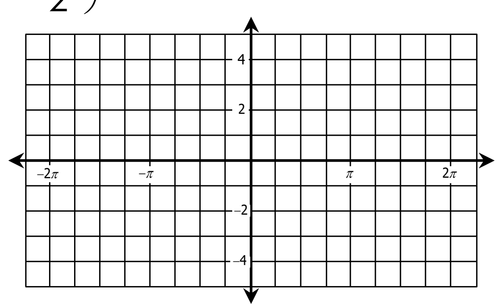


Translating Trigonometric Graphs

Date:

State the amplitude, period, phase shift, vertical shift, and midline for each function. Then graph.

1. $y = 3 \cdot \cos 2\left(\theta + \frac{3\pi}{2}\right) - 2$



Amplitude: _______________

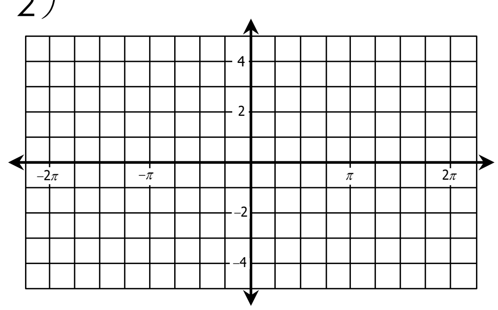
Period: _______________

Phase Shift: _______________

Vertical Shift: _____________

Midline: _______________

2. $y = \frac{1}{3} \cdot \tan\left(\theta - \frac{\pi}{2}\right) + 4$



Amplitude: _______________

Period: _______________

Phase Shift: _______________

Vertical Shift: _____________

Midline: _______________

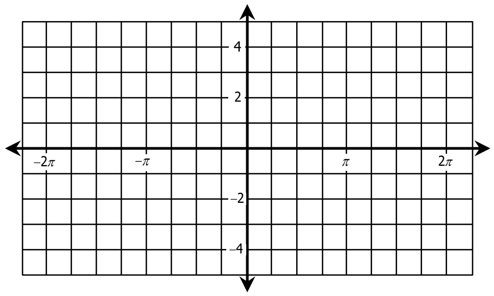
Graphing csc/sec/cot Functions

Date:

Identify the amplitude and period for each function, then graph.

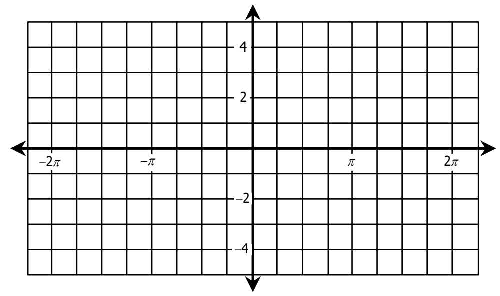
1. $y = \frac{1}{2} \cdot \csc 2\theta$

Amp: _________ Period: __________



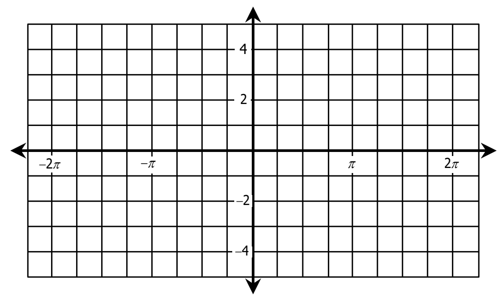
2. $y = 2 \cdot \sec \frac{8}{3}\theta$

Amp: _________ Period: __________



3. $y = \frac{1}{4} \cdot \cot \frac{1}{3}\theta$

Amp: _________ Period: __________



Fundamental Trig Identities

Date:

Find the exact value of each expression if $180° < \theta < 270°$.

1. If $\sin\theta = -\frac{\sqrt{3}}{4}$, find $\sec\theta$.

2. If $\cot\theta = \frac{5}{2}$, find $\sin\theta$.

Simplify each expression.

3. $(\sec\theta + 1)(\sec\theta - 1) \cdot \cot\theta$

4. $\dfrac{\cos^2\theta - \cos\theta \cdot \sec\theta}{\sin^3\theta}$

Sum and Difference of Angles Identities

Date:

Find the exact value of each function.

1. $\cos 255°$

2. $\tan\left(-\frac{7\pi}{12}\right)$

3. Given angles A and B in quadrant II with $\sin A = \frac{4}{5}$ and $\sin B = \frac{12}{13}$ find $\sin(A - B)$.

Double-Angle & Half-Angle Identities

Date:

Find the exact value of each function.

1. $\tan\theta = -\frac{1}{5}$ and $\frac{\pi}{2} < \theta < \pi$; find $\sin 2\theta$.

2. $\cos\theta = -\frac{12}{13}$ and $\pi < \theta < \frac{3\pi}{2}$; find $\tan 2\theta$.

3. $\sin\theta = \frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$; find $\cos\frac{\theta}{2}$.

Proving Identities

Date:

Prove each identity.

1. $1 - \sin\theta \cdot \cot\theta \cdot \cos\theta = \sin^2\theta$

2. $\cot\theta = \dfrac{\cos 2\theta + 1}{\sin 2\theta}$

3. $\dfrac{\csc\theta \cdot \cot^2\theta + \csc\theta}{\cot\theta \cdot \sec\theta} = \sec^2\theta$

4. $\sin(\pi - \theta) \cdot \sin\theta + \cos 2\theta = \cos^2\theta$

Solving Trigonometric Equations

Date:

Solve for the interval $0 < \theta < 360°$. Give all solutions in **degrees**.

1. $4\tan\theta - \sqrt{3} = 7\tan\theta$

2. $2\sin\theta \cdot \tan\theta - \sqrt{3}\tan\theta = 0$

Solve for the interval $\dfrac{\pi}{2} < \theta < \dfrac{3\pi}{2}$. Give all solutions in **radians**.

3. $2\cos^2\theta - 5\cos\theta = 3$

4. $4 - 4\cos^2\theta - \tan\theta \cdot \cot\theta = 0$

Simplifying Radicals & Classifying Numbers

Date:

1. Simplify each expression.

a) $\sqrt{208}$

$\sqrt{16} \cdot \sqrt{13}$

$\boxed{= 4\sqrt{13}}$

b) $-\sqrt{578}$

$-\sqrt{289} \cdot \sqrt{2}$

$\boxed{= -17\sqrt{2}}$

c) $\sqrt{153}$

$= \sqrt{9}\sqrt{17}$

$\boxed{= 3\sqrt{17}}$

2. Name all sets to which each value belongs.

a) $\dfrac{-\sqrt{400}}{\sqrt{16}}$

$= \dfrac{-20}{4}$

$= -5$

Z, Q, R

b) $1.58\overline{3}$

Q, R

c) $\sqrt{95}$

I, R

Order of Ops & Evaluating Expressions

Date:

Simplify each expression.

1. $6^3 \div \left\{(12+5^2)-(|-7|-15)^2\right\}$

$= 216 \div \{37 - 64\}$

$= 216 \div -27$

$= \boxed{-8}$

2. $\dfrac{3^3-6+\sqrt{-40+11^2}}{18-6^2\cdot 2}$

$= \dfrac{27-6+9}{18-36\cdot 2}$

$= \dfrac{30}{-54} = \boxed{\dfrac{-5}{9}}$

Evaluate the expressions below if *a* = 8, *b* = -2 and *c* = -9.

3. $|-a^2-2bc|$

$= |-(8)^2-2(-2)(-9)|$

$= |-64-36|$

$= \boxed{100}$

4. $-\dfrac{7}{6}c+\dfrac{3}{4}ab$

$= -\frac{7}{6}(-9)+\frac{3}{4}(8)(-2)$

$= \frac{63}{6} + \frac{-48}{4}$

$= \boxed{\frac{-3}{2}}$

Multi-Step Equations

Date:

Solve each equation.

1. $14a-(2a+9)=\frac{2}{3}(12a-18)$

$14a-2a-9=8a-12$

$12a-9=8a-12$

$4a=-3$

$\boxed{a=\frac{-3}{4}}$

2. $-3(10-x)=11x-2(4x+15)$

$-30+3x=11x-8x-30$

$-30+3x=3x-30$

$-30=-30$

$\boxed{\mathbb{R};\text{ all real numbers}}$

3. $\frac{3}{8}=\frac{6w-7}{2w+14}$

$3(2w+14)=8(6w-7)$

$6w+42=48w-56$

$98=42w$

$\boxed{\frac{7}{3}=w}$

4. Solve $F=\frac{9}{5}C+32$ for C

$F-32=\frac{9}{5}C$

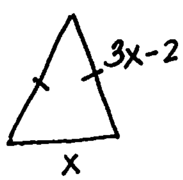
$\frac{5}{9}(F-32)=C$

$\boxed{\frac{5}{9}F-\frac{160}{9}=C}$

Word Problems

Date:

1. The leg of an isosceles triangle is two less than three times the length of its base. If the perimeter of the triangle is 45 meters, find the length of the leg.



let x = base
let 3x-2 = leg

$2(3x-2)+x=45$

$6x-4+x=45$

$7x=49$

$x=7$

$\text{leg}=3(7)-2$

$\boxed{=19\text{ m}}$

2. Find three consecutive numbers such that five times the largest number is 17 less than three times the sum of the smallest and median number.

let x = smallest #
let x+1 = median #
let x+2 = largest #

$5(x+2)=3(x+x+1)-17$

$5x+10=6x+3-17$

$5x+10=6x-14$

$24=x$

$\boxed{24, 25, 26}$

Absolute Value Equations

Date:

Solve each equation. ***Be sure to check for extraneous solutions.***

1. $|4x+6|=26$

$4x+6=26$
$4x=20$
$x=5$ ✓

$4x+6=-26$
$4x=-32$
$x=-8$ ✓

$\boxed{x=\{-8,5\}}$

2. $\dfrac{|-8-5r|}{6}=2$

$|-8-5r|=12$

$-8-5r=12$
$-5r=20$
$r=-4$ ✓

$-8-5r=-12$
$-5r=-4$
$r=\frac{4}{5}$ ✓

$\boxed{r=\{-4, 4/5\}}$

3. $8-|7y-9|=3$

$-|7y-9|=-5$
$|7y-9|=5$

$7y-9=5$
$7y=14$
$y=2$ ✓

$7y-9=-5$
$7y=4$
$y=4/7$ ✓

$\boxed{y=\{\frac{4}{7},2\}}$

4. $|8k-5|=4k-23$

$8k-5=4k-23$
$4k=-18$
$k=\frac{-9}{2}$ x

$8k-5=-4k+23$
$12k=28$
$k=\frac{7}{3}$ x

$\boxed{\emptyset}$

Multi-Step Inequalities

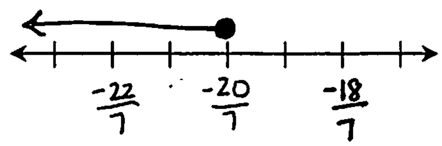
Date:

Solve, graph, and write the solution in interval notation.

1. $\dfrac{3(2x+16)}{-8}\geq x-1$

$6x+48\leq -8x+8$
$14x\leq -40$

$\boxed{x\leq -\frac{20}{7}}$



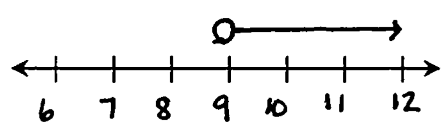
$\boxed{(-\infty, -\frac{20}{7}]}$

2. $-\dfrac{5}{4}(24-6m)>14-\dfrac{1}{2}(16-7m)$

$-30+\frac{15}{2}m>14-8+\frac{7}{2}m$

$-60+15m>28-16+7m$
$-60+15m>12+7m$
$8m>72$

$\boxed{m>9}$



$\boxed{(9,\infty)}$

Compound Inequalities

Date:

Solve, graph, and write the solution in interval notation.

1. $2(2-3c) \le -2$ or $4c+5<-3$

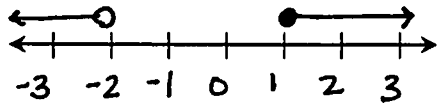
$4-6c \le -2$

$-6c \le -6$

$c \ge 1$

$4c < -8$

$c < -2$



$\boxed{(-\infty,-2) \cup [1,\infty)}$

2. $4-7a \le 67$ and $\dfrac{5a+2}{-9} > 2$

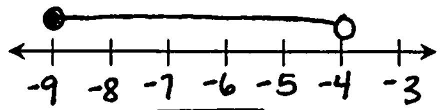
$-7a \le 63$

$a \ge -9$

$5a+2 < -18$

$5a < -20$

$a < -4$

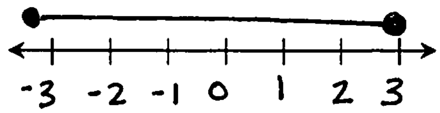


$\boxed{[-9,-4)}$

3. $-50 \le 9n-2 \le 25$

$-48 \le 9n \le 27$

$-\frac{16}{3} \le n \le 3$



$\boxed{[-3,3]}$

More Compound Inequalities

Date:

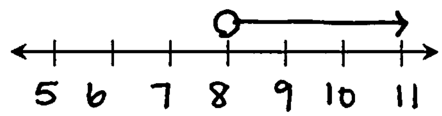
Solve, graph, and write the solution in interval notation.

1. $4-9x<-68$ and $-6x-5 \le -11$

$-9x < -72$

$x > 8$ and $x \ge 1$

$-6x \le -6$



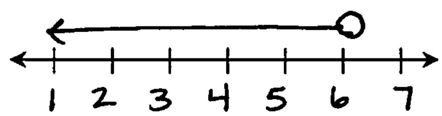
$\boxed{(8,\infty)}$

2. $10n-9<51$ or $7-n \ge 8$

$10n < 60$

$n < 6$ or $n \le 1$

$-n \ge 1$



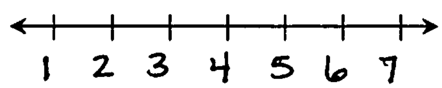
$\boxed{(-\infty,6)}$

3. $-2v-2>-8$ and $4v-7>9$

$-2v > -6$

$v < 3$ and $v > 4$

$4v > 16$



$\boxed{\emptyset}$

Absolute Value Inequalities	Date:

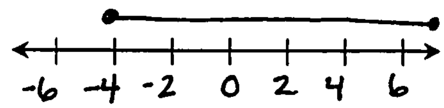
Solve, graph, and write the solution in interval notation.

1. $|6x-10| \le 34$

$6x-10 \le 34$ $6x-10 \ge -34$

$6x \le 44$ $6x \ge -24$

$x \le \frac{22}{3}$ and $x \ge -4$



$\boxed{\left[-4, \frac{22}{3}\right]}$

2. $-6|2v-6|+5 < -79$

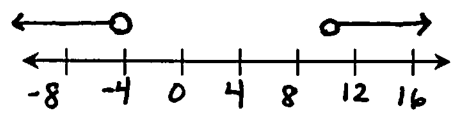
$-6|2v-6| < -84$

$|2v-6| > 14$

$2v-6 > 14$ $2v-6 < -14$

$2v > 20$ $2v < -8$

$v > 10$ or $v < -4$



$\boxed{(-\infty, -4) \cup (10, \infty)}$

Relations & Functions	Date:

For each relation below: a) State the domain and range, and **b)** Determine whether it is a function.

1. {(-1, 4), (0, 5), (1, 4), (2, 7)}

a) D: -1, 0, 1, 2
R: 4, 5, 7

b) Yes, function

2. {(-5, -2), (2, 1), (0, 3), (-5, -4)}

a) D: -5, 0, 2
R: -4, -2, 1, 3

b) Not a function

3. $y = 2x - 7$

a) D: $\mathbb{R}$
R: $\mathbb{R}$

b) Yes, function

4. $y = -x^2 + 3$

a) D: $\mathbb{R}$
R: $y \le 3$

b) Yes, function

Evaluating Functions

Date:

Given $f(x) = 8x - 9$, $g(x) = -x^2 + 7x$ and $h(x) = |2 - 4x|$, find each value.

1. $g(8)$

$g(8) = -(8)^2 + 7(8)$
$= -64 + 56$
$= \boxed{-8}$

2. $h(13) - f(-1)$

$h(13) = |2 - 4(13)| = |-50| = 50$
$f(-1) = 8(-1) - 9 = -17$
$50 - (-17) = \boxed{67}$

3. $f(3y - 1)$

$f(3y-1) = 8(3y-1) - 9$
$= 24y - 8 - 9$
$= \boxed{24y - 17}$

4. $g(a + 2)$

$g(a+2) = -(a+2)^2 + 7(a+2)$
$= -(a^2 + 4a + 4) + 7a + 14$
$= -a^2 - 4a - 4 + 7a + 14$
$= \boxed{-a^2 + 3a + 10}$

5. If $f(x) = -23$, find x.

$8x - 9 = -23$
$8x = -14$
$\boxed{x = -7/4}$

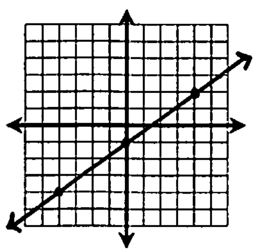
Graphing Linear Functions

Date:

Write the equation in slope-intercept form, if necessary. Then graph.

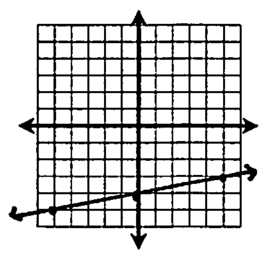
1. $12y = 9x - 12$

$y = \frac{3}{4}x - 1$



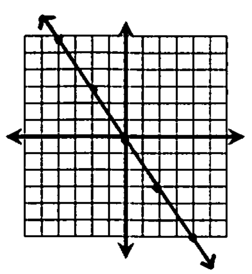
2. $x - 5y = 20$

$-5y = -x + 20$
$y = \frac{1}{5}x - 4$



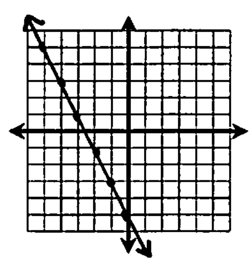
3. $15x = -10y$

$-\frac{3}{2}x = y$

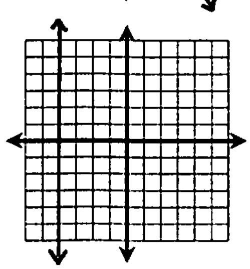


4. $-2x - y = 5$

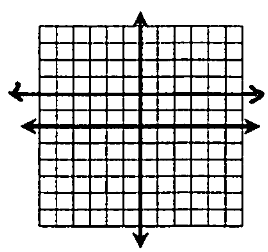
$-y = 2x + 5$
$y = -2x - 5$



5. $x = -4$



6. $y = 2$



x- and y-Intercepts

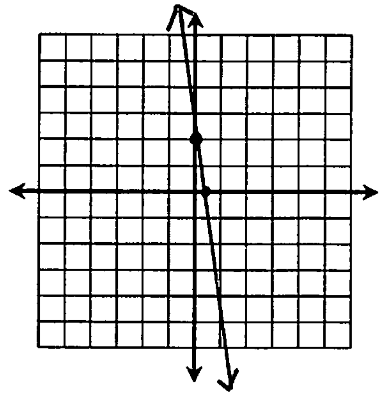
Date:

Find the x- and y-intercepts of each equation, then graph.

1. $y = -6x + 2$

x-int:
$0 = -6x + 2$
$-2 = -6x$
$\frac{1}{3} = x$
$(\frac{1}{3}, 0)$

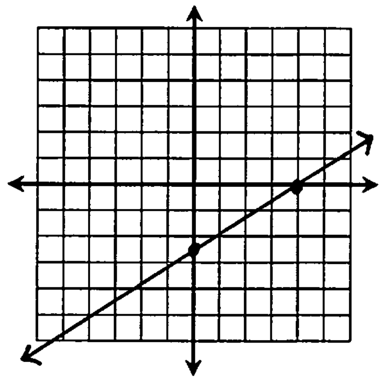
y-int:
$y = -6(0) + 2$
$y = 0 + 2$
$y = 2$
$(0, 2)$



2. $5x = 8y + 20$

x-int:
$5x = 8(0) + 20$
$5x = 20$
$x = 4$
$(4, 0)$

y-int:
$5(0) = 8y + 20$
$0 = 8y + 20$
$-20 = 8y$
$-\frac{5}{2} = y$
$(0, -\frac{5}{2})$



Writing Linear Functions

Date:

Write a linear equation in slope-intercept form with the given information.

1. Passes through (-2, 11) with a slope of -4

$y - 11 = -4(x + 2)$
$y - 11 = -4x - 8$
$\boxed{y = -4x + 3}$

2. Passes through (-6, 4) with a slope of $\frac{1}{2}$

$y - 4 = \frac{1}{2}(x + 6)$
$y - 4 = \frac{1}{2}x + 3$
$\boxed{y = \frac{1}{2}x + 7}$

3. Passes through the points (-4, 8) and (8, -1)

$m = \frac{-1 - 8}{8 + 4} = \frac{-9}{12} = \frac{-3}{4}$

$y - 8 = \frac{-3}{4}(x + 4)$
$y - 8 = \frac{-3}{4}x - 3$
$\boxed{y = \frac{-3}{4}x + 5}$

Parallel vs. Perpendicular Lines

Date:

Determine whether the functions are parallel, perpendicular, or neither.

1. $x - y = 5$ and $y = -x + 9$

$-y = -x + 5$

$y = x - 5$

$\boxed{\text{Perpendicular}}$

2. $4x - 3y = 9$ and $8y = 6x - 8$

$-3y = -4x + 9$

$y = \frac{4}{3}x - 3$

$y = \frac{3}{4}x - 1$

$\boxed{\text{Neither}}$

3. $y = 2$ and $y = -2$

$\boxed{\text{Parallel}}$

Write a linear equation in slope-intercept form with the given information.

4. Passes through (2, -6) and parallel to the line $2x - 3y = 15$.

$y = \frac{2}{3}x - 5$

$y + 6 = \frac{2}{3}(x - 2)$

$y + 6 = \frac{2}{3}x - \frac{4}{3}$

$\boxed{y = \frac{2}{3}x - \frac{22}{3}}$

5. Passes through (8, -5) and perpendicular to the line passing through the points (-3, 9) and (4, -5).

$m = \frac{-5-9}{4+3} = \frac{-14}{7} = -2$

$m = \frac{1}{2}$

$y + 5 = \frac{1}{2}(x - 8)$

$y + 5 = \frac{1}{2}x - 4$

$\boxed{y = \frac{1}{2}x - 9}$

Linear Equations Applications

Date:

Define variables, set up an equation, then solve.

1. Caitlyn is going away to college and will need to rent a truck to help move. The cost of the truck is \$35 plus \$0.79 per mile. If her college is 85 miles away and she budgeted \$100 for the rental, will she have enough money?

C = cost

m = miles

$C = 0.79m + 35$

$C = 0.79(85) + 35$

$= 67.15 + 35$

$= 102.15$

$\boxed{\text{No, Caitlyn does not have enough.}}$

2. Max works two jobs, one at the zoo and the other at the library. He drives 24 miles round-trip to the zoo and 9 miles round-trip to the library. He never works two jobs in the same day. His car has a 15-gallon gas tank and gets 20 miles per gallon. If his tank is now empty and he has worked 12 days at the library since last filling up, how many days did he work at the zoo?

d = days

g = gallons

$g = 24d_1 + 9d_2$

$g = 24d + 9(12)$

$300 = 24d + 108$

$192 = 24d$

$8 = d$

$\boxed{\text{Max worked 8 days at the zoo.}}$

Linear Regression

Date:

On October 3rd, Holly invested $2500 in some stocks. The table below shows the value of her investment on certain days since investing.

	0	3	8	13	17
day	Oct. 3	Oct. 6	Oct. 11	Oct. 16	Oct. 20
value	$2500	$2509.12	$2525.74	$2539.02	$2548.55

1. Use linear regression to find an equation of best fit.

$y = 2.88x + 2500.83$

2. Find the approximate value of her investment on Oct. 31st.

$\downarrow$ $x = 28$

$y = 2.88(28) + 2500.83$

$y = 2581.47$

$\boxed{\$2581.47}$

3. How many days will it take her investment to reach a value of $3,000?

$\downarrow$ $y = 3000$

$3000 = 2.88x + 2500.83$

$499.17 = 2.88x$

$173.32 = x$

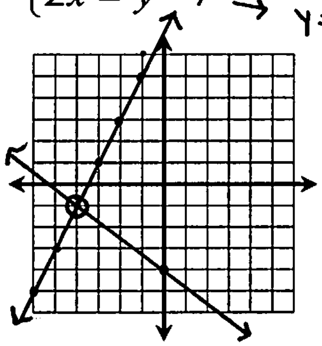
$\boxed{174 \text{ days}}$

Solving Systems: by Graphing & Substitution

Date:

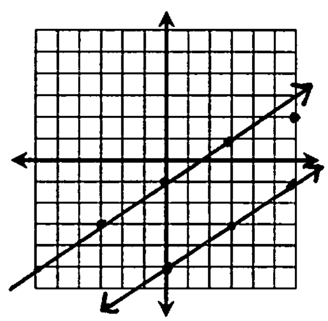
Solve questions 1 and 2 by graphing.

1. $\begin{cases} 3x + 4y = -16 \\ 2x = y - 7 \end{cases}$ $\rightarrow y = -\frac{3}{4}x - 4$, $\rightarrow y = 2x + 7$



$\boxed{(-4, -1)}$

2. $\begin{cases} 2x - 3y = 15 \\ -6y = 6 - 4x \end{cases}$ $\rightarrow y = \frac{2}{3}x - 5$, $\rightarrow y = \frac{2}{3}x - 1$



$\boxed{\emptyset}$

3. Solve by substitution.

$\begin{cases} 2x + 7y = -23 \\ 5x - y = -39 \end{cases}$ $\rightarrow y = 5x + 39$

$2x + 7(5x + 39) = -23$

$2x + 35x +$

Solving Systems: by Elimination	**Date:**

Solve each system by elimination.

1. $\begin{cases} (x-4y=-14)\cdot 2 \\ 6x+8y=-12 \end{cases}$ $\rightarrow$

$$\begin{array}{r} 2x-8y=-28 \\ +\ 6x+8y=-12 \\ \hline 8x=-40 \\ x=-5 \end{array}$$

$$-5-4y=-14$$
$$-4y=-9$$
$$y=9/4$$

$\boxed{(-5, 9/4)}$

2. $\begin{cases} 6y=2x+4 \\ 7x+14=21y \end{cases}$ $\rightarrow$

$$7\cdot(-2x+6y=4)$$
$$2\cdot(7x-21y=-14)$$

$$\begin{array}{r} -14x+42y=28 \\ +\ 14x-42y=-28 \\ \hline 0=0 \end{array}$$

$\boxed{\mathbb{R}}$

Systems Applications	**Date:**

Kate bought 5 pounds of hamburger and 2 pounds of hot dogs and paid $28.50. Eric bought half the amount of hamburger and a fourth of the amount of hot dogs that Kate did and paid $12.50. Find the cost per pound of hamburger and hot dogs.

x = hamburgers
y = hotdogs

$$5x+2y=28.50$$
$$-2\cdot(2.5x+.5y=12.50)$$

$$\begin{array}{r} 5x+2y=28.50 \\ +\ -5x-y=25.00 \\ \hline y=3.50 \end{array}$$

$$5x+2(3.50)=28.50$$
$$5x+7=28.50$$
$$5x=21.50$$
$$x=4.30$$

Hamburger costs $4.30/lb and hotdogs cost $3.50/lb.

Systems with Three Variables

Date:

Solve the following system:

$$\begin{cases} x+2y-4z=7 \\ 3x-y+6z=-7 \\ 2x+3y-z=10 \end{cases}$$

$x+2y-4z=7$
$2\cdot(3x-y+6z=-7)$

$x+2y-4z=7$
$+\ 6x-2y+12z=-14$
$7x+8z=-7$

$3\cdot(3x-y+6z=-7)$
$2x+3y-z=10$

$9x-3y+18z=-21$
$+\ 2x+3y-z=10$
$11x+17z=-11$

$-11\cdot(7x+8z=-7)$
$7\cdot(11x+17z=-11)$

$-77x-88z=77$
$+\ 77x+119z=-77$
$31z=0$
$z=0$

$7x+8(0)=-7$
$7x=-7$
$x=-1$

$-1+2y-4(0)=7$
$2y=8$
$y=4$

$\boxed{(-1,4,0)}$

Three Variable Application

Date:

Jack has 63 pennies, dimes, and quarters worth $6.30. If the number of dimes is three less than the number of quarters, how many of each coin does he have?

x = pennies
y = dimes
z = quarters

$0.01x+0.10y+0.25z=6.30$
$y=z-3$
$x+y+z=63$

$0.01x+0.10(z-3)+0.25z=6.30$
$x+(z-3)+z=63$

$0.01x+0.35z=6$
$-0.01\cdot(x+2z=66)$

$0.01x+0.35z=6$
$+\ -0.01x-0.02z=-0.66$
$0.33z=5.94$
$z=18$

$y=18-3$
$y=15$

$x+15+18=63$
$x=30$

$\boxed{\text{30 pennies, 15 dimes, and 18 quarters}}$

Linear Inequalities & Systems

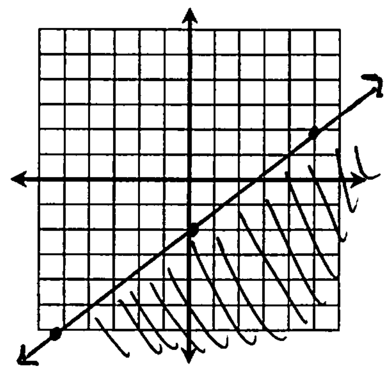
Date:

1. Graph the linear inequality.

$4x - 5y \geq 10$

$-5y \geq -4x + 10$

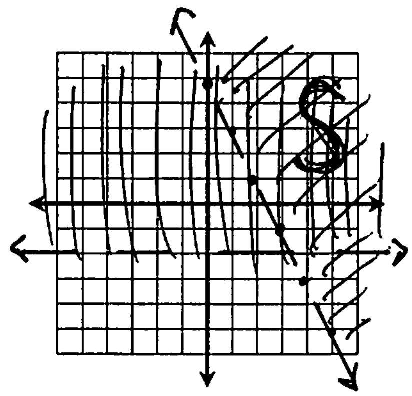
$y \leq \frac{4}{5}x - 2$



2. Graphing the system of linear inequalities.

$\begin{cases} 6x + 3y > 15 \\ y > -2 \end{cases}$ $\rightarrow$ $3y > -6x + 15$

$y > -2x + 5$



Linear Inequalities Application

Date:

It costs \$20 per day to board a dog and \$16 per day to board a cat. The kennel needs to make at least \$800 per day to cover expenses, but can hold no more than 80 total animals. Write a system of inequalities, graph, and give two possible solutions to the number of dogs and cats they can board each day.

x = dog

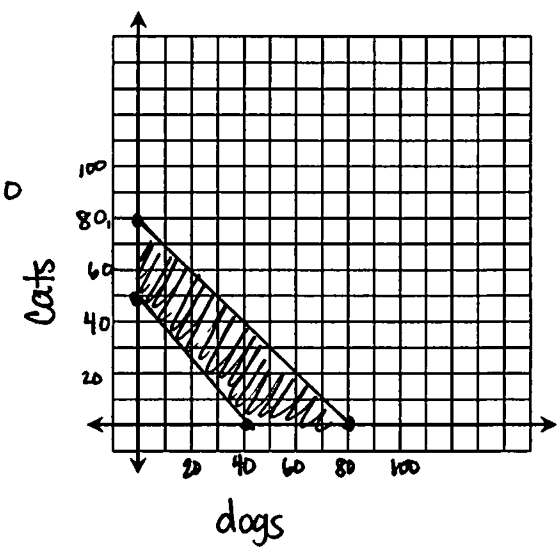
y = cat

$20x + 16y \geq 800 \rightarrow y \geq -\frac{5}{4}x + 50$

$x + y \leq 80 \rightarrow y \leq -x + 80$

(40,20) - 40 dogs + 20 cats

(15,55) - 15 dogs + 55 cats



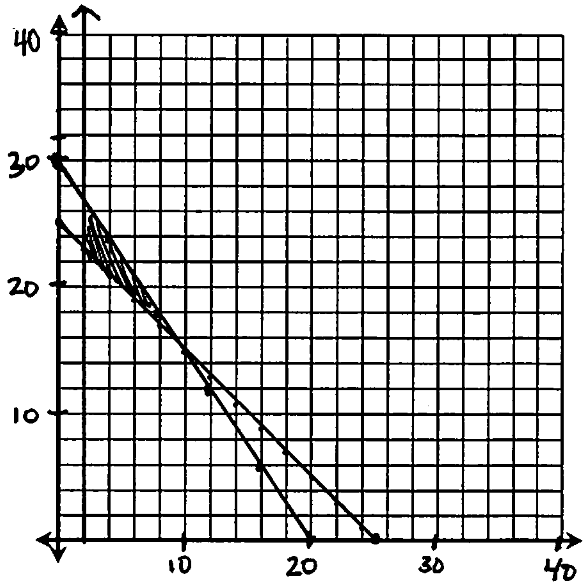
Linear Programming

Date:

Erica is making necklaces and bracelets to sell at the craft show. It takes her 3 hours to make a necklace and 2 hours to make a bracelet. She sells the necklaces for $18 and the bracelets for $15. She has 60 hours available this coming week to make the jewelry and would like to make at least 25 total pieces of jewelry that includes a minimum of 2 necklaces. How many of each should she make to maximize her profit?

x = necklace
y = bracelet

$3x+2y \le 60 \rightarrow y \le -\frac{3}{2}x+30$
$x+y \ge 25 \rightarrow y \ge -x+25$
$x \ge 2$
$y \ge 0$



$P = 18x+15y$ (Max)

$(10,15): 18(10)+15(15) = 405$
$(2,23): 18(2)+15(23) = 381$
$\boxed{(2,27): 18(2)+15(27) = 441}$

Max profit - 2 necklaces + 27 bracelets

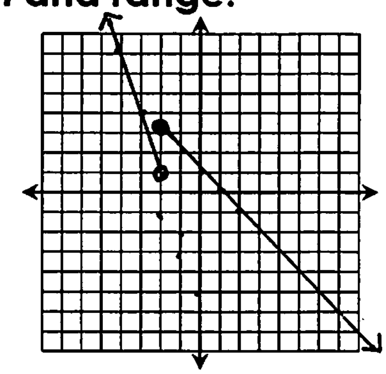
Piecewise Functions

Date:

Graph each function. State the domain and range.

1. $f(x) = \begin{cases} -3x-5 & \text{if } x < -2 \\ -x+1 & \text{if } x \ge -2 \end{cases}$

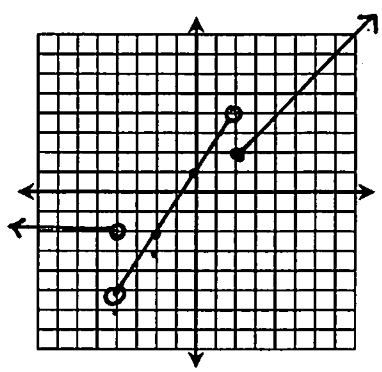
D: $\mathbb{R}$
R: $\mathbb{R}$



2. $g(x) = \begin{cases} -2 & \text{if } x < -4 \\ \frac{3}{2}x+1 & \text{if } -4 < x < 2 \\ x & \text{if } x \ge 2 \end{cases}$

D: $\{x \mid x \ne -4\}$

R: $\{y \mid y > -5\}$



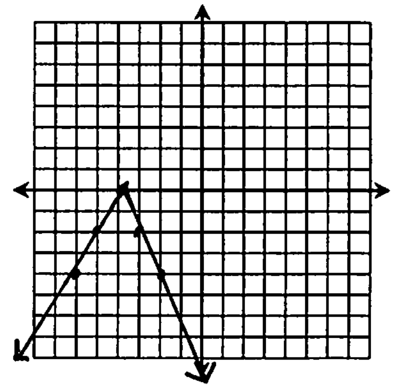
Absolute Value Functions	Date:

Graph function using a table of values.

1. $f(x) = -2|x+4|$

x	f(x)
-6	-4
-5	-2
-4	0
-3	-2
-2	-4

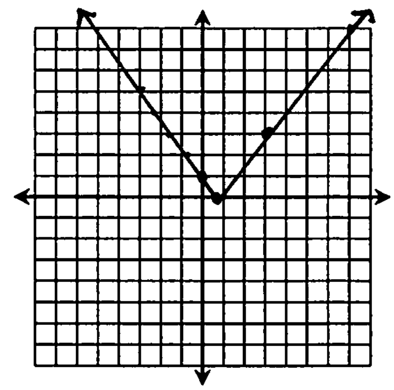
$x+4=0$
$x=-4$



2. $f(x) = \left|\frac{4}{3}x - 1\right|$

x	f(x)
-2	4/3
-1	7/3
0	1
3/4	0
1	1/3
2	5/3
3	3

$\frac{4}{3}x-1=0$
$\frac{4}{3}x=1$
$x=\frac{3}{4}$



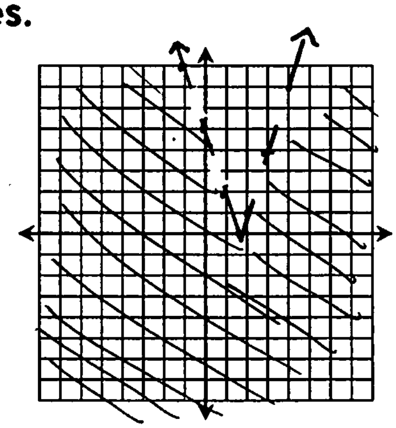
Absolute Value Inequalities	Date:

Graph inequality using a table of values.

1. $y < |3x-5|$

x	f(x)
-1	8
0	5
1	2
5/3	0
2	1
3	4
4	7

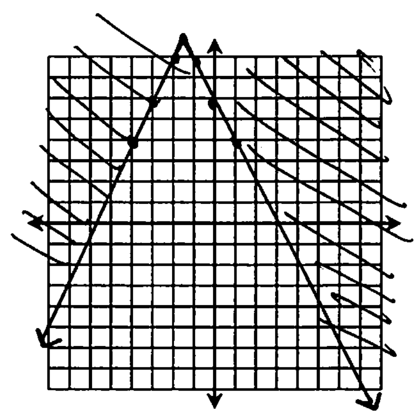
$3x-5=0$
$3x=5$
$x=\frac{5}{3}$



2. $y \geq -|2x+3|+9$

x	f(x)
-4	4
-3	6
-2	8
-3/2	9
-1	8
0	6
1	4

$2x+3=0$
$2x=-3$
$x=-3/2$



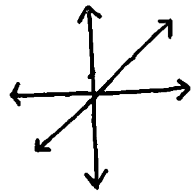
Parent Functions and Transformations

Date:

Identify the parent function for each function family, then sketch the graph.

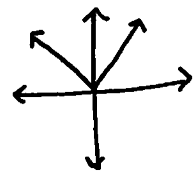
1. Linear

$f(x) = x$



2. Absolute Value

$f(x) = |x|$



Describe the transformations in each function.

3. $f(x) = |x+8|$

Left 8

4. $f(x) = -|x| - 3$

Down 3, Reflect over x-axis

5. $f(x) = 4|x-1|$

Right 1, Vertical Stretch

6. $f(x) = \frac{4}{5}|x+5| + 2$

Left 5, up 2, Vertical Compression

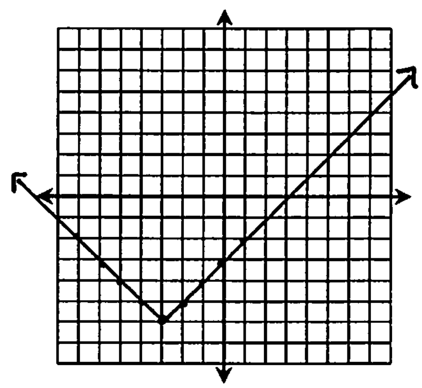
Absolute Value Functions: Vertex Form

Date:

Identify the vertex, then graph.

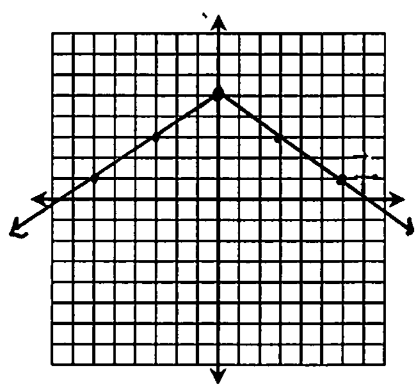
1. $f(x) = |x+3| - 6$

Vertex: $(-3,-6)$



2. $f(x) = -\frac{2}{3}|x| + 5$

Vertex: $(0,5)$



Quadratic Functions: Standard Form

Date:

Give the axis of symmetry and vertex of each function, then graph using a table of values.

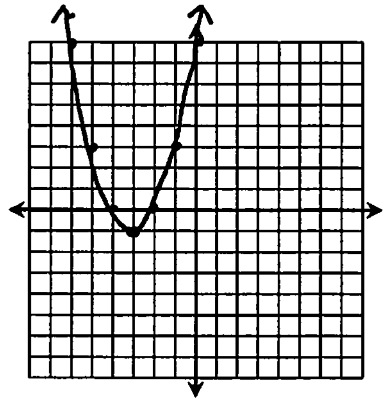
1. $f(x) = x^2 + 6x + 8$

x	f(x)
-6	8
-5	3
-4	0
-3	-1
-2	0
-1	3
0	8

$x = \frac{-6}{2(1)} = -3$

V: (-3, -1)

Axis: x = -3



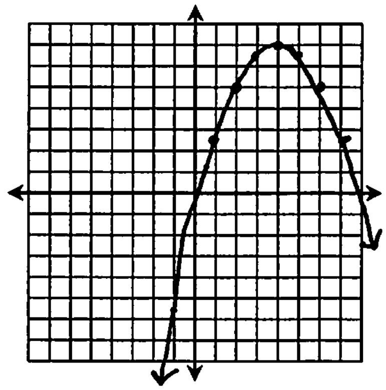
2. $f(x) = -\frac{1}{2}x^2 + 4x - 1$

x	f(x)
1	2.5
2	5
3	6.5
4	7
5	6.5
6	5
7	2.5

$x = \frac{-4}{2(-1/2)} = 4$

V: (4, 7)

Axis: x = 4



Quadratic Inequalities

Date:

Give the axis of symmetry and vertex of each function, then graph using a table of values.

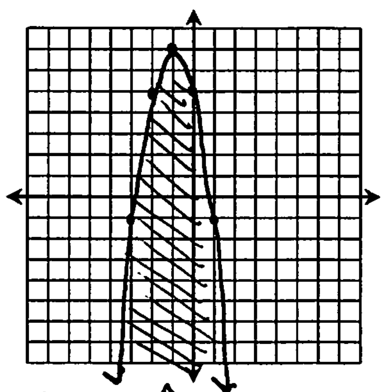
1. $y \le -2x^2 - 4x + 5$

x	y
-4	-11
-3	-1
-2	5
-1	7
0	5
1	-1
2	-11

$x = \frac{4}{2(-2)} = -1$

V: (-1, 7)

Axis: x = -1



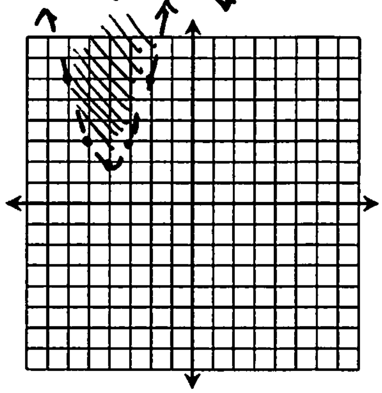
2. $y > x^2 + 8x + 18$

x	y
-7	11
-6	6
-5	3
-4	2
-3	3
-2	6
-1	11

$x = \frac{8}{2(-1)} = -4$

V: (-4, 2)

Axis: x = -4



Quadratic Functions:
Vertex Form

Date:

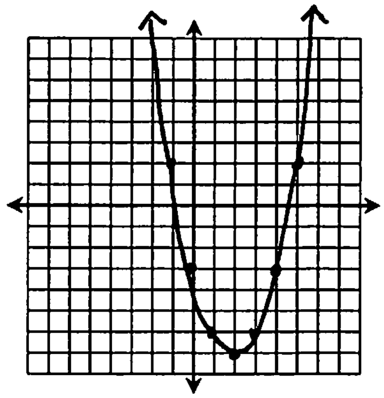
Give the axis of symmetry and vertex of each function. Graph using a table of values.

1. $f(x) = (x-2)^2 - 7$

V: (2, -7)

Axis: x = 2

x	f(x)
-1	2
0	-3
1	-6
2	-7
3	-6
4	-3
5	2

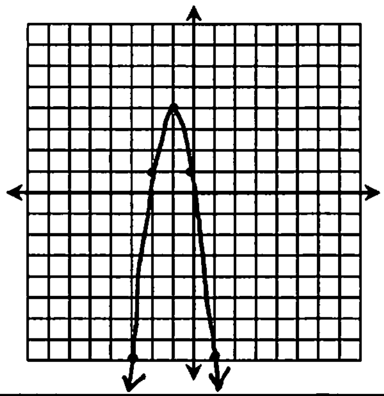


2. $f(x) = -3(x+1)^2 + 4$

V: (-1, 4)

Axis: x = -1

x	f(x)
-4	-23
-3	-8
-2	1
-1	4
0	1
1	-8
2	-23



More Piecewise Functions Practice

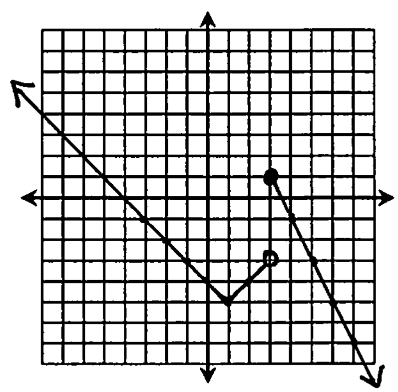
Date:

Graph each function. State the domain and range.

1. $f(x) = \begin{cases} |x-1| - 5 & \text{if } x < 3 \\ -2x + 7 & \text{if } x \geq 3 \end{cases}$

D: ℝ

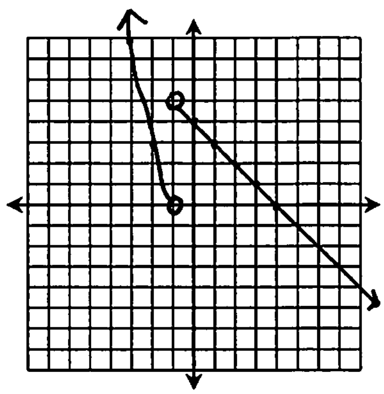
R: ℝ



2. $f(x) = \begin{cases} x^2 - 1 & \text{if } x < -1 \\ -x + 4 & \text{if } x > 1 \end{cases}$

D: {x | x ≠ -1}

R: ℝ



Converting Standard Form to Vertex Form

Date:

Write each function in vertex form. Identify the vertex.

1. $f(x) = x^2 - 8x + 19$ $\quad \frac{-8}{2} = -4$; $\quad (-4)^2 = 16$

$f(x) = (x^2 - 8x + 16) + 19 - 16$

$f(x) = (x-4)(x-4) + 3$

$\boxed{f(x) = (x-4)^2 + 3}$ $\quad$ vertex: $(4, 3)$

2. $f(x) = -2x^2 - 4x + 14$

$f(x) = -2(x^2 + 2x) + 14$ $\quad \frac{2}{2} = 1$; $\quad 1^2 = 1$

$f(x) = -2(x^2 + 2x + 1) + 14 + 2$

$f(x) = -2(x+1)(x+1) + 16$

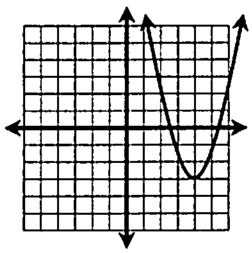
$\boxed{f(x) = -2(x+1)^2 + 16}$ $\quad$ vertex: $(-1, 16)$

End Behavior & Increasing Decreasing Intervals

Date:

Identify the increasing/decreasing intervals and end behavior of each graph.

1.

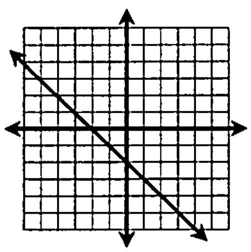


Inc. Intervals: $(4, \infty)$ **Dec. Intervals:** $(-\infty, 4)$

End Behavior: As $x \to \infty$, $f(x) \to$ ∞

As $x \to -\infty$, $f(x) \to$ ∞

2.

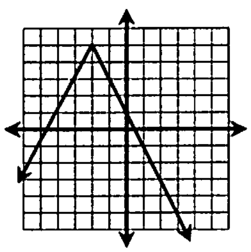


Inc. Intervals: – **Dec. Intervals:** $(-\infty, \infty)$

End Behavior: As $x \to \infty$, $f(x) \to$ $-\infty$

As $x \to -\infty$, $f(x) \to$ ∞

3.



Inc. Intervals: $(-\infty, -2)$ **Dec. Intervals:** $(-2, \infty)$

End Behavior: As $x \to \infty$, $f(x) \to$ $-\infty$

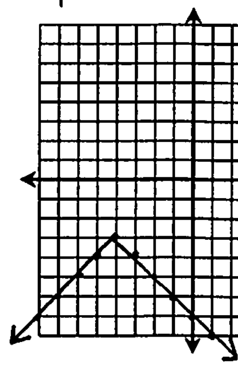
As $x \to -\infty$, $f(x) \to$ $-\infty$

Function Families Review

Date:

Graph each function and give its characteristics.

1. $f(x) = -|x+4| - 3$



Function Family: Absolute Value

D: $\mathbb{R}$; R: $\{y \mid y \le -3\}$

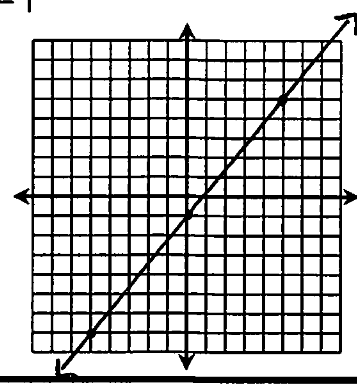
End Behavior: As $x \to \infty$, $f(x) \to$ $-\infty$

As $x \to -\infty$, $f(x) \to$ $-\infty$

Increasing Interval(s): $(-\infty, -4)$

Decreasing Interval(s): $(-4, \infty)$

2. $f(x) = \frac{6}{5}x - 1$



Function Family: Linear

D: $\mathbb{R}$; R: $\mathbb{R}$

End Behavior: As $x \to \infty$, $f(x) \to$ ∞

As $x \to -\infty$, $f(x) \to$ $-\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): —

Greatest Integer Functions

Date:

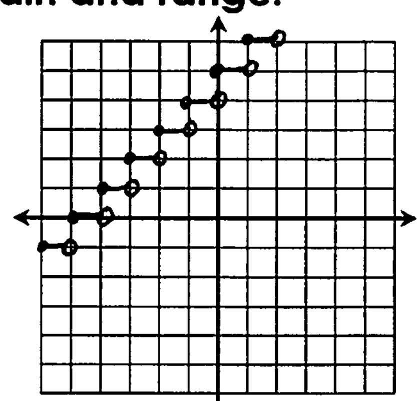
Graph each function. State the domain and range.

1. $f(x) = [\![x+5]\!]$

x	f(x)
0	5
0.25	5
0.5	5
0.75	5
1	6

D: $\mathbb{R}$

R: $\mathbb{Z}$

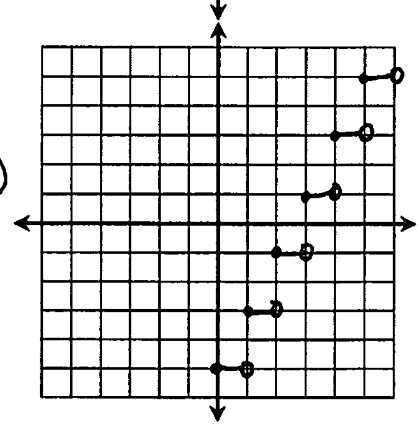


2. $f(x) = 2[\![x-3]\!] + 1$

x	f(x)
0	-5
0.5	-5
1	-3
1.5	-3

D: $\mathbb{R}$

R: $\mathbb{Z}$ (odd only)



Quadratic Roots

Date:

1. What are quadratic roots?

Quadratic roots are where the graph crosses the x-axis

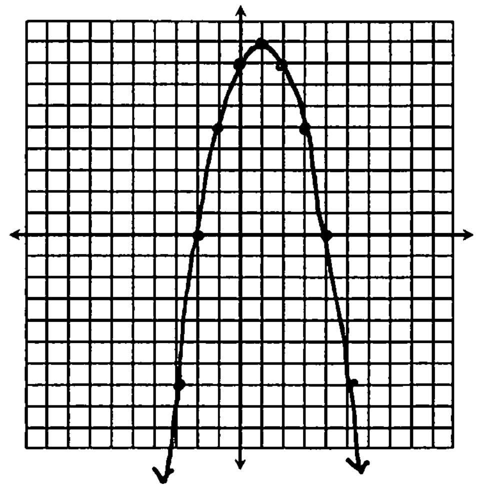
2. What other names are quadratic roots called?

zeros, solutions

3. Find the roots by graphing: $f(x) = -x^2 + 2x + 8$

$x = \frac{-2}{2(-1)} = 1$

x	f(x)
-2	0
-1	5
0	8
1	9
2	8
3	5
4	0



Factoring Review

Date:

Factor each polynomial completely:

1. $x^2 - 14x - 95$

$\boxed{(x-19)(x+5)}$

2. $5x^2 - 40x + 80$

$5(x^2-8x+16)$

$5(x-4)(x-4)$

$\boxed{5(x-4)^2}$

3. $4x^2 - x - 14$

$\boxed{(x-2)(4x+7)}$

4. $16x^2 - 49$

$\boxed{(4x+7)(4x-7)}$

5. $5 - 20x^2$

$5(1-4x^2)$

$\boxed{5(1-2x)(1+2x)}$

6. $24x^2 - 10x$

$\boxed{2x(12x-5)}$

Solving Quadratics by Factoring

Date:

Solve each quadratic equation by factoring:

1. $x^2 + 11x + 30 = 0$

$(x+6)(x+5)=0$

$x+6=0 \quad | \quad x+5=0$

$x=-6 \quad | \quad x=-5$

$\boxed{x=\{-6,-5\}}$

2. $2x^2 + x = 2x + 6$

$2x^2 - x - 6 = 0$

$(x-2)(2x+3)=0$

$x-2=0 \quad | \quad 2x+3=0$

$x=2 \quad | \quad x=-3/2$

$\boxed{x=\{-\frac{3}{2}, 2\}}$

3. $4x^2 - 74 = x^2 + 1$

$3x^2 - 75 = 0$

$3(x^2-25)=0$

$3(x+5)(x-5)=0$

$3 \neq 0 \quad | \quad x+5=0 \quad | \quad x-5=0$

$x=-5 \quad | \quad x=5$

$\boxed{x=\{-5,5\}}$

4. $8x^2 + 43x = 7x$

$8x^2 + 36x = 0$

$4x(2x+9)=0$

$4x=0 \quad | \quad 2x+9=0$

$x=0 \quad | \quad x=-9/2$

$\boxed{x=\{-\frac{9}{2}, 0\}}$

Solving Quadratics by Square Roots

Date:

Solve each quadratic equation by square roots:

1. $x^2 - 10 = 159$

$\sqrt{x^2} = \sqrt{169}$

$x = \pm 13$

$\boxed{x=\{\pm 13\}}$

2. $36x^2 - 1 = 0$

$\sqrt{36x^2} = \sqrt{1}$

$6x = \pm 1$

$x = \pm \frac{1}{6}$

$\boxed{x=\{\pm \frac{1}{6}\}}$

3. $2x^2 + 7 = 41$

$2x^2 = 34$

$\sqrt{x^2} = \sqrt{17}$

$x = \pm \sqrt{17}$

$\boxed{x=\{\pm \sqrt{17}\}}$

4. $-\frac{2}{3}x^2 - 19 = -35$

$-\frac{2}{3}x^2 = -16$

$\sqrt{x^2} = \sqrt{24}$

$x = \pm 2\sqrt{6}$

$\boxed{x=\{\pm 2\sqrt{6}\}}$

Factored Form vs. Vertex Form

Date:

1. What are the roots of the quadratic equation below?

$$f(x) = (3x+2)(x-4)$$

$3x+2=0$, $x=-\frac{2}{3}$ | $x-4=0$, $x=4$

$\boxed{x=\{-\frac{2}{3}, 4\}}$

2. Write the equation below in factored form and vertex form. Give the axis of symmetry, vertex, and roots.

$$f(x) = -x^2 - 12x - 27$$

Factor: $-x^2-12x-27=0$

$x^2+12x+27=0$

$(x+9)(x+3)=0$

$x+9=0$, $x=-9$ | $x+3=0$, $x=-3$

Roots: $x=\{-9,-3\}$

Factored Form: $f(x)=(x+9)(x+3)$

Vertex: $f(x)=-x^2-12x-27$

$f(x)=-1(x^2+12x)-27$

$6^2=36$

$f(x)=-1(x^2+12x+36)-27+36$

$f(x)=-(x+6)^2+9$

Vertex: $(-6,9)$

Axis: $x=-6$

Imaginary Numbers

Date:

Simplify each expression:

1. $\sqrt{-16}$

$=\sqrt{-1}\sqrt{16} = \boxed{4i}$

2. $\sqrt{-5}\cdot\sqrt{-12}\cdot\sqrt{-3}$

$= i\sqrt{5}\cdot i\sqrt{12}\cdot i\sqrt{3} = i^3\sqrt{180}$

$=-i\cdot\sqrt{36}\cdot\sqrt{5}$

$=\boxed{-6i\sqrt{5}}$

3. $14i\cdot 3i$

$=42i^2$

$=42(-1) = \boxed{-42}$

4. i^{67}

$=(i^4)^{16}\cdot i^3$

$=(1)^{16}\cdot i^3 = i^3 = \boxed{-i}$

Solve the equations below:

5. $x^2+100=0$

$\sqrt{x^2}=\sqrt{-100}$

$x=\pm i\sqrt{100}$

$\boxed{x=\{\pm 10i\}}$

6. $7-3x^2=211$

$-3x^2=204$

$\sqrt{x^2}=\sqrt{-68}$

$x=\pm i\sqrt{4}\sqrt{17}$

$\boxed{x=\{\pm 2i\sqrt{17}\}}$

Complex Numbers

Date:

Simplify each expression:

1. $(-9-i)-(4-3i)$

$-9-i-4+3i$

$\boxed{-13+2i}$

2. $(2+5i)(-4-3i)$

$-8-6i-20i-15i^2$

$-8-26i+15$

$\boxed{7-26i}$

3. $\dfrac{14\cdot i}{2i\cdot i}$

$\dfrac{14i}{2i^2}=\dfrac{14i}{-2}=\boxed{-7i}$

4. $\dfrac{(4-5i)(2-4i)}{(2+4i)(2-4i)}=\dfrac{8-16i-10i+20i^2}{4-8i+8i-16i^2}$

$=\dfrac{8-26i-20}{4+16}$

$=\dfrac{-12-26i}{20}=\boxed{\dfrac{-6-13i}{10}}$

Simplify, then name all sets to which the value belongs:

5. $\sqrt{-6}\cdot\sqrt{-3}$

$i\sqrt{6}\cdot i\sqrt{3}$

$i^2\sqrt{18}=\boxed{-3\sqrt{2}}$

I, R, C

6. $i^{22}=(i^{20})\cdot(i^2)$

$=(i^4)^5\cdot i^2$

$=\boxed{-1}$

Completing the Square

Date:

Solve by completing the square:

1. $x^2+12x+47=0$

$x^2+12x=-47$

$x^2+12x+36=-47+36$

$(x+6)^2=-11$

$x+6=\pm\sqrt{-11}$

$\boxed{x=\{-6\pm i\sqrt{11}\}}$

$\frac{12}{2}=6$; $6^2=36$

2. $-4x^2+408=20-8x$

$-4x^2+8x=-388$

$x^2-2x=97$

$x^2-2x+1=97+1$

$(x-1)^2=98$

$x-1=\pm\sqrt{98}$

$\boxed{x=\{1\pm 7\sqrt{2}\}}$

$-\frac{2}{2}=-1$; $(-1)^2=1$

The Quadratic Formula	**Date:**

Solve by the quadratic formula:

1. $10x^2 - 9x = 2x + 6$

$10x^2 - 11x - 6 = 0$

$x = \dfrac{11 \pm \sqrt{(-11)^2 - 4(10)(-6)}}{2(10)}$

$x = \dfrac{11 \pm \sqrt{121 + 240}}{20}$

$x = \dfrac{11 \pm \sqrt{361}}{20}$

$x = \dfrac{11 \pm 19}{20}$ ↗ $\dfrac{30}{20} = \dfrac{3}{2}$ ↘ $\dfrac{-8}{20} = -\dfrac{2}{5}$

$\boxed{x = \left\{-\frac{2}{5}, \frac{3}{2}\right\}}$

2. $-x^2 = 8x + 26$

$-x^2 - 8x - 26 = 0$

$x = \dfrac{8 \pm \sqrt{(-8)^2 - 4(-1)(-26)}}{2(-1)}$

$x = \dfrac{8 \pm \sqrt{64 - 104}}{-2}$

$x = \dfrac{8 \pm \sqrt{-40}}{-2} = \dfrac{8 \pm 2i\sqrt{10}}{-2} = -4 \pm i\sqrt{10}$

$\boxed{x = \{-4 \pm i\sqrt{10}\}}$

The Discriminant	**Date:**

Find the discriminant, then determine the number and type of roots:

1. $-2x^2 + 14x - 23 = 0$

$(14)^2 - 4(-2)(-23)$

$196 - 184$

$\boxed{= 12}$

2 real, irrat roots

2. $16x^2 - 8x + 1 = 0$

$(-8)^2 - 4(16)(1)$

$64 - 64$

$\boxed{= 0}$

1 real, rat root

3. $x^2 - 10x + 31 = 0$

$(-10)^2 - 4(1)(31)$

$100 - 124$

$\boxed{= -24}$

2 imagionary roots

4. $3x^2 - 12 = 0$

$0^2 - 4(3)(-12)$

$0 + 144$

$\boxed{= 144}$

2 real, rat roots

Choosing the Best Method

Date:

Solve one equation by factoring and the other by completing the square. Explain your reasoning.

1. $x^2 - 14x + 37 = 0$

$\frac{-14}{2} = -7$; $(-7)^2 = 49$

$x^2 - 14x + 49 = -37 + 49$

$(x-7)^2 = 12$

$x - 7 = \pm\sqrt{12}$

$\boxed{x = \{7 \pm 2\sqrt{3}\}}$

* Completing the square - even x-term.

2. $-x^2 - 9x + 136 = 0$

$-1(x^2 + 9x - 136) = 0$

$-1(x+17)(x-8) = 0$

$-1 \neq 0$	$x+17=0$	$x-8=0$
	$x=-17$	$x=8$

$\boxed{x = \{-17, 8\}}$

* factorable

Quadratic Applications

Date:

1. The dimensions of a square are altered so that 10 inches is added to one side while 1 inch is subtracted from the other. The area of the resulting rectangle is 80 in². How much larger is the area of the rectangle compared to the square?

$(x+10)(x-1) = 80$

$x^2 + 9x - 10 = 80$

$x^2 + 9x - 90 = 0$

$(x+15)(x-6) = 0$

$\cancel{x=-15}$ | $x = 6$

$A_S = 6 \cdot 6 = 36$

$L = x + 10 = 16$

$W = x - 1 = 5$

$A_R = 16 \cdot 5 = 80$

$80 - 36$

$\boxed{= 44 \text{ in}^2}$

2. Find three consecutive positive even integers such that the product of the median and largest integer is 6 less than 21 times the smallest integer.

#1: x

#2: $x+2$

#3: $x+4$

$(x+2)(x+4) = 21x - 6$

$x^2 + 6x + 8 = 21x - 6$

$x^2 - 15x + 14 = 0$

$(x-14)(x-1) = 0$

$x = 14$ | $\cancel{x=1}$

$\boxed{14, 16, 18}$

Projectile Motion

Date:

The football coach threw a football from a platform to his quarterback below. The height of the football, h, at time t seconds is modeled by the equation $h(t) = -16t^2 + 28t + 15$.

1. What is the maximum height of the ball?

$x = \frac{-28}{2(-16)} = \frac{-28}{-32} = \frac{7}{8}$

$y = -16\left(\frac{7}{8}\right)^2 + 28\left(\frac{7}{8}\right) + 15$

$\boxed{y = 27.25 \text{ ft}}$

2. If the quarterback caught the ball at a height of 6 feet, how many seconds was the ball in the air?

$-16t^2 + 28t + 15 = 6$

$-16t^2 + 28t + 9 = 0$

$x = \frac{-28 \pm \sqrt{(28)^2 - 4(-16)(9)}}{2(-16)}$

$x = \frac{-28 \pm \sqrt{784 + 576}}{-32}$

$x = \frac{-28 \pm \sqrt{1360}}{-32}$

$\nearrow x = -0.277$

$\searrow x = 2.027$

$\boxed{2.027 \text{ sec}}$

3. Give the domain and range of the function.

D: $0 \le x \le 2.027$

R: $6 \le y \le 27.25$

Quadratic Regression

Date:

The table below shows the population in a town from 2018 to 2024, in thousands of people.

	0	1	2	3	4	5	6
year	2018	2019	2020	2021	2022	2023	2024
population	24.8	26.2	27.3	27.8	28.1	27.5	27.2

1. Use quadratic regression to write an equation for the population, y, in years since 2018, x.

$y = -0.21x^2 + 1.62x + 24.81$

2. Find the approximate population of the town in 2027.

$x = 9$

$y = -0.21(9)^2 + 1.62(9) + 24.81$

$y = 22.38$

$\boxed{22{,}380}$

3. If the population continues the same trend, in what year will the population reach 15,000 people?

$-0.21x^2 + 1.62x + 24.81 = 15$

$-0.21x^2 + 1.62x + 9.81 = 0$

$x = \frac{-1.62 \pm \sqrt{(1.62)^2 - 4(-.21)(9.81)}}{2(-.21)}$

$x = \frac{-1.62 \pm \sqrt{10.8648}}{-.42}$

$\nearrow x = \cancel{-3.99}$

$\searrow x = 11.7$

$\boxed{\text{during 2029}}$

Simplifying Monomials

Date:

Simplify the expressions below.

1. $-3xy^3 \cdot (2x^2y^4)^3 + 11x^7y^{15}$

$-3xy^3 \cdot 8x^6y^{12} + 11x^7y^{15}$

$-24x^7y^{15} + 11x^7y^{15}$

$\boxed{-13x^7y^{15}}$

2. $(-9a^3b^5)^2 \cdot \left(\frac{1}{3}a^2b\right)^3$

$81a^6b^{10} \cdot \frac{1}{27}a^6b^3$

$\boxed{3a^{12}b^{13}}$

3. $\frac{5m^3n^7 \cdot 2m^2n}{12m^3n^9}$

$\frac{10m^5n^8}{12m^3n^9}$

$\boxed{\frac{5m^2}{6n}}$

4. $\left(\frac{60c^4d^2}{5cd^3}\right)^2 \cdot \frac{2}{3}c^{-6}d^{-1}$

$\frac{3600c^8d^4}{25c^2d^6} \cdot \frac{2}{3c^6d}$

$\frac{7200c^8d^4}{50c^8d^7} = \boxed{\frac{96}{d^3}}$

Polynomials

Date:

Simplify the expressions below. Classify your answer.

1. $(-5x - 2x^3 - 4x^2) + (x^3 - 7x + 4x^2)$

$\boxed{-x^3 - 12x}$; cubic binomial

2. $(a^4 + 10a^2 - 9a) - (2a^2 - 3a^4 + a - 8a^3)$

$\boxed{4a^4 + 8a^3 + 8a^2 - 10a}$; quartic polynomial

3. $m^2(m-1)^3$

$= m^2(m-1)(m-1)(m-1)$

$= (m^3 - m^2)(m^2 - 2m + 1)$

$= m^5 - 2m^4 + m^3 - m^4 + 2m^3 - m^2$

$= \boxed{m^5 - 3m^4 + 3m^3 - m^2}$; quintic polynomial

4. $\frac{-36k^4 + 8k^3 - 4k^2}{4k^2}$

$\boxed{-9k^2 + 2k - 1}$; quadratic trinomial

Graphing Polynomial Functions

Date:

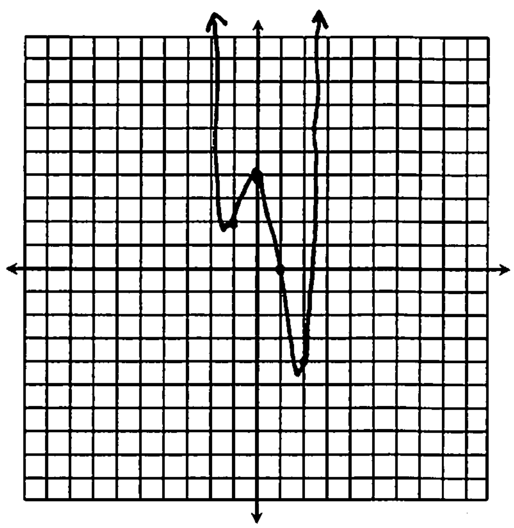
1. Determine the end behavior of an odd-degree polynomial with a negative leading coefficient.

As $x \to \infty$, $f(x) \to -\infty$

As $x \to -\infty$, $f(x) \to \infty$

2. Graph the function below. Identify all key characteristics.

$f(x) = x^4 - x^3 - 4x^2 + 4$



Domain: ℝ **Range:** $\{y \mid y \geq -4.31\}$

End Behavior: As $x \to \infty$, $f(x) \to \infty$

As $x \to -\infty$, $f(x) \to \infty$

Rel. Minimum(s): $(-1.09, 1.95), (1.84, -4.31)$

Rel. Maximum(s): $(0, 4)$

Inc. Intervals: $(-1.09, 0)$ $(1.84, \infty)$

Dec. Intervals: $(-\infty, -1.09)$ $(0, 1.84)$

Zeros of Polynomial Functions

Date:

Give the zeros, their multiplicity, and the effect on the graph for each function below.

1. $f(x) = x(x+5)^4(2x+7)^5$

Zero	Multiplicity	Effect
0	1	intersect
-5	4	bounce
$-\frac{7}{2}$	5	intersect

2. $f(x) = x^5 - 2x^4 + x^3$

$x^3(x^2 - 2x + 1)$

$x^3(x-1)^2$

Zero	Multiplicity	Effect
0	3	intersect
1	2	bounce

3. The graph of a polynomial function has zeros at 4 (multiplicity 2) and 9 (multiplicity 1). Write an equation in standard form to represent this function.

$f(x) = (x-4)^2(x-9)$

$f(x) = (x^2 - 8x + 16)(x-9)$

$f(x) = x^3 - 9x^2 - 8x^2 + 72x + 16x - 144$

$\boxed{f(x) = x^3 - 17x^2 + 88x - 144}$

Factoring Polynomials (2 Terms)

Date:

Factor each expression. Be sure to check for a GCF.

1. $49a^6 - 9b^4$

$\boxed{(7a^3 - 3b^2)(7a^3 + 3b^2)}$

2. $r^3 + 64s^3$

$\boxed{(r + 4s)(r^2 - 4rs + 16s^2)}$

3. $125w^3 - 27$

$\boxed{(5w - 3)(25w^2 + 15w + 9)}$

4. $8n^5 - 32n$

$8n(n^4 - 4)$

$\boxed{8n(n^2 + 2)(n^2 - 2)}$

5. $16x^4 - 128x$

$16x(x^3 - 8)$

$\boxed{16x(x - 2)(x^2 + 2x + 4)}$

6. $7m^5n^2 + 7m^2n^2$

$7m^2n^2(m^3 + 1)$

$\boxed{7m^2n^2(m + 1)(m^2 - m + 1)}$

Factoring Polynomials (3-4 Terms)

Date:

Factor each expression. Be sure to check for a GCF.

1. $3x^3 - 12x^2 - 36x$

$3x(x^2 - 4x - 12)$

$\boxed{3x(x - 6)(x + 2)}$

2. $m^4 + 14m^2 + 45$

$\boxed{(m^2 + 5)(m^2 + 9)}$

3. $y^5 - 20y^3 + 64y$

$y(y^4 - 20y^2 + 64)$

$y(y^2 - 4)(y^2 - 16)$

$\boxed{y(y + 2)(y - 2)(y + 4)(y - 4)}$

4. $4a^4 + 3a^2 - 1$

$(4a^2 - 1)(a^2 + 1)$

$\boxed{(2a + 1)(2a - 1)(a^2 + 1)}$

5. $k^3 - 2k^2 - 7k + 14$

$k^2(k - 2) - 7(k - 2)$

$\boxed{(k^2 - 7)(k - 2)}$

6. $2x^5 + x^4 - 18x - 9$

$x^4(2x + 1) - 9(2x + 1)$

$(x^4 - 9)(2x + 1)$

$\boxed{(x^2 + 3)(x^2 - 3)(2x + 1)}$

Solving Polynomial Equations

Date:

Solve each polynomial equation below by factoring.

1. $15x^3 + 6x^2 = 0$

$3x^2(5x+2)=0$

$3x^2=0 \quad 5x+2=0$

$x^2=0 \quad x=\frac{-2}{5}$

$x=0$

$\boxed{x=\left\{-\frac{2}{5}, 0\right\}}$

2. $20x^3 - 45x = 0$

$5x(4x^2-9)=0$

$5x(2x+3)(2x-3)=0$

$5x=0 \quad 2x+3=0 \quad 2x-3=0$

$x=0 \quad x=\frac{-3}{2} \quad x=\frac{3}{2}$

$\boxed{x=\left\{0, \pm\frac{3}{2}\right\}}$

3. $x^3 + 125 = 0$

$(x+5)(x^2-5x+25)=0$

$x+5=0 \quad x=\frac{5\pm\sqrt{(-5)^2-4(1)(25)}}{2(1)}$

$x=-5 \quad x=\frac{5\pm\sqrt{-75}}{2}$

$x=\frac{5\pm 5i\sqrt{3}}{2}$

$\boxed{x=\left\{-5, \frac{5\pm 5i\sqrt{3}}{2}\right\}}$

Solving Polynomial Equations

Date:

Solve each polynomial equation below by factoring.

1. $x^4 + x^2 - 72 = 0$

$(x^2+9)(x^2-8)=0$

$x^2+9=0 \quad x^2-8=0$

$x^2=-9 \quad x^2=8$

$x=\pm 3i \quad x=\pm 2\sqrt{2}$

$\boxed{x=\left\{\pm 3i, \pm 2\sqrt{2}\right\}}$

2. $6x^3 - 13x^2 - 15x = 0$

$x(6x^2-13x-15)=0$

$x(6x+5)(x-3)=0$

$x=0 \quad 6x+5=0 \quad x-3=0$

$x=\frac{-5}{6} \quad x=3$

$\boxed{x=\left\{\frac{-5}{6}, 0, 3\right\}}$

3. $3x^3 + x^2 - 54x - 18 = 0$

$x^2(3x+1)-18(3x+1)=0$

$(x^2-18)(3x+1)=0$

$x^2-18=0 \quad 3x+1=0$

$x^2=18 \quad x=-\frac{1}{3}$

$x=\pm 3\sqrt{2}$

$\boxed{x=\left\{-\frac{1}{3}, \pm 3\sqrt{2}\right\}}$

Long Division

Date:

Find each quotient using long division.

1. $\dfrac{x^2-3x-20}{x+7}$

$$
\begin{array}{r|l}
 & x \quad -10 \\
x+7 & x^2 - 3x - 20 \\
 & -(x^2 + 7x) \\
\hline
 & \quad -10x - 20 \\
 & \quad -(-10x - 70) \\
\hline
 & \qquad 50
\end{array}
$$

$$\boxed{x - 10 + \frac{50}{x+7}}$$

2. $\dfrac{w^4-8w^3-2w+20}{w-8}$

$$
\begin{array}{r|l}
 & w^3 + 0w^2 + 0w - 2 \\
w-8 & w^4 - 8w^3 + 0w^2 - 2w + 20 \\
 & -(w^4 - 8w^3) \\
\hline
 & \quad 0w^3 + 0w^2 \\
 & \quad -(0w^3 + 0w^2) \\
\hline
 & \qquad 0w^2 - 2w \\
 & \qquad -(0w^2 + 0w) \\
\hline
 & \qquad\quad -2w + 20 \\
 & \qquad\quad -(-2w + 16) \\
\hline
 & \qquad\qquad 4
\end{array}
$$

$$\boxed{w^3 - 2 + \frac{4}{w-8}}$$

Synthetic Division

Date:

Find each quotient using synthetic division.

1. $\dfrac{8n^2-10n+7}{n-1}$

$$
\begin{array}{r|rrr}
1 & 8 & -10 & 7 \\
 & \downarrow & 8 & -2 \\
\hline
 & 8 & -2 & 5
\end{array}
$$

$$\boxed{8x - 2 + \frac{5}{n-1}}$$

2. $\dfrac{x^4-4x^2-15x-9}{x+3}$

$$
\begin{array}{r|rrrrr}
-3 & 1 & 0 & -4 & -15 & -9 \\
 & \downarrow & -3 & 9 & -15 & 90 \\
\hline
 & 1 & -3 & 5 & -30 & 81
\end{array}
$$

$$\boxed{x^3 - 3x^2 + 5x - 30 + \frac{81}{x+3}}$$

Remainder Theorem

Date:

Evaluate each function using the Remainder Theorem.

1. $f(x)=3x^3+7x^2+6x+13;\ f(-2)$

$$\begin{array}{r|rrrr} -2 & 3 & 7 & 6 & 13 \\ & \downarrow & -6 & -2 & -8 \\ \hline & 3 & 1 & 4 & 5 \end{array}$$

$\boxed{f(-2)=5}$

2. $f(x)=x^4-13x^2+10x-1;\ f(3)$

$$\begin{array}{r|rrrrr} 3 & 1 & 0 & -13 & 10 & -1 \\ & \downarrow & 3 & 9 & -12 & -6 \\ \hline & 1 & 3 & -4 & -2 & -7 \end{array}$$

$\boxed{f(3)=-7}$

Function Operations

Date:

Given $f(x)=x^2-x$, $g(x)=x-4$, and $h(x)=x^2+6x-16$, find each function. Indicate any restrictions in the domain.

1. $(f+g)(x)$

$(x^2-x)+(x-4)$

$\boxed{=x^2-4}$ $D=\mathbb{R}$

2. $(g\cdot h)(x)$

$(x-4)(x^2+6x-16)$

$=x^3+6x^2-16x-4x^2-24x+64$

$\boxed{=x^3+2x^2-40x+64}$ $D=\mathbb{R}$

3. $\left(\dfrac{g}{f}\right)(x)$

$\dfrac{x-4}{x^2-x}=\boxed{\dfrac{x-4}{x(x-1)}}$

$D=x\neq 0,1$

4. $(h\circ g)(x)$

$(x-4)^2+6(x-4)-16$

$=x^2-8x+16+6x-24-16$

$\boxed{=x^2-2x-24}$ $D=\mathbb{R}$

Using the same functions, find each function value.

5. $(h-f)(2)$

$h(2)=2^2+6(2)-16=0$

$f(2)=2^2-2=2$

$0-2=\boxed{-2}$

6. $(g\circ f)(-7)$

$f(-7)=(-7)^2-(-7)$

$=49+7=56$

$g(56)=56-4=\boxed{52}$

Regression

Date:

The amount of money in Carmen's savings account on the last day of certain years is given below:

	0	4	6	9	12
Year	2003	2007	2009	2012	2015
Balance	\$4,002	\$3,209	\$2,975	\$3,371	\$4,129

1. Write a **quartic function** to represent the balance of Carmen's savings account.

   $f(x) = -0.64x^4 + 16.28x^3 - 100.27x^2 - 16.41x + 4002$

2. Predict the balance of Carmen's account at the end of 2020. $\rightarrow$ $x = 17$

   $= -0.64(17)^4 + 16.28(17)^3 - 100.27(17)^2 - 16.41(17) + 4002$

   $\boxed{= \$1275.20}$

Simplifying Radicals

Date:

1. $3\sqrt{507}$
2. $-2\sqrt[4]{176}$
3. $\sqrt[3]{\dfrac{54}{128}}$
4. $-3\sqrt{126a^9b^3}$
5. $\sqrt[3]{-448x^{17}y^4}$
6. $\sqrt[4]{625m^{21}n^5}$

Rational Exponents

Date:

Write in radical form. Simplify if possible.

1. $(9x)^{\frac{1}{2}} = \sqrt{9x}$

$\boxed{= 3\sqrt{x}}$

2. $y^{\frac{7}{4}} = \sqrt[4]{y^7}$

$\boxed{= y\sqrt[4]{y^3}}$

3. $m^{\frac{10}{3}}n^{\frac{4}{3}} = \sqrt[3]{m^{10}n^4}$

$\boxed{= m^3 n\sqrt[3]{mn}}$

Write in exponential form:

4. $\sqrt[4]{3k}$

$= (3k)^{1/4}$

$\boxed{= 3^{1/4}k^{1/4}}$

5. $\sqrt[3]{p^{11}}$

$\boxed{= p^{11/3}}$

6. $\sqrt[4]{a^5b^8}$

$a^{5/4}b^{8/4}$

$\boxed{= a^{5/4}b^2}$

Simplify:

7. $y^{\frac{7}{8}} \cdot y^{\frac{3}{8}}$

$= y^{\frac{10}{8}}$

$= y^{5/4}$

$\boxed{= y\sqrt[4]{y}}$

8. $\dfrac{(2m)^{\frac{11}{6}}}{(2m)^{\frac{1}{2}}} = (2m)^{8/6}$

$= (2m)^{4/3}$

$\boxed{= 2m\sqrt[3]{2m}}$

9. $\left(4^{-\frac{3}{4}}\right)^2 = 4^{-\frac{6}{4}}$

$= 4^{-\frac{3}{2}}$

$= \dfrac{1}{4^{\frac{3}{2}}}$

$\boxed{= \dfrac{1}{8}}$

Add, Subtract, & Multiply Radicals

Date:

1. $2\sqrt[4]{768} - 3\sqrt{20} + 7\sqrt[4]{3}$

$= 2\cdot\sqrt[4]{256}\cdot\sqrt[4]{3} - 3\sqrt{4}\sqrt{5} + 7\sqrt[4]{3}$

$= 8\sqrt[4]{3} - 6\sqrt{5} + 7\sqrt[4]{3}$

$\boxed{= 15\sqrt[4]{3} - 6\sqrt{5}}$

2. $2b\sqrt{147a^5} - 3a^2\sqrt{12ab^2}$

$= 2b\cdot\sqrt{49a^4}\sqrt{3a} - 3a^2\sqrt{4b^2}\sqrt{3a}$

$= 14a^2b\sqrt{3a} - 6a^2b\sqrt{3a}$

$\boxed{= 8a^2b\sqrt{3a}}$

3. $3\sqrt[3]{4p^5} \cdot -2\sqrt[3]{14p^6}$

$= -6\sqrt[3]{56p^{11}}$

$= -6\sqrt[3]{8p^9}\ \sqrt[3]{7p^2}$

$\boxed{= -12p^3\sqrt[3]{7p^2}}$

4. $\sqrt{8}(\sqrt{2} - 5\sqrt{3})$

$= \sqrt{16} - 5\sqrt{24}$

$= 4 - 5\sqrt{4}\sqrt{6}$

$\boxed{= 4 - 10\sqrt{6}}$

5. $(6 - 4\sqrt{5})(1 + \sqrt{5})$

$= 6 + 6\sqrt{5} - 4\sqrt{5} - 20$

$\boxed{= -14 + 2\sqrt{5}}$

6. $(2\sqrt{3} - 8)^2$

$= (2\sqrt{3} - 8)(2\sqrt{3} - 8)$

$= 12 - 16\sqrt{3} - 16\sqrt{3} + 64$

$= 12 - 32\sqrt{3} + 64$

$\boxed{= 76 - 32\sqrt{3}}$

Dividing Radicals

Date:

1. $\dfrac{24\sqrt[4]{240}}{3\sqrt[4]{5}} = 8\sqrt[4]{48}$

$= 8\sqrt[4]{16}\,\sqrt[4]{3}$

$\boxed{= 16\sqrt[4]{3}}$

2. $\dfrac{\sqrt[3]{486m^{12}}}{\sqrt[3]{3m}} = \sqrt[3]{162m^{11}}$

$= \sqrt[3]{27m^9}\,\sqrt[3]{6m^2}$

$\boxed{= 3m^3 \cdot \sqrt[3]{6m^2}}$

3. $\sqrt{\dfrac{28x}{3}} = \dfrac{2\sqrt{7x}}{\sqrt{3}} \cdot \dfrac{\sqrt{3}}{\sqrt{3}}$

$= \boxed{\dfrac{2\sqrt{21x}}{3}}$

4. $\dfrac{(3-\sqrt{3})\cdot\sqrt{3}}{\sqrt{12}\cdot\sqrt{3}} = \dfrac{3\sqrt{3}-\sqrt{9}}{\sqrt{36}}$

$= \dfrac{3\sqrt{3}-3}{6}$

$\boxed{= \dfrac{-1+\sqrt{3}}{2}}$

5. $\dfrac{4\,(2+2\sqrt{2})}{(2-2\sqrt{2})(2+2\sqrt{2})}$

$= \dfrac{8+8\sqrt{2}}{4+4\sqrt{2}-4\sqrt{2}-4\sqrt{4}}$

$= \dfrac{8+8\sqrt{2}}{-4} = \boxed{-2-2\sqrt{2}}$

6. $\dfrac{(2-3\sqrt{5})(5+4\sqrt{5})}{(5-4\sqrt{5})(5+4\sqrt{5})}$

$= \dfrac{10+8\sqrt{5}-15\sqrt{5}-12\sqrt{25}}{25+20\sqrt{5}-20\sqrt{5}-16\sqrt{25}}$

$= \dfrac{-50-7\sqrt{5}}{-55} = \boxed{\dfrac{50+7\sqrt{5}}{55}}$

Solving Radical Equations

Date:

Solve each equation. Check for extraneous solutions.

1. $\sqrt[3]{4m+28} - 1 = 3$

$\sqrt[3]{4m+28} = 4$

$4m+28 = 64$

$4m = 36$

$\boxed{m=9}$ ✓

2. $(2-x)^{\frac{1}{2}} = (-4-2x)^{\frac{1}{2}}$

$2-x = -4-2x$

$\boxed{x=-6}$ ✓

3. $\sqrt{54-3k} = k$

$54-3k = k^2$

$0 = k^2+3k-54$

$0 = (k+9)(k-6)$

$\cancel{k=-9}$ | $\boxed{k=6}$ ✓

4. $-1 = \sqrt{5a-9} - a$

$a-1 = \sqrt{5a-9}$

$a^2-2a+1 = 5a-9$

$a^2-7a+10 = 0$

$(a-5)(a-2) = 0$

✓ $\boxed{a=5}$ | $\boxed{a=2}$ ✓

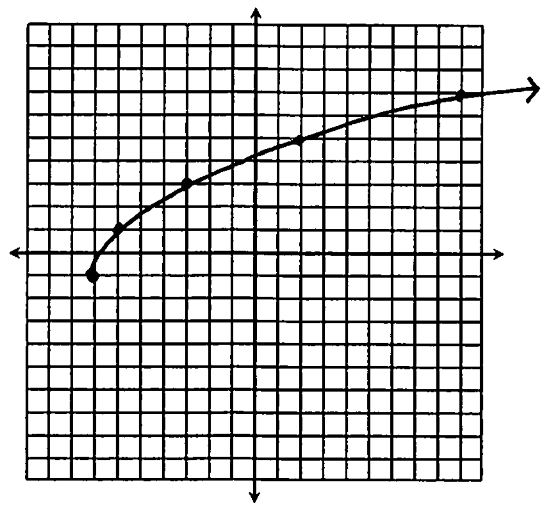
Graphing Square Root Functions

Date:

1. The square root parent function is vertically compressed by a factor of 1/3, reflected in the x-axis, then translated 4 units up. Write an equation to represent this function. Give the coordinates of the endpoint.

$f(x) = -\frac{1}{3}\sqrt{x} + 4$

2. Graph $f(x) = 2\sqrt{x+7} - 1$ and identify all key characteristics.



Domain: $\{x \mid x \geq -7\}$

Range: $\{y \mid y \geq -1\}$

End Behavior:
As $x \to \infty$, $f(x) \to$ ∞
As $x \to -\infty$, $f(x) \to$ -1

Endpoint: $(-7, -1)$

Inc. Interval: $[-7, \infty)$

Dec. Interval: —

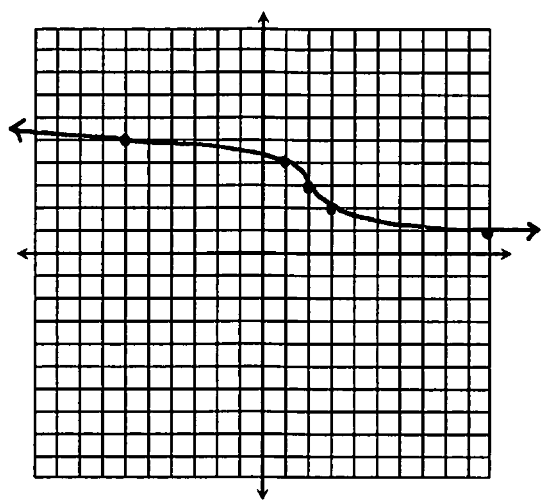
Graphing Cube Root Functions

Date:

1. The cube root parent function is reflected across the x-axis, then translated 9 units right and 2 units down. Write an equation to represent this function. Give the coordinates of the point of inflection.

$f(x) = -\sqrt[3]{x-9} - 2$; $(9, -2)$

2. Graph $f(x) = -\sqrt[3]{x-2} + 3$ and identify all key characteristics.



Domain: $\mathbb{R}$

Range: $\mathbb{R}$

End Behavior:
As $x \to \infty$, $f(x) \to$ $-\infty$
As $x \to -\infty$, $f(x) \to$ ∞

Point of Inflection: $(2, 3)$

Inc. Interval: —

Dec. Interval: $(-\infty, \infty)$

Inverse Relations & Functions

Date:

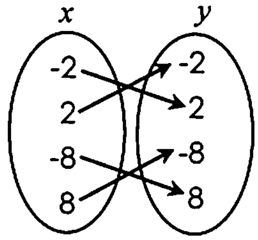
Which relations represent one-to-one functions?

1.

x	y
-9	4
-5	-7
-1	1
0	4

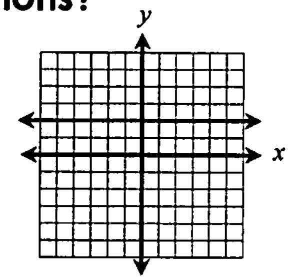
Function, not one-to-one

2.



one-to-one

3.



function, not one-to-one

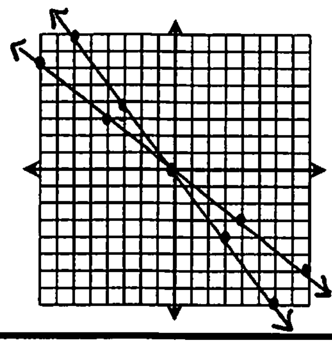
Write the inverse of each function. Graph to verify.

4. $f(x) = -\frac{4}{3}x$

$x = -\frac{4}{3}y$

$-\frac{3}{4}x = y$

$\boxed{f^{-1}(x) = -\frac{3}{4}x}$



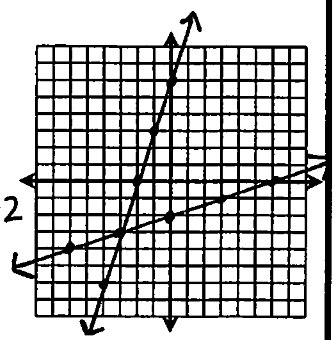
5. $f(x) = 3x + 6$

$x = 3y + 6$

$x - 6 = 3y$

$\frac{1}{3}x - 2 = y$

$f^{-1}(x) = \frac{1}{3}x - 2$



Verifying Inverses

Date:

Determine whether the pair of equations are inverse functions.

1. $f(x) = \frac{-2x-8}{3}$ and $g(x) = -4 - \frac{3}{2}x$

$f(g(x)) = \frac{-2\left(-4 - \frac{3}{2}x\right) - 8}{3}$

$= \frac{8 + 3x - 8}{3} = x$ ✓

$g(f(x)) = -4 - \frac{3}{2}\left(\frac{-2x-8}{3}\right)$

$= -4 + \frac{6x + 24}{6}$

$= x$ ✓

$\boxed{\text{inverses}}$

2. $f(x) = -x - 1$ and $g(x) = 5x + 5$

$f(g(x)) = -(5x + 5) - 1$

$= -5x - 5 - 1$

$= -5x - 6$

$\boxed{\text{Not inverses}}$

3. $f(x) = \sqrt{x} + 3$ and $g(x) = x^2 - 6x + 9$ (if $x \geq 3$)

$\hookrightarrow (x-3)^2$

$f(g(x)) = \sqrt{x^2 - 6x + 9} + 3$

$= \sqrt{(x-3)^2} + 3$

$= x - 3 + 3$

$= x$ ✓

$g(f(x)) = \left[(\sqrt{x} + 3) - 3\right]^2$

$= \sqrt{x}^2$

$= x$ ✓

$\boxed{\text{inverses}}$

Graphing Exponential Functions

Date:

1. Determine whether the function below represents an exponential growth or decay:

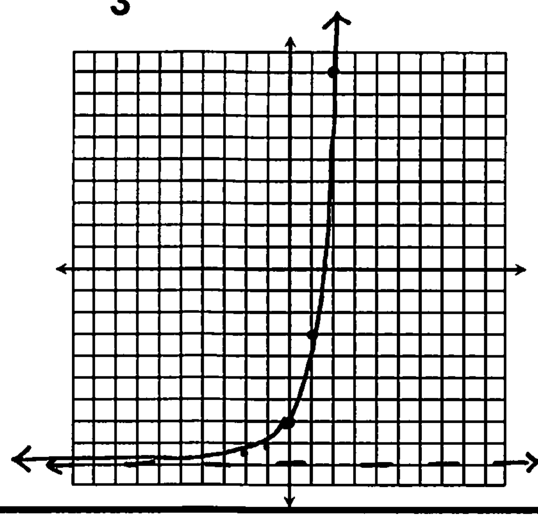
a. $f(x) = 2 \cdot \left(\frac{5}{6}\right)^x$

decay ↳ < 1

b. $f(x) = 0.8 \cdot \left(\frac{4}{3}\right)^x$

growth ↳ > 1

2. Graph $f(x) = \frac{2}{3} \cdot 3^{x+1} - 9$ and identify all key characteristics.



Domain: $\mathbb{R}$

Range: $y > -9$

End Behavior:

As $x \to \infty$, $f(x) \to$ ∞

As $x \to -\infty$, $f(x) \to$ -9

***y*-intercept:** $(0, -7)$

Asymptote: $y = -9$

Solving Exponential Functions (with a Common Base)

Date:

Solve each equation.

1. $6^{7+3x} = 6^{x-11}$

$7 + 3x = x - 11$

$2x = -18$

$\boxed{x = -9}$

2. $2^1 \cdot 2^{a+7} = 2^{5a-4}$

$2^{a+8} = 2^{5a-4}$

$a + 8 = 5a - 4$

$12 = 4a$

$\boxed{3 = a}$

3. $\left(\frac{1}{9}\right) = 3^{5n-12}$

$3^{-2} = 3^{5n-12}$

$-2 = 5n - 12$

$10 = 5n$

$\boxed{2 = n}$

4. $4^{3p} \cdot \frac{1}{64} = 16^{3p-9}$

$4^{3p} \cdot 4^{-3} = 4^{2(3p-9)}$

$3p - 3 = 6p - 18$

$15 = 3p$

$\boxed{5 = p}$

Logarithms

Date:

Write in exponential form:

1. $\log_3 729 = 6$

$3^6 = 729$

2. $\log_2 x = 7$

$2^7 = x$

3. $\log 65 = x + 2$

$10^{x+2} = 65$

Write in logarithmic form:

4. $2^9 = 512$

$\log_2 512 = 9$

5. $8^{x+3} = 36$

$\log_8 36 = x+3$

6. $\sqrt{49} = 7$ $\quad 49^{1/2} = 7$

$\log_{49} 7 = \frac{1}{2}$

Evaluate. Round to the nearest ten-thousands if necessary.

7. $\log_6 216$

$= 3$

8. $\log_5 1$

$= 0$

9. $\log_{16} 64$

$= 1.5$

10. $\log 100$

$= 2$

11. $\log_9 5$

$= 0.7325$

12. $\log_4 97$

$= 3.3$

Properties of Logarithms

Date:

Condense each expression into a single logarithm:

1. $\frac{1}{2} \cdot \log_3 16 + \log_3 5$

$= \log_3 16^{1/2} + \log_3 5$

$\boxed{= \log_3 20}$

2. $7 \cdot \log_5 a - 2 \cdot \log_5 b^4$

$= \log_5 a^7 - \log_5 b^{4 \cdot 2}$

$\boxed{= \log_5 \frac{a^7}{b^8}}$

3. $2 \cdot \log 3 + \log(x-8)$

$= \log 3^2 + \log(x-8)$

$= \log 9(x-8)$

$\boxed{= \log 9x-72}$

4. $\frac{1}{2}(\log_8 48 - \log_4 3) + 3 \cdot \log_8 3$

$= \log_8 16^{1/2} + \log_8 3^3$

$= \log_8 4 \cdot 27$

$\boxed{= \log_8 108}$

Expand:

5. $\log_4 \left(\frac{p^5}{q^2}\right)^3 = \log_4 \frac{p^{15}}{q^6}$

$\boxed{= 15 \cdot \log_4 p - 6 \cdot \log_4 q}$

6. $\log \sqrt[4]{m^2 n^5} = \log m^{1/2} n^{5/4}$

$\boxed{= \frac{1}{2} \log m + \frac{5}{4} \log n}$

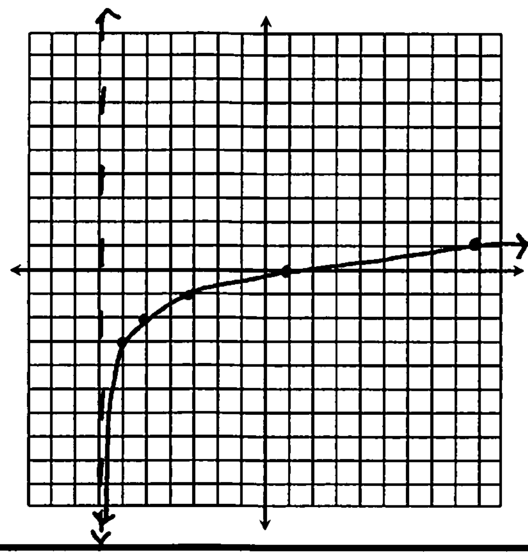
Graphing Logarithmic Functions

Date:

1. What is the relationship between exponential and logarithmic functions?

They are inverses of one another.

2. Graph $f(x) = \log_2(x+7) - 3$ and identify all key characteristics.



Domain: $x > -7$

Range: $\mathbb{R}$

End Behavior:

As $x \to \infty$, $f(x) \to \infty$

As $x \to -7$, $f(x) \to -\infty$

x-intercept: $(1, 0)$

Asymptote: $x = -7$

Solving Logarithmic Equations

Date:

Solve each equation. Check all solutions.

1. $\log_7(4x-9) = \log_7(2x+19)$

$4x - 9 = 2x + 19$

$2x = 28$

$\boxed{x = 14}$

2. $2 \cdot \log_5 k = \log_5(k+42)$

$k^2 = k + 42$

$k^2 - k - 42 = 0$

$(k-7)(k+6) = 0$

$\boxed{k = 7}$ | $\cancel{k = -6}$

3. $\log 84 - \log 3 = \log 2 + \log(m-5)$

$\log \frac{84}{3} = \log 2(m-5)$

$28 = 2m - 10$

$38 = 2m$

$\boxed{19 = m}$

4. $\log_3(8y-23) = 4$

$3^4 = 8y - 23$

$81 = 8y - 23$

$104 = 8y$

$\boxed{13 = y}$

Solving Exponential Functions
(with Logarithms)

Date:

Solve each equation.

1. $3^{x-2} = 40$

$\log_3 40 = x-2$

$3.3578 = x-2$

$\boxed{5.3578 = x}$

2. $-4 \cdot 5^{2x-1} = -128$

$5^{2x-1} = 32$

$\log_5 32 = 2x-1$

$2.1534 = 2x-1$

$3.1534 = 2x$

$\boxed{1.5767 = x}$

3. $\frac{1}{2} \cdot 9^{w} - 3 = 14$

$\frac{1}{2} \cdot 9^{w} = 17$

$9^{w} = 34$

$\log_9 34 = w$

$\boxed{1.6049 = w}$

4. $2 \cdot 3^{p+5} + 27 = 163$

$2 \cdot 3^{p+5} = 136$

$3^{p+5} = 68$

$\log_3 68 = p+5$

$3.8408 = p+5$

$\boxed{-1.1592 = p}$

Exp/Logs Equations Review

Date:

Solve each equation.

1. $2 \cdot \log_4(m+6) = \log_4 1$

$\log_4 (m+6)^2 = \log_4 1$

$(m+6)^2 = 1$

$m^2 + 12m + 36 = 1$

$m^2 + 12m + 35 = 0$

$(m+5)(m+7) = 0$

$\boxed{m = -5}$ | $\cancel{m = -7}$

2. $4 = \log_2(3y - 26)$

$2^4 = 3y - 26$

$16 = 3y - 26$

$42 = 3y$

$\boxed{y = 14}$

3. $8^{2w-1} = 32^{2w+5}$

$(2^3)^{2w-1} = (2^5)^{2w+5}$

$6w - 3 = 10w + 25$

$-28 = 4w$

$\boxed{-7 = w}$

4. $4 \cdot 5^{p-2} - 10 = 82$

$4 \cdot 5^{p-2} = 92$

$5^{p-2} = 23$

$\log_5 23 = p-2$

$1.9482 = p-2$

$\boxed{3.9482 = p}$

Base e & Natural Logs

Date:

Write in logarithmic form:

1. $e^7 = x$

$\boxed{\ln x = 7}$

2. $e^{2x+1} = 9$

$\boxed{\ln 9 = 2x+1}$

Write in exponential form:

3. $\ln 10 = 2$

$\boxed{e^2 = 10}$

4. $\ln 71 = x$

$\boxed{e^x = 71}$

Solve.

5. $3 \cdot \ln 2 + \ln(m-1) = 5$

$\ln 2^3 \cdot (m-1) = 5$

$8(m-1) = e^5$

$8m - 8 = 148.4132$

$8m = 156.4132$

$\boxed{m = 19.5516}$

6. $2 \cdot e^{m+7} - 1 = 103$

$2 \cdot e^{m+7} = 104$

$e^{m+7} = 52$

$m+7 = \ln 52$

$m+7 = 3.9512$

$\boxed{m = -3.0488}$

Exponential Growth & Decay

Date:

1. The current population of a town is 8,200. If the population increases by 15% each year, find the population of the town in 12 years.

$f(x) = 8200(1.15)^x$

$f(12) = 8200(1.15)^{12}$

$\boxed{= 43{,}872 \text{ people}}$

2. Emmanuel bought a new boat in 2011 for $36,000. Each year, it depreciates at a rate of 6.5%. Find the value of the boat in 2020.

$f(x) = 36000(0.935)^x$

$f(9) = 36000(0.935)^9$

$\boxed{= \$19{,}661.06}$

Compound Interest

Date:

1. Mr. Jameson invested $3,200 in a savings account that earns 4% interest compounded semiannually. Find the total amount of money he will have in 5 years.

$A = 3200\left(1 + \frac{0.04}{2}\right)^{2 \cdot 5}$ $(n=2)$

$= 3200(1.02)^{10}$

$\boxed{= \$3,900.78}$

2. Kate moved $20,000 from her savings to a new account that earns 7.5% interest compounded monthly. How much will the account earn in interest after 10 years? $(n=12)$

$A = 20000\left(1 + \frac{0.075}{12}\right)^{12 \cdot 10}$

$= 20000(1.00625)^{120}$

$= 42241.29$

$\boxed{\text{Interest} = \$22,241.29}$

Exponential & Logarithmic Regression

Date:

1. The table below shows the revenue of a company in thousands of dollars since 2005. Use an **exponential function** to predict the revenue of the company in 2025.

	Year	Revenue
0	2005	261
1	2006	318
2	2007	404
3	2008	492
4	2009	575

$f(x) = 263.21 \cdot (1.22)^x$

$f(20) = 263.21 \cdot (1.22)^{20}$

$\boxed{= \$14,044.26}$

2. The table below shows the average freshman class GPA of a university during certain years. Use a **logarithmic function** to predict the year the GPA will reach 4.0.

	Year	Revenue
1	1996	3.12
5	2000	3.49
9	2004	3.68
13	2008	3.71
17	2012	3.82

$f(x) = 3.12 + 0.24 \ln x$

$4.0 = 3.12 + 0.24 \ln x$

$0.88 = 0.24 \ln x$

$3.67 = \ln x$

$e^{3.67} = x$

$x = 39 \rightarrow \boxed{2035}$

Choosing the Best Model

Date:

1. The table below shows the number of trees (in thousands) planted at a tree farm. Which function is the better model: quadratic, cubic, or exponential? Write an equation and approximate the number of trees planted in 2015.

	Year	Trees
0	1992	0.8
2	1994	10.2
4	1996	45.1
6	1998	115.2
8	2000	241.9

Cubic

$f(x) = 0.32x^3 + 0.93x^2 + 1.96x + 0.63$

$f(13)$ $\boxed{886.32 \text{ trees}}$

2. The table below shows the number of days it takes to build a house based on the number of workers. Which function is the better model: linear, quadratic, or logarithmic? Write an equation and approximate the number of days it will take to build a house if there are 40 workers.

Workers	Days
6	178
10	121
12	115
18	100
24	86

logarithmic

$f(x) = 278.70 - 62.74 \ln x$

$f(40) = 278.70 - 62.74 \ln(40)$
$= 47.26 \rightarrow \boxed{48 \text{ days}}$

Simplifying Rational Expressions

Date:

Simplify.

1. $\dfrac{16mn^3}{2m \cdot 5n^7}$

$= \dfrac{16mn^3}{10mn^7} = \boxed{\dfrac{8}{5n^4}}$

2. $\dfrac{18x^4 + 27x^3}{24x + 36}$

$= \dfrac{9x^3(2x+3)}{12(2x+3)}$

$= \dfrac{9x^3}{12} = \boxed{\dfrac{3x^3}{4}}$

3. $\dfrac{35k - 14}{4 - 25k^2}$

$= \dfrac{7(5k-2)}{(2-5k)(2+5k)}$

$= \boxed{\dfrac{-7}{2+5k}}$

4. $\dfrac{p^2 - p - 72}{3p^2 - 28p + 9}$

$= \dfrac{(p+8)(p-9)}{(3p-1)(p-9)}$

$= \boxed{\dfrac{p+8}{3p-1}}$

Mult/Div Rational Expressions

Date:

Find each product or quotient.

1. $\dfrac{4ab}{3a^2b} \cdot \dfrac{9b^3}{10a}$

$= \dfrac{36ab^4}{30a^3b} = \boxed{\dfrac{6b^3}{5a^2}}$

2. $\dfrac{9-3w}{4w} \cdot \dfrac{20w^2}{w^2-11w+24}$

$= \dfrac{3(3-w)}{4w} \cdot \dfrac{20w^2}{(w-3)(w-8)}$

$= \dfrac{-60w^2}{4w(w-8)} = \boxed{\dfrac{-15w}{w-8}}$

3. $\dfrac{15pq}{2q^2} \div (3p^5)$

$= \dfrac{15pq}{2q^2} \cdot \dfrac{1}{3p^5}$

$= \dfrac{15pq}{6p^5q^2} = \boxed{\dfrac{5}{2p^4q}}$

4. $\dfrac{2n^2-11n+14}{6n^2-24} \div \dfrac{3n-9}{2n^2-2n-12}$

$\dfrac{(2n-7)(n-2)}{6(n+2)(n-2)} \cdot \dfrac{2(n-3)(n+2)}{3(n-3)}$

$= \dfrac{2(2n-7)}{18} = \boxed{\dfrac{2n-7}{9}}$

Add/Sub Rational Expressions

Date:

Find each sum or difference.

1. $\dfrac{13x+1}{2x^3-6x^2} - \dfrac{x+37}{2x^3-6x^2}$

$\dfrac{12x-36}{2x^3-6x^2} = \dfrac{12(x-3)}{2x^2(x-3)}$

$= \boxed{\dfrac{6}{x^2}}$

2. $\dfrac{3 \cdot 3}{3 \cdot 2a^2} + \dfrac{5 \cdot 2a}{3a \cdot 2a}$

$\boxed{\dfrac{9+10a}{6a^2}}$

3. $\dfrac{(y-2)\,7}{(y-2)\,y-5} + \dfrac{4\,(y-5)}{y-2\,(y-5)}$

$\dfrac{7y-14+4y-20}{(y-2)(y-5)}$

$\boxed{= \dfrac{11y-34}{(y-2)(y-5)}}$

4. $\dfrac{r^2-6r+19}{2r^2-15r+7} - \dfrac{2\,(2r-1)}{r-7\,(2r-1)}$

$= \dfrac{r^2-6r+19-4r+2}{(r-7)(2r-1)}$

$= \dfrac{r^2-10r+21}{(r-7)(2r-1)} = \dfrac{(r-7)(r-3)}{(r-7)(2r-1)} = \boxed{\dfrac{r-3}{2r-1}}$

Complex Fractions

Date:

Simplify.

1. $\dfrac{\dfrac{18w}{4w^3}}{\dfrac{15w^5}{28w^2}}$

$= \dfrac{18w}{4w^3} \cdot \dfrac{28w^2}{15w^5} = \dfrac{504w^3}{60w^8}$

$\boxed{= \dfrac{42}{5w^5}}$

2. $\dfrac{\dfrac{4}{3k} + 2 \cdot \dfrac{3k}{3k}}{\dfrac{3k+2}{9}}$

$= \dfrac{4+6k}{3k} \cdot \dfrac{9}{3k+2}$

$= \dfrac{2(2+3k)}{3k} \cdot \dfrac{9}{3k+2} = \dfrac{18}{3k} = \boxed{\dfrac{6}{k}}$

3. $\dfrac{\dfrac{2p}{q} - 2 \cdot \dfrac{q}{q}}{\dfrac{2p}{q} + 2 \cdot \dfrac{q}{q}}$

$\dfrac{2p-2q}{q} \cdot \dfrac{q}{2p+2q} = \dfrac{2(p-q)}{2(p+q)}$

$= \boxed{\dfrac{p-q}{p+q}}$

4. $\dfrac{\dfrac{x}{x} \cdot \dfrac{x}{4} - \dfrac{25}{x} \cdot \dfrac{4}{4}}{\dfrac{x}{2} + 5 \cdot \dfrac{2}{2}}$

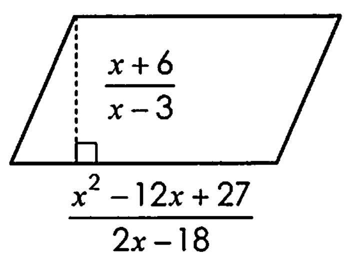
$\dfrac{x^2-100}{4x} \cdot \dfrac{2}{x+10} = \dfrac{(x+10)(x-10)}{4x} \cdot \dfrac{2}{x+10}$

$= \boxed{\dfrac{x-10}{2x}}$

Applications

Date:

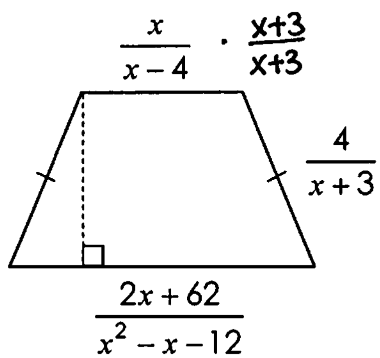
1. Find the **area** of the parallelogram below.



$\dfrac{x+6}{x-3} \cdot \dfrac{(x-9)(x-3)}{2(x-9)}$

$= \boxed{\dfrac{x+6}{2}}$

2. Find the **perimeter** of the trapezoid below.



$= \dfrac{x^2+3x+4x-16+4x-16+2x+62}{(x-4)(x+3)}$

$= \dfrac{x^2+13x+30}{(x-4)(x+3)}$

$= \dfrac{(x+3)(x+10)}{(x-4)(x+3)} = \boxed{\dfrac{x+10}{x-4}}$

Reciprocal Functions

Date:

1. The reciprocal parent function is reflected across the x-axis, then translated four units left. Write an equation that could represent this function, then identify the vertical and horizontal asymptotes.

$f(x) = \frac{-1}{x+4}$

HA: $y=0$

VA: $x=-4$

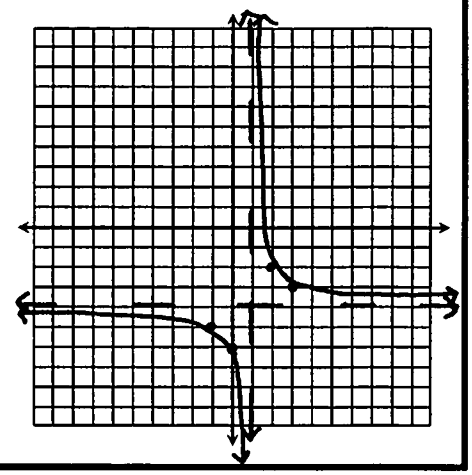
2. Graph $f(x) = \frac{2}{x-1} - 4$ and identify all key characteristics.

Domain: $\{x \mid x \neq 1\}$

Range: $\{y \mid y \neq -4\}$

VA: $x=1$

HA: $y=-4$



Rational Functions

Date:

1. Identify the key characteristics of the function below.

$$f(x) = \frac{x^3}{x^2+2x}$$

$$\frac{x^3}{x(x+2)} = \frac{x^2}{x+2}$$

x-int: none

Holes: $(0,0)$

VA: $x=-2$

HA: none

D: $\{x \mid x \neq -2\}$

R: $\{y \mid y \leq -8 \cup y > 0\}$

2. Graph $f(x) = \frac{x^2-36}{x^2+7x+6}$ and all key characteristics.

$$\frac{(x+6)(x-6)}{(x+1)(x+6)} = \frac{x-6}{x+1}$$

x-int: $(6,0)$

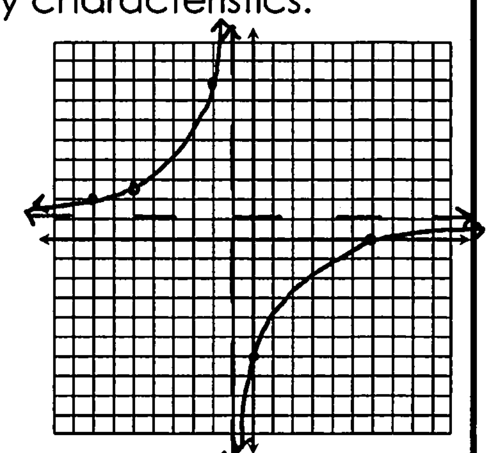
Holes: $(-6, \frac{12}{5})$

VA: $x=-1$

HA: $y=1$

D: $\{x \mid x \neq -6, -1\}$

R: $\{y \mid y \neq 1, \frac{12}{5}\}$



Solving Rational Functions

Date:

Solve. Be sure to check all solutions.

1. $\frac{x-2}{x+2}=\frac{6}{x+9}$

$6x+12 = x^2+7x-18$

$0 = x^2+x-30$

$0 = (x+6)(x-5)$

$x+6=0$ | $x-5=0$

$x=-6$ | $x=5$

$\boxed{x=\{-6,5\}}$

2. $\left[\frac{a+1}{2a}-\frac{a-1}{4a}=\frac{1}{3}\right]\cdot 12a$

$6(a+1) - 3(a-1) = 4a$

$6a+6 - 3a+3 = 4a$

$3a+9 = 4a$

$\boxed{9=a}$

2. $\left[\frac{5}{w+2}+\frac{3}{w-2}=\frac{12}{w^2-4}\right](w+2)(w-2)$

$5(w-2) + 3(w+2) = 12$

$5w-10 + 3w+6 = 12$

$8w-4 = 12$

$8w = 16$

$\cancel{w=2}$

$\boxed{\varnothing}$

4. $\frac{3}{r+1}=\frac{2r-1}{r^2-5r+6}+\frac{3\cdot(r-3)}{r-2\cdot(r-3)}$

$\frac{3}{r+1} = \frac{5r-10}{(r-2)(r-3)}$

$\frac{3}{r+1} = \frac{5(r-2)}{(r-2)(r-3)}$

$\frac{3}{r+1} = \frac{5}{r-3}$

$5r+5 = 3r-9$

$2r = -14$

$\boxed{r=-7}$

Variation

Date:

1. Determine whether the equations represent a direct, joint, or inverse variation. Identify the constant.

a) $\frac{6y}{z}=10x$

$6y = 10xz$

$y = \frac{5}{3}xz$

$\boxed{\text{joint; } k=\frac{5}{3}}$

b) $\frac{8y}{x}=4$

$8y = 4x$

$y = \frac{1}{2}x$

$\boxed{\text{direct; } x=\frac{1}{2}}$

c) $\frac{12}{x}=\frac{y}{6}$

$72 = xy$

$y = \frac{72}{x}$

$\boxed{\text{inverse; } k=72}$

2. The time, t, it takes a child to go down a slide varies directly with the length, l, of the slide and inversely with the square root of the height, h, of the slide. If it takes 12 seconds to go down a slide 15 feet long and 9 feet high, how long will it take a child to go down a slide 25 feet long and 16 feet high?

$t = \frac{k\cdot l}{\sqrt{h}}$

$12 = \frac{k\cdot 15}{\sqrt{9}}$

$15k = 36$

$k = \frac{12}{5}$

$t = \frac{\frac{12}{5}\cdot 25}{4}$

$\boxed{t = 15\text{ sec}}$

Circles

Date:

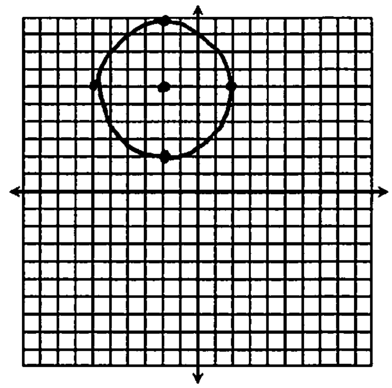
Write in standard form, then graph the circle:

1. $x^2 + y^2 + 4x - 12y = -24$

$x^2 + 4x + y^2 - 12y = -24$

$(x^2 + 4x + \underline{4}) + (y^2 - 12y + \underline{36}) = -24 + \underline{4} + \underline{36}$

$(x+2)^2 + (y-6)^2 = 16$



Center: $(-2, 6)$ Radius: 4

Write the equation of the following circles:

2. center: (-9, 0);
point on circle: (-5, 3)

$r = \sqrt{(-9+5)^2 + (0-3)^2}$

$= \sqrt{16 + 9}$

$= \sqrt{25} = 5$

$\boxed{(x+9)^2 + y^2 = 25}$

3. Endpoints of diameter:
(4, -7) and (-2, -1)

Center $= (1, -4)$

$r = \sqrt{(-2-1)^2 + (-1+4)^2}$

$= \sqrt{9 + 9}$

$= \sqrt{18} = 3\sqrt{2}$

$\boxed{(x-1)^2 + (y+4)^2 = 18}$

Ellipses (Graphing)

Date:

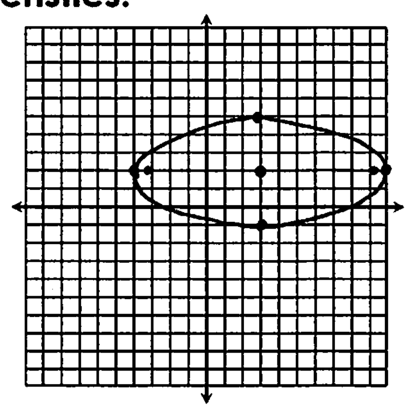
Graph each ellipse. Identify key characteristics.

1. $\dfrac{(x-3)^2}{49} + \dfrac{(y-2)^2}{9} = 1$

$c^2 = 49 - 9$

$c^2 = 40$

$c = 6.32$



Center: $(3, 2)$

Vertices: $(10, 2)$ $(-4, 2)$

Co-Vertices: $(3, 5)$ $(3, -1)$

Foci: $(9.32, 2)$ $(-3.32, 2)$

2. $16x^2 + y^2 + 10y = -9$

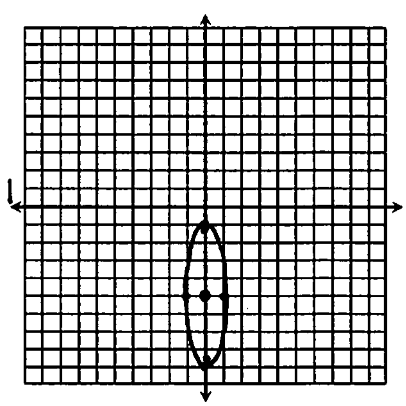
$16x^2 + y^2 + 10y + \underline{25} = -9 + \underline{25}$

$16x^2 + (y+5)^2 = 16 \longrightarrow x^2 + \dfrac{(y+5)^2}{16} = 1$

$c^2 = 16 - 1$

$c^2 = 15$

$c = 3.87$



Center: $(0, -5)$

Vertices: $(0, -9)$ $(0, -1)$

Co-Vertices: $(-1, -5)$ $(1, -5)$

Foci: $(0, -1.13)$ $(0, -8.87)$

Ellipses
(Writing Equations)

Date:

Write the equation of the ellipse in standard form with the given characteristics.

1. Vertices: (-8, 9), (-8, -5)
Co-Vertices: (-5, 2), (-11, 2)

Center = (-8, 2)
a = 7, b = 3 vert.

$$\boxed{\frac{(x+8)^2}{9} + \frac{(y-2)^2}{49} = 1}$$

2. Vertices: (-3, 12), (-3, -6)
Co-Vertices: (2, 3), (-8, 3)

Center = (-3, 3)
a = 9 b = 5 vert.

$$\boxed{\frac{(x+3)^2}{25} + \frac{(y-3)^2}{81} = 1}$$

3. Vertices: (18, -7), (-8, -7)
Foci: (10, -7), (0, -7)

Center = (5, -7) Horiz.
a = 13, c = 5

$25 = 169 - b^2$
$-144 = -b^2$
$b = 12$

$$\boxed{\frac{(x-5)^2}{169} + \frac{(y+7)^2}{144} = 1}$$

4. Co-Vertices: (3, 4), (-1, 4)
Foci: $(1, 4 \pm 2\sqrt{3})$

Center = (1, 4) vert.
b = 2, $c = 2\sqrt{3}$

$12 = a^2 - 4$
$16 = a^2$
$4 = a$

$$\boxed{\frac{(x-1)^2}{4} + \frac{(y-4)^2}{16} = 1}$$

Hyperbolas
(Graphing)

Date:

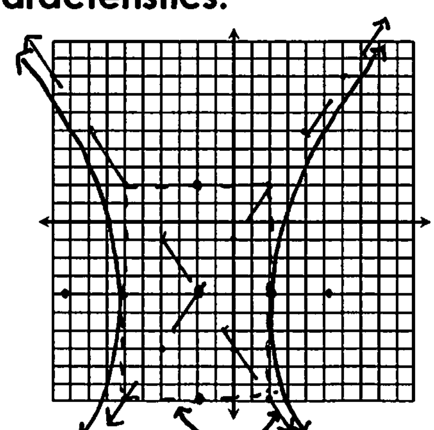
Graph each hyperbola. Identify key characteristics.

1. $\frac{(x+2)^2}{16} - \frac{(y+4)^2}{36} = 1$

$c^2 = 16 + 32$
$c^2 = 52$
$c = 7.21$

$y = \pm \frac{3}{2}(x+2) - 4$

Center: (-2, -4)
Vertices: (2, -4) (-6, -4)
Co-Vertices: (-2, -10) (-2, 2)
Foci: (5.21, -4) (-9.21, -4)
Asymptotes: $y = \frac{3}{2}x - 1$
$y = -\frac{3}{2}x - 7$



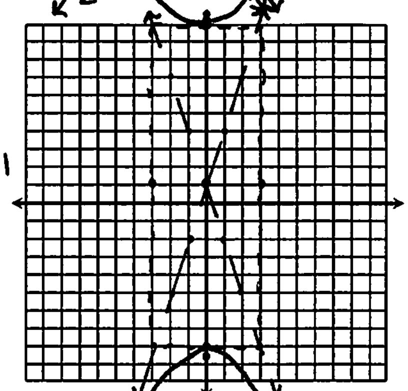
2. $-9x^2 + y^2 - 2y - 80 = 0$

$-9x^2 + y^2 - 2y + 1 = 80 + 1$
$-9x^2 + (y-1)^2 = 81$
$-\frac{x^2}{9} + \frac{(y-1)^2}{81} = 1$
$\frac{(y-1)^2}{81} - \frac{x^2}{9} = 1$

$c^2 = 81 + 9$
$c^2 = 90$
$c = 9.49$

$y = \pm 3(x-0) + 1$

Center: (0, 1)
Vertices: (0, 10) (0, -8)
Co-Vertices: (3, 1) (-3, 1)
Foci: (0, 10.49) (0, -8.49)
Asymptotes: $y = 3x + 1$
$y = -3x + 1$



Hyperbolas
(Writing Equations)

Date:

Write the equation of the hyperbola in standard form with the given characteristics.

1. Vertices: (10, 5), (-4, 5)
Co-Vertices: (3, 15), (3, -5)

Center (3, 5) $a=7,\ b=10$ Horiz.

$$\boxed{\frac{(x-3)^2}{49}-\frac{(y-5)^2}{100}=1}$$

2. Vertices: (8, 4), (8, -12)
Co-Vertices: (13, -4), (3, -4)

Center (8, -4) $a=8$, $b=5$ Vert.

$$\boxed{\frac{(y+4)^2}{64}-\frac{(x-8)^2}{25}=1}$$

3. Vertices: (3, 10), (3, 2)
Foci: (3, 11), (3, 1)

Center (3, 6) $a=4$, $c=5$ vert.

$25=16+b^2$
$9=b^2$ $(b=3)$

$$\boxed{\frac{(y-6)^2}{16}-\frac{(x-3)^2}{9}=1}$$

4. Co-Vertices: (-7, 10), (-7, -8)
Foci: $(-7\pm\sqrt{130}, 1)$

center (-7, 1) $b=9$, $c=\sqrt{130}$ Horiz.

$130=a^2+81$
$49=a^2$ $(a=7)$

$$\boxed{\frac{(x+7)^2}{49}-\frac{(y-1)^2}{81}=1}$$

Parabolas
(Graphing)

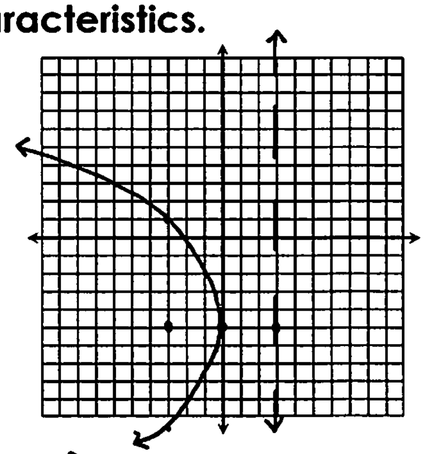
Date:

Graph each parabola. Identify key characteristics.

1. $-\frac{1}{12}(y+5)^2=x$

$(y+5)^2=-12x$

$-12=4p$
$-3=p$



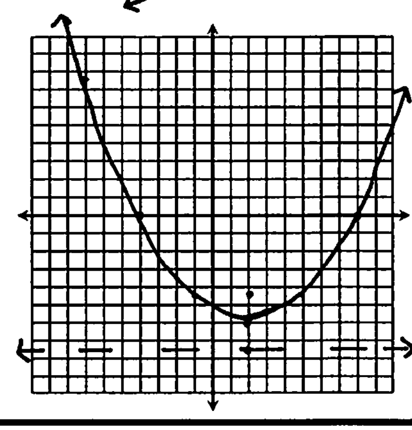
Vertex: $(0,-5)$ Axis of Sym.: $y=-5$

Focus: $(-3,-5)$ Directrix: $x=3$

2. $x^2-4x-6y=32$

$x^2-4x=6y+32$
$x^2-4x+\underline{4}=6y+32+\underline{4}$
$(x-2)^2=6y+36$
$(x-2)^2=6(y+6)$

$6=4p$
$\frac{3}{2}=p$



Vertex: $(2,-6)$ Axis of Sym.: $x=2$

Focus: $(2,-\frac{9}{2})$ Directrix: $y=\frac{-15}{2}$

Parabolas
(Writing Equations)

Date:

Write the equation of the parabola in standard form with the given characteristics.

1. Vertex: (1, -4), Focus: (1, -1)

vert / up

$p = 3$

$\boxed{(x-1)^2 = 12(y+4)}$

2. Vertex: (-3, -8) Directrix: $x = 2$

horiz. / left

$p = -5$

$\boxed{(y+8)^2 = -20(x+3)}$

3. Vertex: (0, 6) Directrix: $y = 6.5$

vert. / down

$p = -\frac{1}{2}$

$\boxed{x^2 = -2(y-6)}$

4. Focus: (5, 2) Directrix: $x = -9$

horiz. / right

$p = 7$

vertex $(-2, 2)$

$\boxed{(y-2)^2 = 28(x+2)}$

Identifying Conics

Date:

Classify the conic, then write the equation in standard form.

1. $-16x^2 + y^2 + 14y + 33 = 0$

Hyperbola

$-16x^2 + y^2 + 14y = -33$

$-16x^2 + y^2 + 14y + 49 = -33 + 49$

$-16x^2 + (y+7)^2 = 16$

$-x^2 + \frac{(y+7)^2}{16} = 1$

$\boxed{\frac{(y+7)^2}{16} - x^2 = 1}$

2. $y^2 + 8x + 6y + 49 = 0$

Parabola

$y^2 + 6y = -8x - 49$

$y^2 + 6y + \underline{9} = -8x - 49 + \underline{9}$

$(y+3)^2 = -8x - 40$

$\boxed{(y+3)^2 = -8(x+5)}$

3. $x^2 + y^2 - 8x - 18y + 72 = 0$

Circle

$x^2 - 8x + y^2 - 18y = -72$

$x^2 - 8x + \underline{16} + y^2 - 18y + \underline{81} = -72 + \underline{16} + \underline{81}$

$\boxed{(x-4)^2 + (y-9)^2 = 25}$

4. $9x^2 + 4y^2 - 18x + 16y - 11 = 0$

Ellipse

$9x^2 - 18x + 4y^2 + 16y = 11$

$9(x^2 - 2x + \underline{1}) + 4(y^2 + 4y + \underline{4}) = 11 + \underline{9} + \underline{16}$

$9(x-1)^2 + 4(y+2)^2 = 36$

$\boxed{\frac{(x-1)^2}{4} + \frac{(y+2)^2}{9} = 1}$

Non-Linear Systems
(Solve by Graphing)

Date:

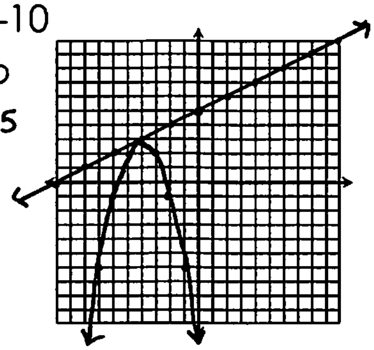
Solve each system by graphing.

1. $\begin{cases} y = -x^2 - 8x - 13 \\ x - 2y = -10 \end{cases}$

$-2y = -x - 10$

$y = \frac{1}{2}x + 5$

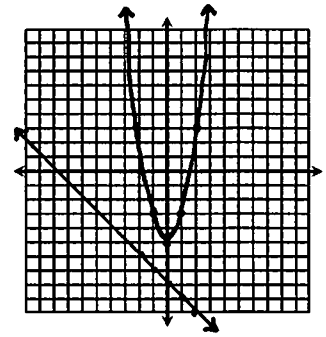
$\boxed{(-4, 3)}$



2. $\begin{cases} y = 2x^2 - 5 \\ x + y = -8 \end{cases}$

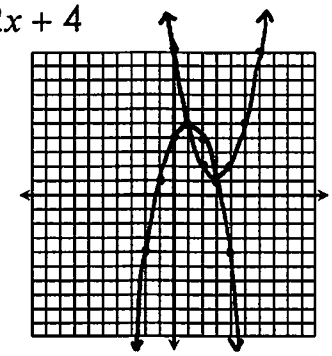
$y = -x - 8$

$\boxed{\emptyset}$



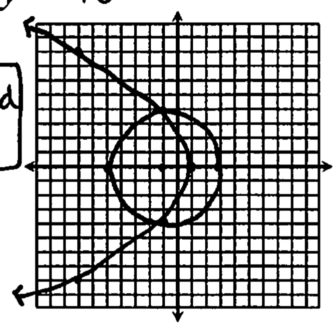
3. $\begin{cases} y = x^2 - 6x + 10 \\ y = -x^2 + 2x + 4 \end{cases}$

$\boxed{(1,5) \text{ and } (3,1)}$



4. $\begin{cases} y^2 = -8(x-1) \\ (x+1)^2 + y^2 = 16 \end{cases}$

$\boxed{(-1,4) \text{ and } (1,4)}$



Non-Linear Systems
(Solve Algebraically)

Date:

Solve each system algebraically.

1. $\begin{cases} y = x^2 - 9x + 15 \\ 3x + y = 7 \rightarrow y = -3x + 7 \end{cases}$

$x^2 - 9x + 15 = -3x + 7$

$x^2 - 6x + 8 = 0$

$(x-4)(x-2) = 0$

$x = 4$ | $x = 2$

$y = -3(4) + 7$ | $y = -3(2) + 7$

$y = -5$ | $y = 1$

$\boxed{(2,1) \text{ and } (4,-5)}$

2. $\begin{cases} y = x^2 + 3x - 18 \\ y = x^2 - 2x + 2 \end{cases}$

$x^2 + 3x - 18 = x^2 - 2x + 2$

$3x - 18 = -2x + 2$

$5x = 20$

$x = 4$

$y = 4^2 + 3(4) - 18$

$y = 10$

$\boxed{(4, 10)}$

3. $\begin{cases} y = 2x^2 + 10x + 7 \\ y = -x^2 - 5x - 5 \end{cases}$

$2x^2 + 10x + 7 = -x^2 - 5x - 5$

$3x^2 + 15x + 12 = 0$

$3(x^2 + 5x + 4) = 0$

$3(x+4)(x+1) = 0$

$3 \neq 0$ | $x = -4$ | $x = -1 \rightarrow y = -(-1)^2 - 5(-1) - 5$, $y = -1$

$y = -(-4)^2 - 5(-4) - 5$

$y = -1$

$\boxed{(-4,-1) \text{ and } (-1,-1)}$

4. $\begin{cases} 5x^2 + 5y^2 = 85 \\ x - y = 3 \rightarrow y = x - 3 \end{cases}$

$\boxed{(4,-1) \text{ and } (1,-2)}$

$5x^2 + 5(x-3)^2 = 85$

$5x^2 + 5x^2 - 30x + 45 = 85$

$10x^2 - 30x - 40 = 0$

$10(x-4)(x+1) = 0$

$10 \neq 0$ | $x = 4$ | $x = -1$

$y = 4 - 3 = 1$ | $y = -1 - 3 = -4$

Sequences

Date:

Write the first five terms of each sequence.

1. $a_1 = -2; a_n = 5a_{n-1} + 3$ (for $n \geq 2$) $\boxed{-2, -7, -32, -167, -782}$

2. $a_1 = 3, a_2 = 7; a_n = a_{n-1} + 2a_{n-2}$ (for $n \geq 3$) $\boxed{3, 7, 13, 27, 53}$

3. $a_n = n^3 - 4n^2$ $\boxed{-3, -8, -9, 0, 25}$

4. $a_n = \frac{1}{n} + n$ $\boxed{2, \frac{5}{2}, \frac{10}{3}, \frac{17}{4}, \frac{26}{25}}$

5. Write a recursive formula: $\left\{1, 2, \frac{1}{2}, 4, \frac{1}{8}, 32, \ldots\right\}$

$\boxed{a_n = \frac{a_{n-2}}{a_{n-1}} \text{ (for } n \geq 3)}$

6. Write an explicit formula: $\{4, 10, 28, 82, 244, \ldots\}$

$\boxed{a_n = 3^n - 1}$

Series & Summations

Date:

Find S_7 for each series.

1. $\{2 + 3 + 6 + 18 + 108 + \ldots\}$
$+1944 + 209952$

$\boxed{S_7 = 212{,}033}$

2. $\{-2 + 6 - 18 + 54 - 162 + \ldots\}$

Expand and evaluate each series.

3. $\sum_{k=1}^{10}(3 - 4k) = -1 - 5 - 9 - 13 - 17 - 21 - 25 - 29 - 33 - 37$

$\boxed{= -190}$

4. $\sum_{m=2}^{18}(m + 1)^2 = 9 + 16 + 25 + 36 + 49 + 64 + 81 + 100 + 121 + 144 + 169$
$+ 196 + 225 + 256 + 289 + 324 + 361$

$\boxed{= 2465}$

5. $\frac{2}{3} \cdot \sum_{x=8}^{24}\left(\frac{1}{2}x + 7\right) = \frac{2}{3}(11 + 11.5 + 12 + 12.5 + 13 + 13.5 + 14 + 14.5$
$+ 15 + 15.5 + 16 + 16.5 + 17 + 17.5 + 18$
$+ 18.5 + 19)$

$= \frac{2}{3}(255) = \boxed{170}$

Arithmetic Sequences

$a_n = d(n-1) + a_1$

Date:

Write a rule for each sequence, then find the indicated term.

1. $\{19, 12, 5, -2, -9, \ldots\}; a_{32}$

$a_n = -7(n-1) + 19$

$= -7n + 7 + 19$

$\boxed{= -7n + 26}$

$a_{32} = -7(32) + 26 = \boxed{-198}$

2. $\left\{1, \frac{8}{3}, \frac{13}{3}, 6, \frac{23}{3}, \ldots\right\}; a_{25}$

$a_n = \frac{5}{3}(n-1) + 1$

$= \frac{5}{3}n - \frac{5}{3} + 1$

$\boxed{= \frac{5}{3}n - \frac{2}{3}}$

$a_{25} = \frac{5}{3}(25) - \frac{2}{3} = \boxed{\frac{158}{3}}$

If the terms belong to an arithmetic sequence, write the rule and find the indicated value.

3. $a_1 = -37$ and $a_{15} = 19$; Find a_7

$19 = d(15-1) - 37$

$19 = 14d - 37$

$56 = 14d$

$4 = d$

$a_7 = 4(7-1) - 37$

$= 4(6) - 37$

$\boxed{= -13}$

Arithmetic Series

$S_n = n\left(\frac{a_1 + a_n}{2}\right)$

Date:

Find S_9 for each series.

1. $\{26 + 34 + 42 + 50 + \ldots\}$

$a_9 = 8(9-1) + 26 = 90$

$S_9 = 9\left(\frac{26+90}{2}\right) = \boxed{522}$

2. $\{(-5) + (-8) + (-11) + (-14) + \ldots\}$

$a_9 = -3(9-1) - 5 = -29$

$S_9 = 9\left(\frac{-5-29}{2}\right)$

$\boxed{= -153}$

Evaluate each series.

3. $\sum_{n=1}^{19}(5n - 17)$

$S_n = 19\left(\frac{-12+78}{2}\right)$

$\boxed{= 627}$

4. $\sum_{y=1}^{20}\left(8 - \frac{3}{2}y\right)$

$S_n = 20\left(\frac{6.5 - 22}{2}\right)$

$\boxed{= -155}$

5. $\sum_{p=2}^{37}(9 - n)$

$S_n = 36\left(\frac{7-28}{2}\right)$

$= \boxed{-378}$

6. $\sum_{i=5}^{24}\left(\frac{1}{4}i + 3\right)$

$S_n = 20\left(\frac{4.25+9}{2}\right)$

$\boxed{= 132.5}$

Geometric Sequences

$a_n = a_1 \cdot r^{n-1}$

Date:

Write a rule for each sequence, then find the indicated term.

1. $\{7, 21, 63, 189, \ldots\}; a_{10}$

$a_n = 7(3)^{n-1}$

$a_{10} = 7(3)^9$

$\boxed{= 137{,}781}$

2. $\{5120, -1280, -320, 80, \ldots\}; a_8$

$a_n = 5120\left(-\frac{1}{4}\right)^{n-1}$

$a_8 = 5120\left(-\frac{1}{4}\right)^7$

$\boxed{= \frac{-5}{16}}$

If the terms belong to a geometric sequence, write the rule and find the indicated value.

3. $a_3 = -35.2$ and $a_7 = -9011.2$; Find a_1

$a_7 = a_3 \cdot r^4$

$-9011.2 = -35.2\,(r)^4$

$256 = r^4$

$4 = r$

$-35.2 = a_1 \cdot 4^2$

$-35.2 = 16a_1$

$\boxed{-2.2 = a_1}$

Geometric Series

$S_n = \frac{a_1(1-r^n)}{1-r}$

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Date:

Find S_7 for each series.

1. $\{480 - 240 + 120 - 60 + \ldots\}$

$S_n = \frac{480\left(1-\left(-\frac{1}{2}\right)^7\right)}{1-(-1/2)}$

$\boxed{= 322.5}$

2. $\left\{\frac{1}{2} + \frac{3}{2} + \frac{9}{2} + \frac{27}{2} + \ldots\right\}$

$S_n = \frac{\frac{1}{2}(1-3^7)}{1-3}$

$\boxed{= 546.5}$

Evaluate each series.

3. $\sum_{k=1}^{9} 4^{k-1}$

$S_n = \frac{1(1-4^9)}{1-4}$

$\boxed{= 87{,}381}$

4. $\sum_{m=1}^{7} -2187 \cdot \left(\frac{1}{3}\right)^{m-1}$

$S_n = \frac{-2187\left(1-\left(\frac{1}{3}\right)^7\right)}{1-1/3}$

$\boxed{= -3279}$

5. $\sum_{i=3}^{10} -\frac{1}{8} \cdot (-2)^{i-1}$

$S_n = \frac{-\frac{1}{2}\left(1-(-2)^8\right)}{1-(-2)} = \boxed{42.5}$

6. $\sum_{k=2}^{8} 2.56 \cdot \left(\frac{5}{2}\right)^{k-1}$

$S_n = \frac{6.4\left(1-(5/2)^7\right)}{1-5/2} = \boxed{2599.9}$

Date:

Infinite Geometric Series

$S_n = \frac{a_1}{1-r}$

Determine if the series is convergent or divergent. Find the sum when possible.

1. $\{1-2+4-8+\ldots\}$

$r=-2$, divergent

2. $\{80+60+45+33.75+\ldots\}$

$r=3/4$, convergent

$S_n = \frac{80}{1-3/4} = \boxed{320}$

3. $\sum_{n=1}^{\infty} 30\cdot\left(\frac{3}{5}\right)^{k-1}$

$r=3/5$, convergent

$S_n = \frac{30}{1-3/5} = \boxed{75}$

4. $\sum_{y=1}^{\infty} \frac{2}{5}\cdot(-4)^{y-1}$

$r=-4$, divergent

5. $\sum_{x=3}^{\infty} -6\cdot\left(\frac{4}{3}\right)^{x-1}$

$r=\frac{4}{3}$, divergent

6. $\sum_{m=1}^{\infty} -\frac{27}{8}\cdot\left(-\frac{2}{3}\right)^{m-1}$

$r=-2/3$, convergent

$S_n = \frac{-27/8}{1-(-2/3)} = \boxed{\frac{-81}{40}}$

Date:

Arithmetic vs. Geometric

Determine whether the sequence is arithmetic, geometric, or neither. Identify the common difference or ratio.

1. $\left\{\frac{5}{32}, \frac{5}{8}, \frac{5}{2}, 10, \ldots\right\}$

Geometric, $r=4$

2. $\{-14, -15, -16, -17, \ldots\}$

Arithmetic, $d=-1$

3. $\left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \ldots\right\}$

Neither

4. $\left\{600, -180, 54, -\frac{81}{5}, \ldots\right\}$

Geometric, $r=-0.3$

Determine arithmetic or geometric, then find the sum.

5. $\{(-9)+(-1)+7+15+\ldots\}; S_{29}$

Arith; $a_{29} = 8(29-1)-9$

$= 215$

$S_n = 29\left(\frac{-9+215}{2}\right) = \boxed{2987}$

6. $\left\{\frac{1}{2}-\frac{1}{4}+\frac{1}{8}-\frac{1}{16}+\ldots\right\}; S_8$

Geometric

$S_n = \frac{\frac{1}{2}\left(1-(-1/2)^8\right)}{1-(-1/2)} = \boxed{\frac{85}{256}}$

Sequences Applications

Date:

1. Dan invested \$2500 in stock that is estimated to increase in value by 6% each year. Determine whether this represents an arithmetic or geometric situation, then write a formula and find the value of the stock in 30 years.

Geometric

$a_n = 2500(1.06)^{29}$

$\boxed{= \$13,545.97}$

2. Angela is at the library signing and giving away 400 books to promote a book she recently published. After the first hour, she had 362 books left. After the second hour, she had 324 books left. Determine whether this represents an arithmetic or geometric situation, then write a formula to find the number of books left after 8 hours.

0 hr - 400
1 hr - 362
2 hr - 324

$a_n = -38(8-1) + 362$

$\boxed{= 96 \text{ books}}$

Arithmetic

Series Applications

Date:

1. The table below shows the number of miles that Sophie ran for the first three days of her new exercise routine. Determine whether this represents an arithmetic or geometric situation, then find the total miles ran after 16 days.

Day	Miles
1	1.8
2	2.2
3	2.6

Arithmetic

$a_{16} = 0.4(15) + 1.8 = 7.8$

$S_n = 16\left(\frac{1.8 + 7.8}{2}\right) = \boxed{76.8 \text{ mi}}$

2. During a flu outbreak, the hospital recorded 12 cases the first week, 30 cases the second week, and 75 cases the third week. Determine whether this represents an arithmetic or geometric sequence, then find the total number of cases reported after 6 weeks.

1 wk - 12
2 wk - 30
3 wk - 75

$S_n = \frac{12(1 - 2.5^6)}{1 - 2.5}$

$= 1945.125$

$\boxed{1945 \text{ cases}}$

Geometric

Counting Principle, Permutations & Combinations

Date:

1. Initials are created by the first letter of the first name, middle name, and last name. How many 3-letter initials are possible?

$26 \cdot 26 \cdot 26 = \boxed{17{,}576}$

Determine if the situation represents a permutation or combination, then solve.

2. How many ways are there to select five employees from 24 to send to a conference?

${}_{24}C_5 = \boxed{42{,}504}$

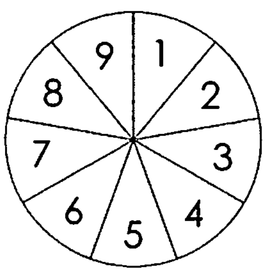
3. There are 14 violinists in the orchestra. How many ways can the conductor choose first, second, and third chair?

${}_{14}P_3 = \boxed{2{,}184}$

4. How many different ways can you arrange the letters in the word FIREFIGHTER?

$\frac{11!}{2!2!2!2!} = \boxed{2{,}494{,}800}$

Theoretical Probability



Date:

1. The spinner to the left is spun then a card is randomly selected from a standard deck. Find the probability of spinning a number less than 5, then getting a jack or a diamond.

$\frac{4}{9} \cdot \frac{16}{52} = \boxed{\frac{16}{117}}$

2. A bag of Skittles has 5 red, 11 orange, 7 yellow, 8 green, 14 blue, and 9 purple skittles. A skittle is randomly selected, eaten, then another is selected. Find each probability.

a) P(yellow then blue)

$\frac{7}{54} \cdot \frac{14}{53} = \boxed{\frac{49}{1431}}$

b) P(purple then red)

$\frac{9}{54} \cdot \frac{5}{53} = \boxed{\frac{5}{318}}$

c) P(two green)

$\frac{8}{54} \cdot \frac{7}{53} = \boxed{\frac{28}{1431}}$

d) P(two orange)

$\frac{11}{54} \cdot \frac{10}{53} = \boxed{\frac{55}{1431}}$

Conditional Probability

Date:

1. A number 1-25 is randomly chosen. What is the probability that it is a multiple of 3, given that it is at most 16?

$\boxed{\frac{5}{16}}$

2. Out of 32 people surveyed, 15 own a dog and 8 own a cat. Four of those who own a dog also own a cat. If a person is chosen at random, find the probability that they own a dog, if it is known that they do not own a cat.

$\boxed{\frac{11}{24}}$

3. A taste test was conducted at the mall to see whether people prefer Coke or Pepsi. The results are shown in the table below. Find each probability.

	Males	Females
Coke	108	92
Pepsi	76	124

a) P(male | prefers Pepsi)

$\frac{76}{200} = \boxed{\frac{19}{50}}$

b) P(prefers Coke | female)

$\frac{92}{216} = \boxed{\frac{23}{54}}$

Frequencies

Date:

Four math classes took the same unit test. The joint relative frequencies are shown in the table to the left. Give the marginal joint frequencies. Then find each probability.

	Pass	Fail	Total
Class A	0.192	0.064	0.256
Class B	0.256	0.024	0.280
Class C	0.184	0.04	0.224
Class D	0.208	0.032	0.240
Total	0.840	0.260	1

1. P(failed | Class B)

$\frac{0.024}{0.280} = 0.0857$

$\boxed{8.57\%}$

2. P(Class D | passed)

$\frac{0.208}{0.840} = 0.2476$

$\boxed{24.76\%}$

The Binomial Theorem

Date:

Expand each binomial:

1. $(x-4)^5$

$= {}_5C_0 x^5(-4)^0 + {}_5C_1 x^4(-4)^1 + {}_5C_2 x^3(-4)^2 + {}_5C_3 x^2(-4)^3 + {}_5C_4 x^1(-4)^4 + {}_5C_5 x^0(-4)^5$

$\boxed{= x^5 - 20x^4 + 160x^3 - 640x^2 + 1280x - 1024}$

2. $(3k-1)^6$

$= {}_6C_0 (3k)^6(-1)^0 + {}_6C_1 (3k)^5(-1)^1 + {}_6C_2 (3k)^4(-1)^2 + {}_6C_3 (3k)^3(-1)^3 + {}_6C_4 (3k)^2(-1)^4 + {}_6C_5 (3k)^1(-1)^5 + {}_6C_6 (3k)^0(-1)^6$

$\boxed{= 729k^6 - 1458k^5 + 1215k^4 - 540k^3 + 135k^2 - 18k + 1}$

3. Find the third term of $(2m + n)^8$.

${}_8C_2 (2m)^6(n)^2 = \boxed{1792m^6n^2}$

Binomial Probability

Date:

1. Type O blood is the most frequently occurring blood type found in 43% of the population. If 12 people are randomly selected, find the probability that exactly 7 people have this blood type.

${}_{12}C_7 (0.43)^7(0.57)^5 = 0.1295$ $\boxed{12.95\%}$

2. A coin is tossed four times. What is the probability of getting tails no more than one time?

$\left.\begin{array}{l} {}_4C_1 (\frac{1}{2})^1(\frac{1}{2})^3 = 0.25 \\ {}_4C_0 (\frac{1}{2})^0(\frac{1}{2})^4 = 0.0625 \end{array}\right\} 0.3125$ $\boxed{31.25\%}$

3. A die is rolled eight times. What is the probability that an odd number shows up at least six times?

$\left.\begin{array}{l} {}_8C_6 (\frac{1}{2})^6(\frac{1}{2})^2 = 0.1094 \\ {}_8C_7 (\frac{1}{2})^7(\frac{1}{2}) = 0.0313 \\ {}_8C_8 (\frac{1}{2})^8(\frac{1}{2})^0 = 0.0039 \end{array}\right\} 0.1446$ $\boxed{14.46\%}$

Measures of Variation	**Date:**

Find the mean absolute deviation, standard deviation, and variance for each data set.

1. {16, 4, 18, 7, 9, 23, 5, 19, 11, 8} $\mu = 12$

$$= \frac{4+8+6+5+3+11+7+7+1+4}{10}$$

$$= \frac{56}{10} = 5.6$$

MAD = 5.6

Variance: σ^2 = 38.56

Standard Deviation: σ = 6.21

2. {146, 150, 144, 152, 147, 143} $\mu = 147$

$$= \frac{1+3+3+5+0+4}{6}$$

$$= \frac{16}{6} = 2.67$$

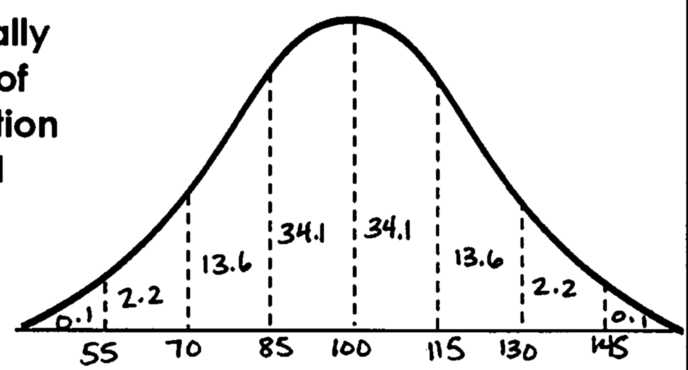
MAD = 2.67

Variance: σ^2 = 9.99

Standard Deviation: σ = 3.16

Normal Distribution	**Date:**

Adult IQ scores are normally distributed with a mean of 100 and a standard deviation of 15. Label the normal distribution curve, then answer the questions



1. What percent of adults have an IQ no more than 85?

15.9%

2. What is the probability that an adult will have an IQ between 70 and 100? 47.7%

3. Approximately how many adults out of 500 have an IQ of at least 115?

15.9% → About 79 adults

z-Scores

Date:

1. A group of test scores are normally distributed with a mean of 85 and a standard deviation of 4. Find each z-score.

a) 82 $\frac{82-85}{4} = \boxed{-0.75}$

b) 96 $\frac{96-85}{4} = \boxed{2.75}$

c) 73 $\frac{73-85}{4} = \boxed{-3}$

2. Human body temperature is normally distributed with a mean of 98.2° and a standard deviation of 0.5°. If Nate's z-score is 2.2, find his temperature.

$2.2 = \frac{x-98.2}{0.5}$

$1.1 = x-98.2$

$x = \boxed{99.3°}$

3. The number of pages in a book is normally distributed with a standard deviation of 25. If a particular book has 175 pages with a z-score of -0.4, find the mean number of pages.

$-0.4 = \frac{175-\mu}{25}$

$-10 = 175-\mu$

$-185 = -\mu$

$\boxed{\mu = 185}$

Standard Normal Distribution

Date:

Credit scores for a group of 60 people are normally distributed with a mean of 680 and a standard deviation of 42.

1. If X represents the credit score, find each probability:

a. $P(X < 600)$

$z < -1.90$

$\boxed{2.87\%}$

b. $P(X > 750)$

$z > 1.67$

$\boxed{4.75\%}$

c. $P(700 < X < 825)$

$0.48 < z < 3.45$

$\boxed{31.53\%}$

2. Approximately how many people have a credit score of at least 720?

$z > 0.95$

17.11%

$\boxed{\text{About 10 people}}$

3. Approximately how many people have a credit score between 625 and 790?

$-1.31 < z < 2.62$

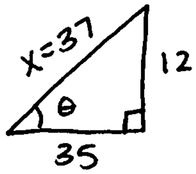
90.05%

$\boxed{\text{About 54 people}}$

Trigonometric Functions

Date:

1. If $\tan\theta = \dfrac{12}{35}$, find the remaining function values.



$12^2 + 35^2 = x^2$
$1369 = x^2$
$37 = x$

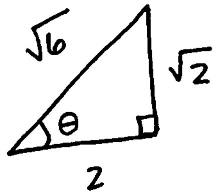
$\sin\theta = \underline{\frac{12}{37}}$ $\csc\theta = \underline{\frac{37}{12}}$

$\cos\theta = \underline{\frac{35}{37}}$ $\sec\theta = \underline{\frac{37}{35}}$

$\tan\theta = \underline{\frac{12}{35}}$ $\cot\theta = \underline{\frac{35}{12}}$

2. If $\sec\theta = \dfrac{\sqrt{6}}{2}$, find the remaining function values.

$\cos\theta = \frac{2}{\sqrt{6}}$



$2^2 + x^2 = \sqrt{6}^2$
$4 + x^2 = 6$
$x^2 = 2$
$x = \sqrt{2}$

$\sin\theta = \underline{\frac{\sqrt{2}}{\sqrt{6}} = \frac{\sqrt{3}}{3}}$ $\csc\theta = \underline{\frac{\sqrt{6}}{\sqrt{2}} = \sqrt{3}}$

$\cos\theta = \underline{\frac{2}{\sqrt{6}} = \frac{\sqrt{6}}{3}}$ $\sec\theta = \underline{\frac{\sqrt{6}}{2}}$

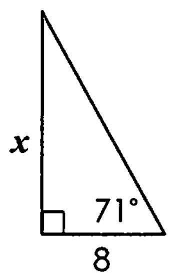
$\tan\theta = \underline{\frac{\sqrt{2}}{2}}$ $\cot\theta = \underline{\frac{2}{\sqrt{2}} = \sqrt{2}}$

Finding Missing Sides & Angles

Date:

Find each missing side or angle:

1.

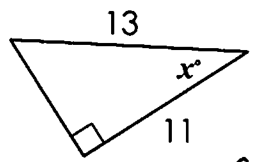


$\tan 71 = \frac{x}{8}$

$8 \cdot \tan 71 = x$

$\boxed{x = 23.2}$

2.

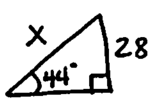


$\cos x = \frac{11}{13}$

$x = \cos^{-1}\left(\frac{11}{13}\right)$

$\boxed{x = 32.2°}$

3. A ladder placed against a building reaches a window that is 28 feet above the ground. If the angle of elevation from the bottom of the ladder to the window is 44°, find the length of the ladder.

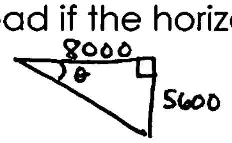


$\sin 44 = \frac{28}{x}$

$x = \frac{28}{\sin 44}$

$\boxed{x = 40.3 \text{ ft}}$

4. A helicopter flying at an altitude of 5,600 feet spots a landing pad below. Find the angle of depression from the helicopter to the pad if the horizontal distance between them is 8,000 feet.



$\tan\theta = \frac{5600}{8000}$

$\theta = \tan^{-1}\left(\frac{5600}{8000}\right)$ $\theta = 35°$

Degrees & Radians

Date:

Convert each degree measure to radians and radian measure to degrees:

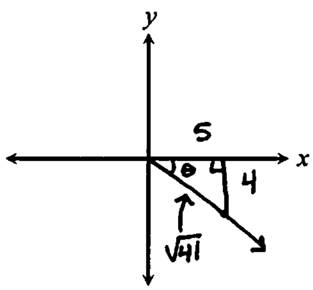
1. $63° \cdot \frac{\pi}{180} = \boxed{\frac{7\pi}{20}}$ 2. $-320° \cdot \frac{\pi}{180} = \boxed{\frac{-16\pi}{9}}$ 3. $\frac{13\pi}{10} \cdot \frac{180}{\pi} = \boxed{234°}$ 4. $-\frac{7\pi}{15} \cdot \frac{180}{\pi} = \boxed{-84°}$

Give two coterminal angles (one positive and one negative) and a reference angle for each angle:

5. $155°$ $515°, -205°$

6. $-\frac{2\pi}{9}$ $\frac{16\pi}{9}, -\frac{20\pi}{9}$

7. Sketch angle θ in standard form with a terminal side that passes through the point (5, -4). Find the exact value for each function.



$x=5$
$y=4$
$r=\sqrt{41}$

$\sin\theta = \frac{-4\sqrt{41}}{41}$ $\csc\theta = \frac{-\sqrt{41}}{4}$

$\cos\theta = \frac{5\sqrt{41}}{41}$ $\sec\theta = \frac{\sqrt{41}}{5}$

$\tan\theta = \frac{-4}{5}$ $\cot\theta = \frac{-5}{4}$

The Unit Circle

Date:

1. Which trigonometric functions are negative for angles terminating in Quadrant IV?

Sine, tangent

2. If $\csc\theta > 0$ and $\tan\theta < 0$, which quadrant(s) could the terminal side of θ lie? Quadrant II

Find the exact value of each trigonometric function:

3. $\cos 225°$ $-\frac{\sqrt{2}}{2}$

4. $\tan 30°$ $\frac{\sqrt{3}}{3}$

5. $\sec 180°$ -1

6. $\sin\frac{5\pi}{6}$ $\frac{1}{2}$

7. $\cot\frac{3\pi}{2}$ 0

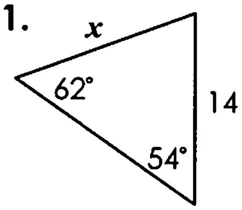
8. $\csc\frac{4\pi}{3}$ $\frac{-2\sqrt{3}}{3}$

Law of Sines

Date:

Find each missing measure to the nearest tenth:

1.

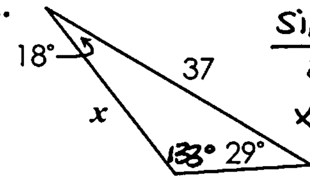


$\frac{\sin 62}{14} = \frac{\sin 54}{x}$

$x \cdot \sin 62 = 14 \sin 54$

$\boxed{x = 12.8}$

2.



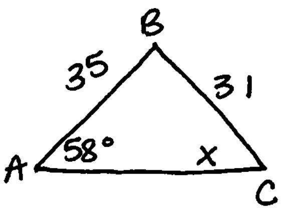
$\frac{\sin 133}{37} = \frac{\sin 29}{x}$

$x \cdot \sin 133 = 37 \cdot \sin 29$

$\boxed{x = 24.5}$

Information about ΔABC is given below. Give all possibilities.

3. $m\angle A = 58°$, $AB = 35$, $BC = 31$, find $m\angle C$.



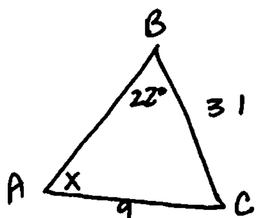
$\frac{\sin 58}{31} = \frac{\sin x}{35}$

$31 \cdot \sin x = 35 \cdot \sin 58$

$\sin x = 0.95747$

$\boxed{x = 73.2°, 106.8°}$

4. $m\angle B = 22°$, $BC = 31$, $AC = 9$, find $m\angle A$.



$\frac{\sin 22}{9} = \frac{\sin x}{31}$

$31 \cdot \sin 22 = 9 \cdot \sin x$

$1.2903 = \sin x$

$x = \varnothing$

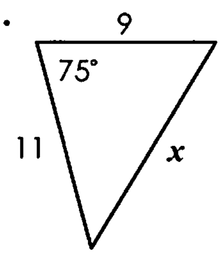
$\boxed{\text{No Solution}}$

Law of Cosines

Date:

Find each missing measure to the nearest tenth:

1.



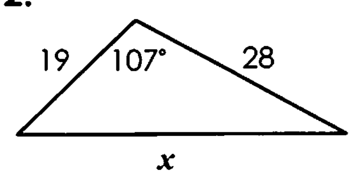
$x^2 = 11^2 + 9^2 - 2(11)(9)\cos 75$

$x^2 = 202 - 198 \cdot \cos 75$

$x^2 = 150.7538$

$\boxed{x = 12.3}$

2.



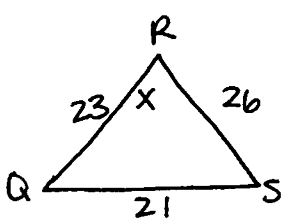
$x^2 = 19^2 + 28^2 - 2(19)(28) \cdot \cos 107$

$x^2 = 1145 - 1064 \cdot \cos 107$

$x^2 = 1456.0835$

$\boxed{x = 38.2}$

3. In ΔQRS, $QR = 23$, $RS = 26$, and $QS = 21$. Find $m\angle R$.



$21^2 = 23^2 + 26^2 - 2(23)(26)\cos x$

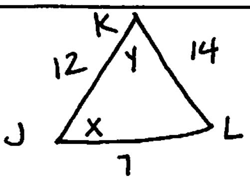
$441 = 1205 - 1196 \cdot \cos x$

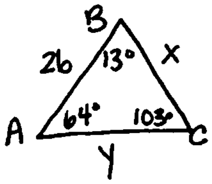
$-764 = -1196 \cos x$

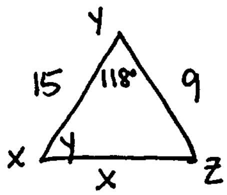
$0.6388 = \cos x$

$\boxed{x = 50.3°}$

Solving Triangles







Date:

Use the information given to solve each triangle:

1. In ΔJKL, $JK = 12$, $KL = 14$, and $JL = 7$.

$14^2 = 12^2 + 7^2 - 2(12)(7)\cos X$

$196 = 193 - 168\cos X$

$-0.018 = \cos X$

$X = 91°$

$\frac{\sin 91}{14} = \frac{\sin Y}{7}$

$0.4999 = \sin Y$

$Y = 30°$

$m\angle J =$ 91°

$m\angle K =$ 30°

$m\angle L =$ 59°

2. In ΔABC, $m\angle C = 103°$, $m\angle A = 64°$, and $AB = 26$.

$\frac{\sin 64}{x} = \frac{\sin 103}{26}$

$x = 24$

$\frac{\sin 103}{26} = \frac{\sin 13}{y}$

$y = 6$

$m\angle B =$ 13°

$AC =$ 6

$BC =$ 24

3. In ΔXYZ, $m\angle Y = 118°$, $YZ = 9$ and $XY = 15$.

$x^2 = 15^2 + 9^2 - 2(15)(9)\cos 118$

$x^2 = 306 - 270 \cdot \cos 118$

$x^2 = 432.7573$

$x = 20.8$

$\frac{\sin 118}{20.8} = \frac{\sin y}{9}$

$\sin y = 0.3820$

$y = 22.5°$

$m\angle X =$ 22.5°

$m\angle Z =$ 39.5°

$XZ =$ 20.8

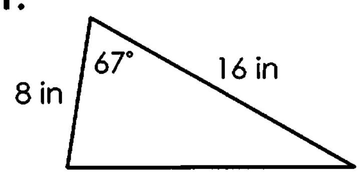
Area of Triangles

$A = \frac{1}{2} a \cdot b \cdot \sin C$

Date:

Find the area of each triangle to the nearest tenth:

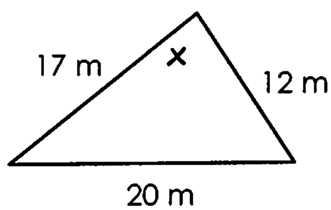
1.



$A = \frac{1}{2}(8)(16)\sin 67$

$\boxed{= 58.9 \text{ in}^2}$

2.



$20^2 = 17^2 + 12^2 - 2(17)(12)\cos x$

$400 = 433 - 408\cos x$

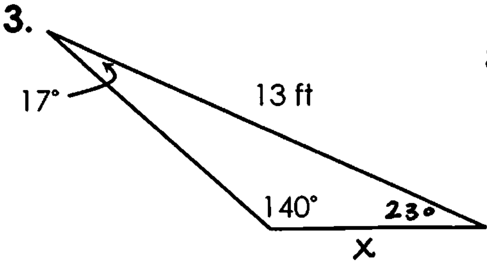
$0.0809 = \cos x$

$x = 85.4°$

$A = \frac{1}{2}(17)(12)\sin 85.4$

$\boxed{= 101.7 \text{ m}^2}$

3.



$\frac{\sin 17}{x} = \frac{\sin 140}{13}$

$x = 5.9$

$A = \frac{1}{2}(13)(5.9)\sin 23$

$\boxed{= 15 \text{ ft}^2}$

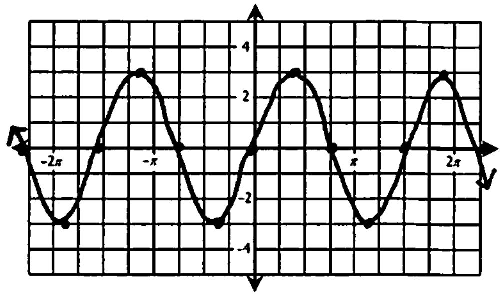
Graphing sin/cos/tan Functions

Date:

Identify the amplitude and period for each function, then graph.

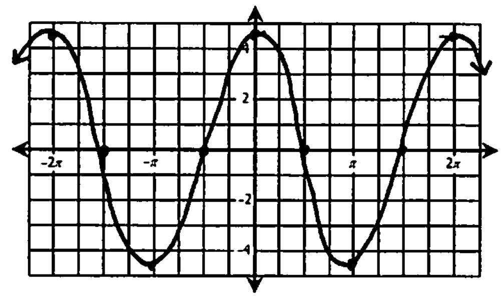
1. $y = 3 \cdot \sin\frac{4}{3}\theta$

Amp: 3 **Period:** $\frac{3\pi}{2}$



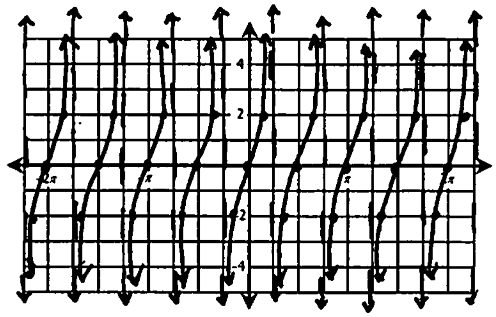
2. $y = \frac{9}{2} \cdot \cos\theta$

Amp: $\frac{9}{2}$ **Period:** 2π



3. $y = 2 \cdot \tan 2\theta$

Amp: undef **Period:** $\pi/2$

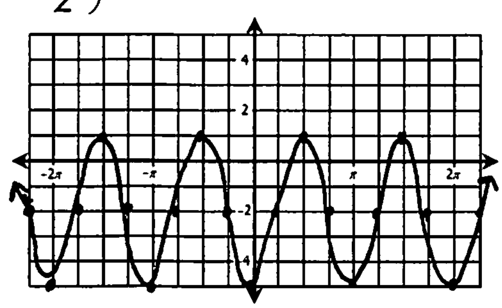


Translating Trigonometric Graphs

Date:

State the amplitude, period, phase shift, vertical shift, and midline for each function. Then graph.

1. $y = 3 \cdot \cos 2\left(\theta + \frac{3\pi}{2}\right) - 2$



Amplitude: 3

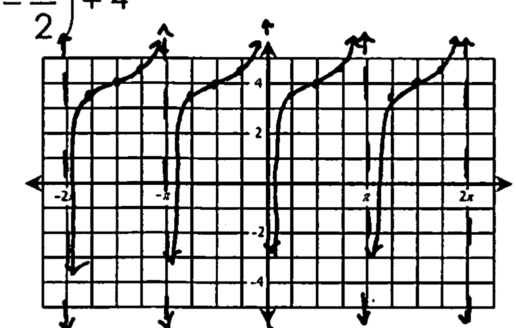
Period: π

Phase Shift: $\frac{3\pi}{2}$, left

Vertical Shift: down 2

Midline: $y = -2$

2. $y = \frac{1}{3} \cdot \tan\left(\theta - \frac{\pi}{2}\right) + 4$



Amplitude: undef.

Period: π

Phase Shift: $\frac{\pi}{2}$, right

Vertical Shift: up 4

Midline: $y = 4$

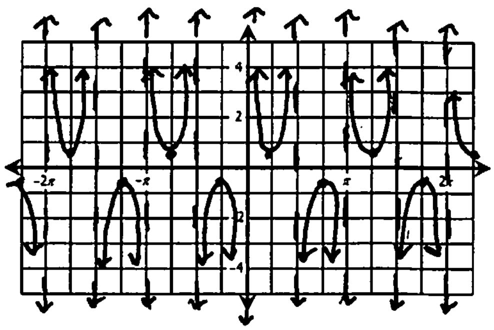
Graphing csc/sec/cot Functions

Date:

Identify the amplitude and period for each function, then graph.

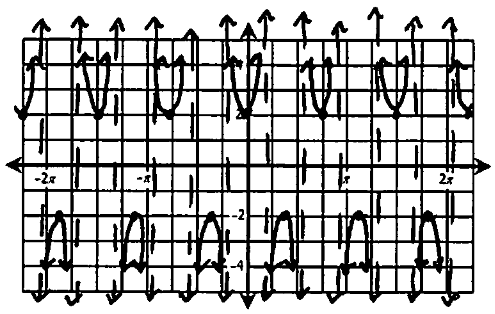
1. $y = \frac{1}{2} \cdot \csc 2\theta$

Amp: Undef Period: π



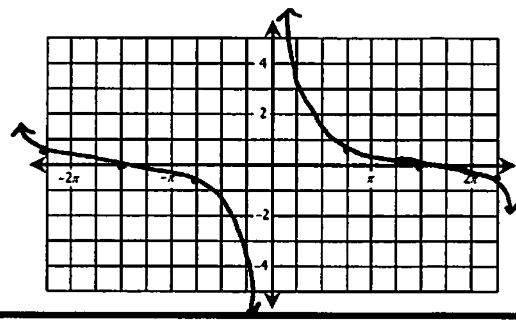
2. $y = 2 \cdot \sec \frac{8}{3}\theta$

Amp: Undef. Period: $\frac{3\pi}{4}$



3. $y = \frac{1}{4} \cdot \cot \frac{1}{3}\theta$

Amp: Undef Period: 3π



Fundamental Trig Identities

Date:

Find the exact value of each expression if $180° < \theta < 270°$.

1. If $\sin\theta = -\frac{\sqrt{3}}{4}$, find $\sec\theta$.

$\cos^2\theta + \left(-\frac{\sqrt{3}}{4}\right)^2 = 1$

$\cos^2\theta + \frac{3}{16} = 1$

$\cos^2\theta = \frac{13}{16}$

$\cos\theta = \frac{\sqrt{13}}{4} \rightarrow \boxed{\sec\theta = \frac{-4\sqrt{13}}{13}}$

2. If $\cot\theta = \frac{5}{2}$, find $\sin\theta$.

$(5/2)^2 + 1 = \csc^2\theta$

$25/4 + 1 = \csc^2\theta$

$\frac{29}{4} = \csc^2\theta$

$-\frac{\sqrt{29}}{2} = \csc\theta$

$\boxed{\sin\theta = \frac{-2\sqrt{29}}{29}}$

Simplify each expression.

3. $(\sec\theta + 1)(\sec\theta - 1) \cdot \cot\theta$

$= \sec^2\theta - \sec\theta + \sec\theta - 1 \cdot \cot\theta$

$= \underbrace{\sec^2\theta - 1}_{} \cdot \cot\theta$

$= \tan^2\theta \cdot \cot\theta$

$\boxed{= \tan\theta}$

4. $\frac{\cos^2\theta - \cos\theta \cdot \sec\theta}{\sin^3\theta}$

$= \frac{\cos^2\theta - 1}{\sin^3\theta}$

$= \frac{-\sin^2\theta}{\sin^3\theta}$

$= -\frac{1}{\sin\theta} = \boxed{-\csc\theta}$

Sum and Difference of Angles Identities

Date:

Find the exact value of each function.

1. $\cos 255°$

$= \cos 255 \cos 30 - \sin 255 \sin 30$

$= \left(-\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) - \left(-\frac{\sqrt{2}}{2}\right)\left(\frac{1}{2}\right)$

$= \boxed{\frac{-\sqrt{6}+\sqrt{2}}{4}}$

2. $\tan\left(-\frac{7\pi}{12}\right)$

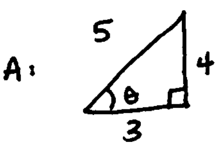
$= \frac{\tan \pi/4 - \tan 5\pi/6}{1 + \tan \pi/4 \cdot \tan 5\pi/6}$

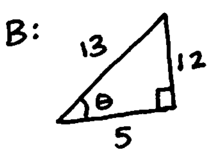
$= \frac{1-\left(-\frac{\sqrt{3}}{3}\right)}{1+1\left(-\frac{\sqrt{3}}{3}\right)} = \frac{\frac{3+\sqrt{3}}{3}}{\frac{3-\sqrt{3}}{3}} = \frac{3+\sqrt{3}}{3-\sqrt{3}}$

$= \frac{12+6\sqrt{3}}{6}$

$\boxed{= 2+\sqrt{3}}$

3. Given angles A and B in quadrant II with $\sin A = \frac{4}{5}$ and $\sin B = \frac{12}{13}$ find $\sin(A-B)$.





$\sin A \cos B - \cos A \sin B$

$= (4/5)(-5/13) - (-3/5)(12/13)$

$= \frac{-20}{65} + \frac{36}{65}$

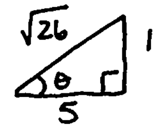
$= \boxed{\frac{16}{65}}$

Double-Angle & Half-Angle Identities

Date:

Find the exact value of each function.

1. $\tan\theta = -\frac{1}{5}$ and $\frac{\pi}{2} < \theta < \pi$; find $\sin 2\theta$.



$2\left(\frac{1}{\sqrt{26}}\right)\left(\frac{-5}{\sqrt{26}}\right) = \frac{-10}{26} = \boxed{\frac{-5}{13}}$

2. $\cos\theta = -\frac{12}{13}$ and $\pi < \theta < \frac{3\pi}{2}$; find $\tan 2\theta$.

$\frac{2\left(\frac{5}{12}\right)}{1-\left(\frac{5}{12}\right)^2} = \frac{\frac{5}{6}}{\frac{119}{144}} = \frac{5}{6} \cdot \frac{144}{119} = \boxed{\frac{120}{119}}$

3. $\sin\theta = \frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$; find $\cos\frac{\theta}{2}$.

$-\sqrt{\frac{1+(-4/5)}{2}} = -\sqrt{\frac{1/5}{2}} = -\sqrt{\frac{1}{10}} = \boxed{\frac{-\sqrt{10}}{10}}$

Proving Identities

Date:

Prove each identity.

1. $1-\sin\theta\cdot\cot\theta\cdot\cos\theta=\sin^2\theta$

$1-\sin\theta\cdot\frac{\cos\theta}{\sin\theta}\cdot\cos\theta=\sin^2\theta$

$1-\cos^2\theta=\sin^2\theta$

$\boxed{\sin^2\theta=\sin^2\theta}$

2. $\cot\theta=\dfrac{\cos2\theta+1}{\sin2\theta}$

$\cot\theta=\frac{2\cos^2\theta-1+1}{2\sin\theta\cos\theta}$

$\cot\theta=\frac{\cos\theta}{\sin\theta}$

$\boxed{\cot\theta=\cot\theta}$

3. $\dfrac{\csc\theta\cdot\cot^2\theta+\csc\theta}{\cot\theta\cdot\sec\theta}=\sec^2\theta$

$\dfrac{\frac{1}{\sin\theta}\cdot\frac{\cos^2\theta}{\sin^2\theta}+\frac{1}{\sin\theta}}{\frac{\cos\theta}{\sin\theta}\cdot\frac{1}{\cos\theta}}$

$=\dfrac{\frac{\cos^2\theta+\sin^2\theta}{\sin^3\theta}}{\frac{1}{\sin\theta}}=\sec^2\theta$

$=\boxed{\sec^2\theta=\sec^2\theta}$

4. $\sin(\pi-\theta)\cdot\sin\theta+\cos2\theta=\cos^2\theta$

$[\sin\pi\cos\theta-\cos\pi\sin\theta]\cdot\sin\theta+\cos2\theta=\cos^2\theta$

$[0-(-\sin\theta)]\cdot\sin\theta+\cos2\theta=\cos^2\theta$

$\sin^2\theta+\cos2\theta=\cos^2\theta$

$\sin^2\theta+(\cos^2\theta-\sin^2\theta)=\cos^2\theta$

$\boxed{\cos^2\theta=\cos^2\theta}$

Solving Trigonometric Equations

Date:

Solve for the interval $0<\theta<360°$. Give all solutions in **degrees**.

1. $4\tan\theta-\sqrt{3}=7\tan\theta$

$-\sqrt{3}=3\tan\theta$

$-\frac{\sqrt{3}}{3}=\tan\theta$

$\boxed{\theta=150°,330°}$

2. $2\sin\theta\cdot\tan\theta-\sqrt{3}\tan\theta=0$

$\tan\theta(2\sin\theta-\sqrt{3})=0$

$\tan\theta=0$ $\qquad$ $\sin\theta=\frac{\sqrt{3}}{2}$

$\theta=0°,180°,360°$ $\qquad$ $\theta=60°,120°$

$\boxed{\theta=0°,60°,120°,180°,360°}$

Solve for the interval $\frac{\pi}{2}<\theta<\frac{3\pi}{2}$. Give all solutions in **radians**.

3. $2\cos^2\theta-5\cos\theta=3$

$2\cos^2\theta-5\cos\theta-3=0$

$(2\cos\theta+1)(\cos\theta-3)=0$

$\cos\theta=-\frac{1}{2}$ $\qquad$ $\cos\theta=3$

$\boxed{\theta=\frac{2\pi}{3},\frac{4\pi}{3}}$ $\qquad$ $\theta=\varnothing$

4. $4-4\cos^2\theta-\tan\theta\cdot\cot\theta=0$

$4-4\cos^2\theta-1=0$

$-4\cos^2\theta=-3$

$\cos^2\theta=\frac{3}{4}$

$\cos\theta=\frac{\sqrt{3}}{4}$

$\theta=\cancel{\frac{\pi}{6}},\cancel{\frac{11\pi}{6}}$

$\boxed{\text{No Solution}}$

CREDITS

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Art with Jenny K

Many thanks to these talented artists!